Mathematical Simulation of Blood Flow

A Literature Review and Idealized Stenosis Study

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Abstract.

This report specifies and audits a reduced-order workflow for comparing velocity outputs in an idealized stenotic vessel. It is not a physiological or clinical validation study. The implemented solver is the Julia `StenosisHemodynamics` realization of an $R_{\max}$-normalized Canic-derived one-dimensional area–flow model, with physical area and flow mapped to solver variables $a = R^2$ and $q = Q_{\text{phys}}/\pi$. The principal C23/C40 comparison uses MUSCL finite volume with minmod-limited Rusanov fluxes and native SSPRK3 time integration; first-order and Richtmyer/Lax–Wendroff finite-volume paths, modal DG degrees $p = 0, 1, 2$, and generated stationary-Stokes finite-element solves provide implementation and sensitivity context.

The numerical record combines manufactured-solution rows, zero-forcing rest-state drift diagnostics, CFL, characteristic-speed, mass, and positivity metadata, sampled self-convergence, native/SciML cross-integrator checks, stationary-Stokes sensitivity, and resolved-velocity output-operator checks. These records expose an important limitation: the present Rusanov/MUSCL flux–source realization is not well balanced for the geometry-rest state, and zero-input rest-state runs on the C23/C40 geometries retain artificial flow of the same order as the reported comparison-flow scale at $t = 1.0$ s. Section-averaged axial velocities are then compared with two resolved three-dimensional velocity datasets using plane–tetrahedron quadrature. Because the wall models, boundary conditions, transient histories, and production grid/time sensitivities are not fully matched, the reported C23/C40 metrics are single-realization cross-model descriptors under this operator, not converged accuracy estimates and not support for pressure-ratio, FFR, physiological, or clinical claims.

Keywords:

computational hemodynamics, blood flow simulation, stenotic vessels, Navier–Stokes, reduced-order models, idealized stenosis

Contents

1 Introduction

Coronary and arterial stenoses motivate a basic computational question: how do geometry, wall properties, rheology, and boundary data affect blood flow through a narrowed vessel? Anatomical narrowing alone does not determine functional obstruction, so a useful computational report must distinguish the physical quantity being predicted, the model tier that produces it, and the evidence used to check the computation. This report stays on the mathematical and numerical side of that problem. It studies a C^∞ idealized stenosis geometry, an extended one-dimensional area–flow realization, and a bounded comparison with resolved 3D velocity data.

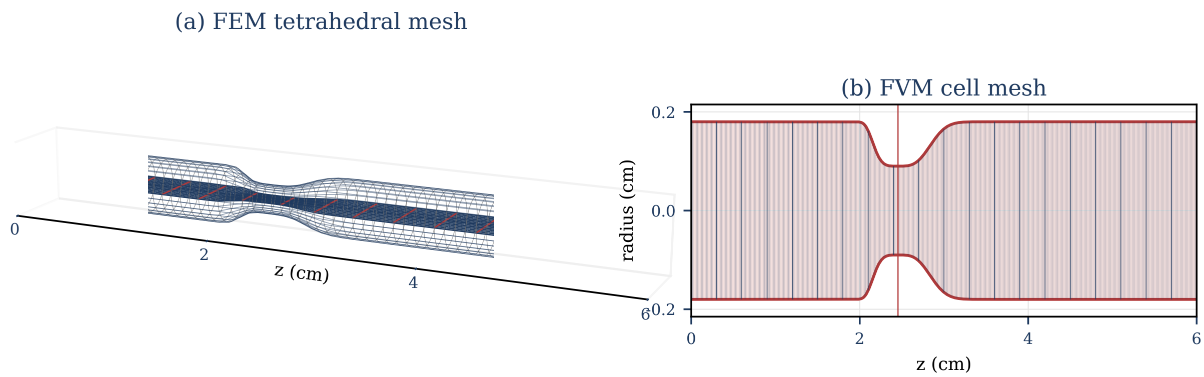

Figure 1: Finite-element and finite-volume views of the same C^∞ idealized stenosis geometry used in this report. Panel (a) shows the generated tetrahedral Gridap mesh for the stationary-Stokes initializer; panel (b) shows the one-dimensional finite-volume grid sampled along the reference radius profile $R_0(z)$. The takeaway is that one declared smooth geometry can feed distinct discretizations and controlled local refinement studies; the panels show geometry only, not velocity, pressure-drop, pressure-ratio, structural deformation, or patient-specific clinical data.

1.1 Clinical and Computational Motivation

Anatomy–function distinction. Coronary stenosis is the motivating lesion class because a visible narrowing does not automatically imply a flow-limiting obstruction. Clinical assessment therefore distinguishes anatomical stenosis detection from functional pressure–flow assessment [1, p. 4401] [2, Chs. 1–2 and 5] [3, Sec. 1]. Coronary lesion-imaging studies can describe local geometry and lesion length, but those anatomical descriptors do not by themselves define a pressure–flow output [4, Abstract, Methods]. This anatomy–function split motivates reporting pressure drop and reduced pressure ratio under declared observation and reference conventions, rather than treating stenosis severity $s_{\max}$ alone as the reported reduced output. Section 5 reports velocity diagnostics only; the pressure-drop and pressure-ratio conventions fix reduced-output vocabulary for the model framework. Figure 2 summarizes this motivation as a schematic distinction between visible narrowing and pressure–flow loss.

Clinical fractional flow reserve (FFR) is an invasive pressure-wire index, conventionally interpreted in its measurement setting as distal coronary pressure divided by aortic pressure during maximal hyperemia [2, Ch. 5, Sec. 5.5] [5, Abstract]. FAME lesion-level evidence documented the discordance between angiographic and functional severity and is the clinical source for the familiar FFR ≤ 0.80 decision context [6, Abstract]. Computed tomography-derived FFR (CT-FFR) studies supply noninvasive comparison vocabulary by inferring pressure–flow quantities from coronary computed tomography angiography (CCTA)-derived anatomy and comparing them against invasive FFR or related reference measurements [7, Abstract, Methods] [3, Secs. 1.1–1.2]. Those clinical settings motivate the output vocabulary but are not reproduced by the completed numerical study in this report. The reduced pressure ratio PR_{1D} is therefore a convention-governed model output, not a reported FFR, CT-FFR, or pressure-measurement comparison.

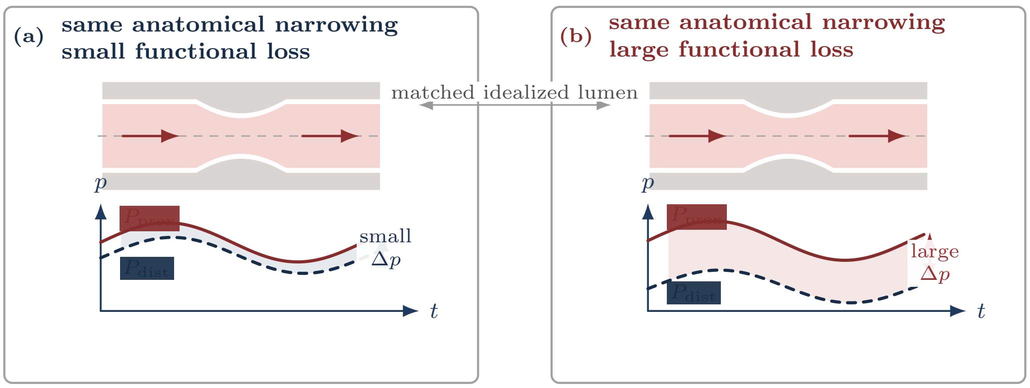

Figure 2: Pressure–flow motivation for reduced pressure outputs. The same anatomical narrowing can correspond to either a small functional pressure loss in panel (a) or a larger loss in panel (b). The reduced-output target is therefore the declared pressure-loss signature, not stenosis severity alone.

For the selected model gauge-pressure field p_{g}, define the observed proximal and distal pressure traces through declared scalar observation maps:

$$P_{\mathrm{prox}}(t) \;:=\; \mathcal{O}_{\mathrm{prox}}[p_{\mathrm{g}}(\cdot,t)], \qquad P_{\mathrm{dist}}(t) \;:=\; \mathcal{O}_{\mathrm{dist}}[p_{\mathrm{g}}(\cdot,t)].$$

The pressure-drop trace is invariant under common additive pressure shifts:

$$\Delta p(t) \;:=\; P_{\mathrm{prox}}(t) - P_{\mathrm{dist}}(t).$$

Unlike $\Delta p(t)$, a pressure ratio depends on the pressure reference used before forming the numerator and denominator. A reduced pressure-ratio output must therefore declare its proximal and distal observation maps, pressure reference, averaging window, and denominator admissibility. Convention C.3 records those requirements and the trace-level definition of $\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$. Its high-flow averaging window is a recorded output window, not a clinical hyperemia model unless an external measurement comparison is separately specified. Figure 3 shows the corresponding axial-tap shorthand for an idealized 1D stenosis.

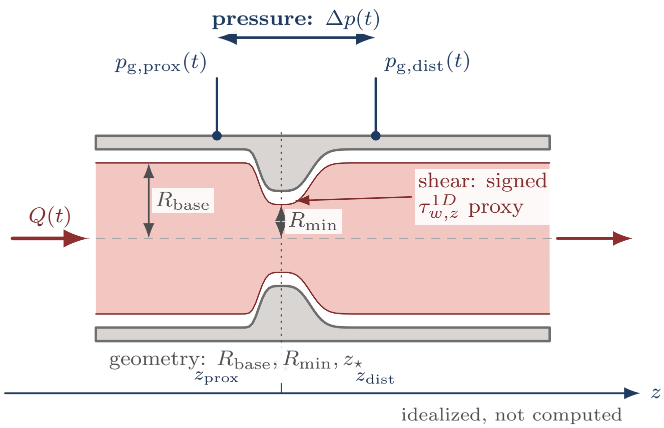

Figure 3: Schematic tap and output convention for an idealized stenosis, not drawn from a computed case. The blue pressure taps define $\Delta p(t)$, the gray geometry landmarks locate the stenosis region, and the red/orange cue marks the reduced shear-proxy convention. Convention C.3 supplies the pressure-ratio observation maps, averaging window, pressure reference, and admissibility rule for $\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$.

Shear diagnostics. Pressure drops and pressure ratios are lesion-scale pressure-output conventions. A resolved wall-shear-stress output belongs to a resolved stress state and must declare the stress tensor, wall normal, wall location, output operator, and signed tangent direction when a signed component is reported [8, Abstract, Sec. 2, pp. 8–10]. Reduced shear proxies are model-tier diagnostics rather than clinical anatomy–function outputs; Convention C.4 records the declaration requirements for comparisons with traction-based wall shear stress.

1.2 Hemodynamics and Modeling Requirements in Stenotic Arteries

This subsection fixes the macroscopic assumptions needed before the model classes are compared. It introduces only the continuum vocabulary that later supports the 3D PDE tier, the reduced 1D model, and the rheology/closure catalog; proof-level transport details are deferred to Appendix E.

1.2.1 Physiological scale and continuum idealization

The present work is posed at the macrocirculatory arterial scale. At this scale, blood is commonly modeled as a viscous fluid whose bulk behavior can be described by continuum mechanics. The point is not to resolve individual red blood cells, platelets, or plasma constituents. Instead, the model aims to predict macroscopic observables such as flow rate, pressure, pressure drop, pressure ratio, and velocity profiles under declared assumptions. This scale separation is standard in large-vessel hemodynamic modeling, while the small-vessel and low-shear regimes require additional rheological caution [9, 10, 11].

For a final time $T_{\mathrm{final}} > 0$, let $I_T := (0, T_{\mathrm{final}})$ denote the open interior-time interval and $\overline{I}_T := [0, T_{\mathrm{final}}]$ denote the closed interval used for initial data and flow maps. Define the time-dependent blood domain and its spatial-temporal domain as:

$$\begin{cases} \Omega_B(t) \ := \ \{\boldsymbol{x} \in \mathbb{R}^3 : \boldsymbol{x} \text{ lies inside the vessel at time } t\} \\ \mathcal{Q}_T \ := \ \{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}. \end{cases}$$

The reference and current configurations are denoted $\Omega_B(0)$ and $\Omega_B(t)$, respectively, with $\Omega_B(t)$ depending on time $t \in \overline{I}_T$. This domain notation is defined in Appendix B; boundary and function-space conventions are summarized in Appendix D. In this report, the continuum assumption is a large-vessel idealization: individual blood cells, platelets, and plasma constituents are not resolved, and blood velocity, pressure, density, and Cauchy stress are represented by sufficiently regular fields on $\mathcal{Q}_T$. Cellular-scale effects may enter only through later constitutive choices [10, Part I, Ch. 1, Secs. 1.1–1.2, pp. 4–5] [9, Secs. 1.1.3 and 3.2, pp. 12–14 and 46–47]. The definitions below fix the continuum contract used by the later model hierarchy.

Assumption 1.1 (Blood-dynamics kinematic setting)**.** For $t \in \overline{I}_T$, blood motion is represented by a sufficiently regular orientation-preserving diffeomorphism

$$\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t), \quad \det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t > 0$$

It carries each material label $\boldsymbol{\xi}$ to its current location $\boldsymbol{x}(t, \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field $\boldsymbol{u} : \mathcal{Q}_T \to \mathbb{R}^3$ satisfies

$$\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi})).$$

The accompanying continuum fields are pressure $p : \mathcal{Q}_T \to \mathbb{R}$, density $\rho : \mathcal{Q}_T \to \mathbb{R}^+$, and Cauchy stress $\boldsymbol{T} : \mathcal{Q}_T \to \mathbb{R}^{3\times 3}$.

Figure 4 summarizes the change of viewpoint from material labels to Eulerian field evaluation.

Remark 1.2 (Eulerian vs. Lagrangian)*.* The Eulerian description views fields such as $\phi(t, \boldsymbol{x})$ or $\boldsymbol{u}(t, \boldsymbol{x})$ at fixed spatial locations. The Lagrangian description follows material particles $\boldsymbol{\xi}$ along trajectories generated by the velocity field and views fields as functions of the material label $\boldsymbol{\xi}$ and time t, such as $\widehat{\phi}(t, \boldsymbol{\xi}) = \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$ or $\widehat{\boldsymbol{u}}(t, \boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$. The flow map $\mathcal{L}_t$ is the change of variables between these descriptions, and the trajectory of a particle with label $\boldsymbol{\xi}$ is the characteristic curve $t \mapsto \boldsymbol{x}(t; \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. Where defined, the inverse $\mathcal{L}_t^{-1}$ maps an Eulerian location back to its material label.

Classical identities in this report are read under the standing regularity convention in Convention D.1. For trajectory existence and uniqueness, the standard ODE hypothesis is that the prescribed velocity field is continuous in time and uniformly Lipschitz in space; under that condition, particle trajectories exist uniquely and depend continuously on their initial labels [12, Ch. 2].

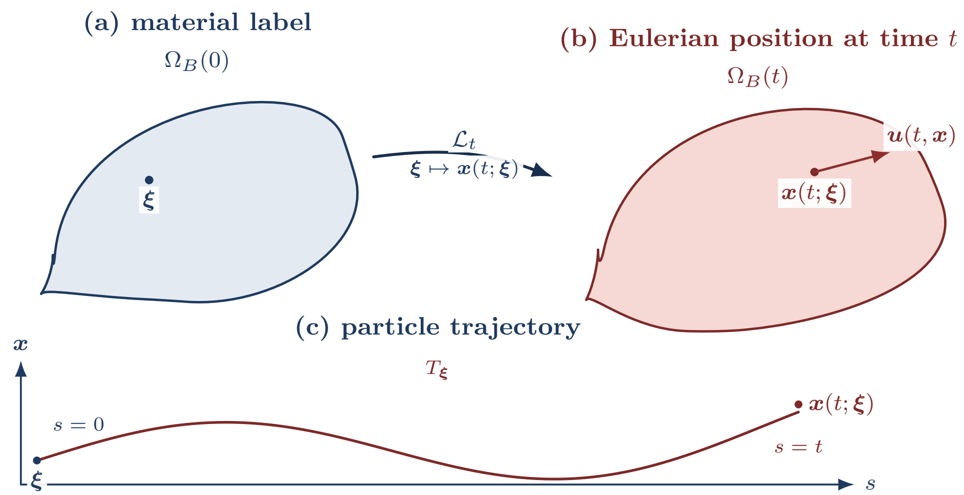

Figure 4: Lagrangian-to-Eulerian flow-map notation. The map $\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t)$ sends a material label $\boldsymbol{\xi}$ to the current spatial point $\boldsymbol{x}(t;\boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field is then evaluated at the occupied point, so the particle velocity satisfies $\partial_t \boldsymbol{x}(t;\boldsymbol{\xi}) = \boldsymbol{u}(t, \boldsymbol{x}(t;\boldsymbol{\xi})) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$. The lower curve $T_{\boldsymbol{\xi}}$ records the same labelled particle over $s \in \overline{I}_T$, from $\boldsymbol{\xi}$ at $s=0$ to $\boldsymbol{x}(t;\boldsymbol{\xi})$ at $s=t$.

1.2.2 Material derivative, transport, and balance laws

Definition 1.3 (Material derivative)**.** For $\phi \in C^1(\mathcal{Q}_T)$, the material derivative is the derivative along a particle trajectory:

$$D_t\phi(t,\boldsymbol{x}) \ := \ \frac{d}{dt}\phi(t,\mathcal{L}_t(\boldsymbol{\xi})) = \partial_t\phi(t,\boldsymbol{x}) + \boldsymbol{u}(t,\boldsymbol{x}) \cdot \nabla\phi(t,\boldsymbol{x}), \qquad \boldsymbol{\xi} = \mathcal{L}_t^{-1}(\boldsymbol{x}).$$

For vector fields, the material derivative is applied componentwise. Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}}\mathcal{L}_t$ and $\widehat{J}_t = \det \widehat{\boldsymbol{F}}_t$. In Eulerian notation this is written $\boldsymbol{F}_t(\boldsymbol{x}) = \widehat{\boldsymbol{F}}_t(\mathcal{L}_t^{-1}(\boldsymbol{x}))$ and $J_t(\boldsymbol{x}) = \det \boldsymbol{F}_t(\boldsymbol{x}) = \det \widehat{\boldsymbol{F}}_t(\mathcal{L}_t^{-1}(\boldsymbol{x}))$. The standard Jacobian evolution law is

$$D_t J_t = J_t \,\mathrm{div}\boldsymbol{u}.$$

This Jacobian identity supports the Reynolds transport theorem. Mass conservation is the separate balance law that gives the continuity equation used below; in the constant-density specialization, that equation reduces to the divergence-free constraint. Derivation details are deferred to Appendix E.

Theorem 1.4 (Reynolds Transport Theorem)**.** *Let $V(t) = \mathcal{L}_t(V(0))$ be a bounded material volume with Lipschitz boundary. If $\phi \in C^1(\mathcal{Q}_T)$, then*

$$\begin{aligned} \frac{d}{dt}\int_{V(t)} \phi \; d\boldsymbol{x} &= \int_{V(t)} (D_t\phi + \phi \; div\boldsymbol{u}) \; d\boldsymbol{x} \\ &= \int_{V(t)} (\partial_t\phi + div(\phi\boldsymbol{u})) \; d\boldsymbol{x}. \end{aligned}$$

Proof-level details for Theorem 1.4 and the momentum balance below are found in Appendix E. Assumption 1.1 supplies the 3D velocity, pressure, density, and stress fields used below. Mass conservation and linear momentum balance are imposed on material volumes and then localized to obtain the balance equations used in the model hierarchy.

Definition 1.5 (Material density preservation)**.** A flow has material density preservation if density is preserved along particle trajectories:

$$D_t\rho = 0.$$

Definition 1.6 (Incompressible flow)**.** A flow is incompressible, or volume preserving, if material volumes are preserved. With $\mathcal{L}_0 = \mathrm{Id}$ this is equivalent to

$$J_t \equiv 1 \quad \Longleftrightarrow \quad \mathrm{div}\boldsymbol{u} = 0.$$

Remark 1.7 (Divergence-free condition)*.* Mass conservation gives

$$\partial_t\rho + \mathrm{div}(\rho\boldsymbol{u}) = 0 \quad \Longleftrightarrow \quad D_t\rho + \rho\mathrm{div}\boldsymbol{u} = 0.$$

In the constant-density model used for incompressible Navier–Stokes, $\rho \equiv \rho_0 > 0$, so mass conservation reduces to $\mathrm{div}\boldsymbol{u} = 0$ in $\mathcal{Q}_T$.

Definition 1.8 (Linear momentum balance)**.** For a material fluid element $V(t) \subset \Omega_B(t)$, the linear momentum balance is

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\,dV = \int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\,dS + \int_{V(t)} \rho\boldsymbol{f_b}\,dV,$$

where $\boldsymbol{T}$ is the Cauchy stress tensor and $\boldsymbol{f_b}$ is body force per unit mass.

Definition 1.9 (Strong momentum balance)**.** Under the regularity and mass-balance assumptions above, the local momentum balance is

$$\rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}, \qquad \text{in } \Omega_B(t).$$

The stress law closes the momentum equation. Let the rate-of-deformation tensor be

$$\boldsymbol{D}(\boldsymbol{u}) \;:=\; \frac{1}{2}\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\top\right).$$

Definition 1.10. A fluid is Newtonian if, after separating the pressure contribution, its viscous stress depends linearly on the rate-of-deformation tensor $\boldsymbol{D}(\boldsymbol{u})$.

Remark 1.11 (Large-vessel continuum closure). Blood is often modeled as an isotropic continuum fluid at large-vessel scales. This is a closure assumption for averaged behavior, not a microscopic statement about cells suspended in plasma. The cited sources support the scale and rheology caveat: nearly constant large-vessel relative viscosity versus diameter- and hematocrit-dependent apparent viscosity in small tubes, not a direct cellular-level isotropy claim [9, Sec. 1.1.3, Fig. 1.14, pp. 12–14, Sec. 3.2, pp. 46–47] [11, Abstract, pp. H1770–H1772].

Remark 1.12 (Isotropic constitutive response). The stress law is first required to be objective, so it is not an artifact of the observer or coordinate frame. Assuming in addition material isotropy and linear dependence on $\boldsymbol{D}(\boldsymbol{u})$ gives the Newtonian isotropic form below; isotropy is a material-symmetry assumption, not the same statement as coordinate invariance.

Definition 1.13 (Newtonian, isotropic constitutive law)**.** For a Newtonian, isotropic fluid,

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta\,\boldsymbol{D}(\boldsymbol{u}) + \lambda\,\mathrm{div}(\boldsymbol{u})\,\mathbf{I},$$

where η is the dynamic viscosity and λ is the coefficient of the volumetric-viscous term.

Definition 1.14 (Incompressible stress tensor)**.** When $\mathrm{div}\boldsymbol{u} = 0$, the Newtonian incompressible Cauchy stress is

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta\,\boldsymbol{D}(\boldsymbol{u}).$$

Generalized-Newtonian blood models keep the stress-level constitutive form but allow the dynamic viscosity to depend on the local deformation-rate magnitude [10, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19]. With the shear-rate convention

$$\dot{\gamma} \ := \ \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})},$$

the incompressible generalized-Newtonian Cauchy stress is

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta_{\mathrm{eff}}(\dot{\gamma})\boldsymbol{D}(\boldsymbol{u}).$$

The pointwise Frobenius norm convention is

$$\|\boldsymbol{D}(\boldsymbol{u})\|^2 \ := \ \boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u}) = \sum_{i,j} D_{ij}(\boldsymbol{u})^2$$

When a source writes the viscosity as a function of $D : D$ or $|D|_{\mathrm{F}}^2$, this report translates it through the scalar shear-rate convention above. The coefficient η_{eff} is an effective dynamic viscosity; dividing by density gives the corresponding kinematic viscosity. The Newtonian incompressible model is recovered from this generalized Newtonian form when η_{eff} is constant. Positivity of a displayed scalar viscosity law alone is not used here as a standalone parabolicity theorem. Any theorem-level classification of the generalized-Newtonian PDE would additionally require stated lower-bound, growth, and monotonicity hypotheses for $2\eta_{\text{eff}}(\sqrt{2D:D})D$.

Definition 1.15 (Implemented effective-viscosity closure catalog)**.** A computed 1D realization uses the rheology descriptor $\mathcal{R}$ to choose an implemented scalar effective dynamic-viscosity law $\eta^{\mathcal{R}}_{\text{eff}}(\dot\gamma)$. The admitted descriptor values are `newtonian`, `carreau`, `carreau-yasuda`, `casson`, and `power-law`. The displayed superscripts abbreviate these descriptors, with newt, CY, and power denoting `newtonian`, `carreau-yasuda`, and `power-law`. This catalog defines the scalar dynamic-viscosity laws. A computed 1D realization evaluates the selected law on the shear-rate proxy in Appendix G and forms the closure-indexed kinematic viscosity $\nu^{\mathcal{R}}_{\text{eff}} = \eta^{\mathcal{R}}_{\text{eff}}/\rho$. Assumption 3.5 records that the velocity profile, friction law, rheology descriptor, and wall closure are part of the closed model specification. For computed non-Newtonian realizations, the laws are evaluated at the positive regularized shear rate

$$\dot\gamma_\epsilon \;:=\; \max\{|\dot\gamma|, \epsilon_\gamma\}, \qquad \epsilon_\gamma > 0, \qquad [\epsilon_\gamma] = \mathrm{s}^{-1}.$$

This positive regularization is required for the Casson and power-law families when their zero-shear limits are singular or degenerate. For descriptors that declare dynamic-viscosity bounds, the bounds satisfy $0 \le \eta_{\min} \le \eta_{\max} \le \infty$ with $\eta_{\max} > 0$ and define the interval projection

$$\Pi_{[\eta_{\min},\eta_{\max}]}(\xi) \;:=\; \min\{\eta_{\max}, \max\{\eta_{\min}, \xi\}\}.$$

The bounded viscosity is then $\Pi_{[\eta_{\min},\eta_{\max}]}(\eta^{\mathcal{R}}_{\text{eff}}(\dot\gamma))$; without declared bounds, this projection is the identity on the displayed model law.

$$\begin{aligned}
\eta^{\text{newt}}_{\text{eff}} &\equiv \eta,\\
\eta^{\text{carreau}}_{\text{eff}}(\dot\gamma) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\text{C}}\dot\gamma_\epsilon)^2\right)^{(n_{\text{C}}-1)/2},\\
\eta^{\text{CY}}_{\text{eff}}(\dot\gamma) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\text{CY}}\dot\gamma_\epsilon)^{a_{\text{Y}}}\right)^{(n_{\text{CY}}-1)/a_{\text{Y}}},\\
\eta^{\text{casson}}_{\text{eff}}(\dot\gamma) &= \left(\sqrt{\tau_y/\dot\gamma_\epsilon} + \sqrt{\eta_p}\right)^2,\\
\eta^{\text{power}}_{\text{eff}}(\dot\gamma) &= K_{\text{pl}}\dot\gamma_\epsilon^{\,n_{\text{pl}}-1}.
\end{aligned}$$

In the computed Newtonian path, the constant dynamic viscosity is $\eta = \rho\nu$, so the closure-indexed kinematic viscosity reduces to ν. The parameter domains are

$$\eta_0 > 0, \quad \eta_\infty \ge 0, \quad \eta_0 \ge \eta_\infty, \quad \lambda_{\text{C}}, \lambda_{\text{CY}} \ge 0, \quad n_{\text{C}}, n_{\text{CY}} > 0, \quad a_{\text{Y}} > 0,$$

for the Carreau and Carreau–Yasuda families,

$$\tau_y \geq 0, \qquad \eta_p > 0$$

for Casson, and

$$K_{\mathrm{pl}} > 0, \qquad n_{\mathrm{pl}} > 0$$

for the power-law closure. Under the SI convention, η, η_0, η_∞, η_p, $\eta_{\min}$, and $\eta_{\max}$ have units $\mathrm{Pa} \cdot \mathrm{s}$, λ_{C} and λ_{CY} have units s, τ_y has units Pa, and K_{pl} has units $\mathrm{Pa} \cdot \mathrm{s}^{n_{\mathrm{pl}}}$. In the cgs numerical convention of Convention C.2, the corresponding dynamic-viscosity parameters are in $\mathrm{g}/(\mathrm{cm\,s})$, τ_y is in $\mathrm{dyn/cm^2}$, and K_{pl} is in $\mathrm{dyn\,s}^{n_{\mathrm{pl}}}/\mathrm{cm}^2$. The Yasuda transition exponent is denoted a_{Y} here so that it is not confused with the solver coordinate $a = R^2$ used in the reduced numerical realization. The listed families are the implemented generalized-Newtonian viscosity options used by this report's closure catalog; their parameter values are model-record data rather than universal constants [10, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19] [13, 14].

Remark 1.16 (Newtonian blood justification). For large arteries, empirical studies suggest that apparent viscosity is often approximately constant across the relevant operating range, motivating the Newtonian approximation in many reduced-order hemodynamic models. However, diameter-dependent, hematocrit-dependent, and shear-dependent effects become increasingly important in smaller vessels and low-shear regions [9, Sec. 1.1.3, pp. 12–14] [11].

Together, Assumption 1.1 and the balance-law and stress-closure definitions in this subsection form the continuum contract used by the following hierarchy: material volumes and Reynolds transport support the mass and momentum balances, while incompressibility and the selected constitutive law specialize those balances into the PDE tiers that are later reduced by cross-sectional averaging.

1.3 Full-Order Blood-Flow Models: Three-Dimensional CFD and FSI

The continuum assumptions of Section 1.2 lead to a layered modeling hierarchy. In particular, Assumption 1.1 supplies the flow map, material volumes, and continuum fields on which the conservation laws act. The high-fidelity starting point is the incompressible Navier–Stokes or generalized-Newtonian system on a three-dimensional lumen. This subsection keeps that setting in three dimensions: it records the PDE forms, boundary-data vocabulary, and well-posedness context needed before the model hierarchy in Section 2.2 descends to reduced and lumped tiers.

1.3.1 Conservation Laws and Navier–Stokes Specialization

The governing equations originate from conservation of mass and linear momentum applied to material volumes under Assumption 1.1, followed by the constant-density incompressible specialization. The derivation details and row-divergence convention are recorded in Appendix E and Definition D.6. Figure 5 summarizes the route from these balance laws to the forms used in the report [10, Part I, Ch. 1, Secs. 2.2–3.1, pp. 11–19, Part IV, Ch. 1, Sec. 1.1, pp. 380–383] [15, Sec. 6].

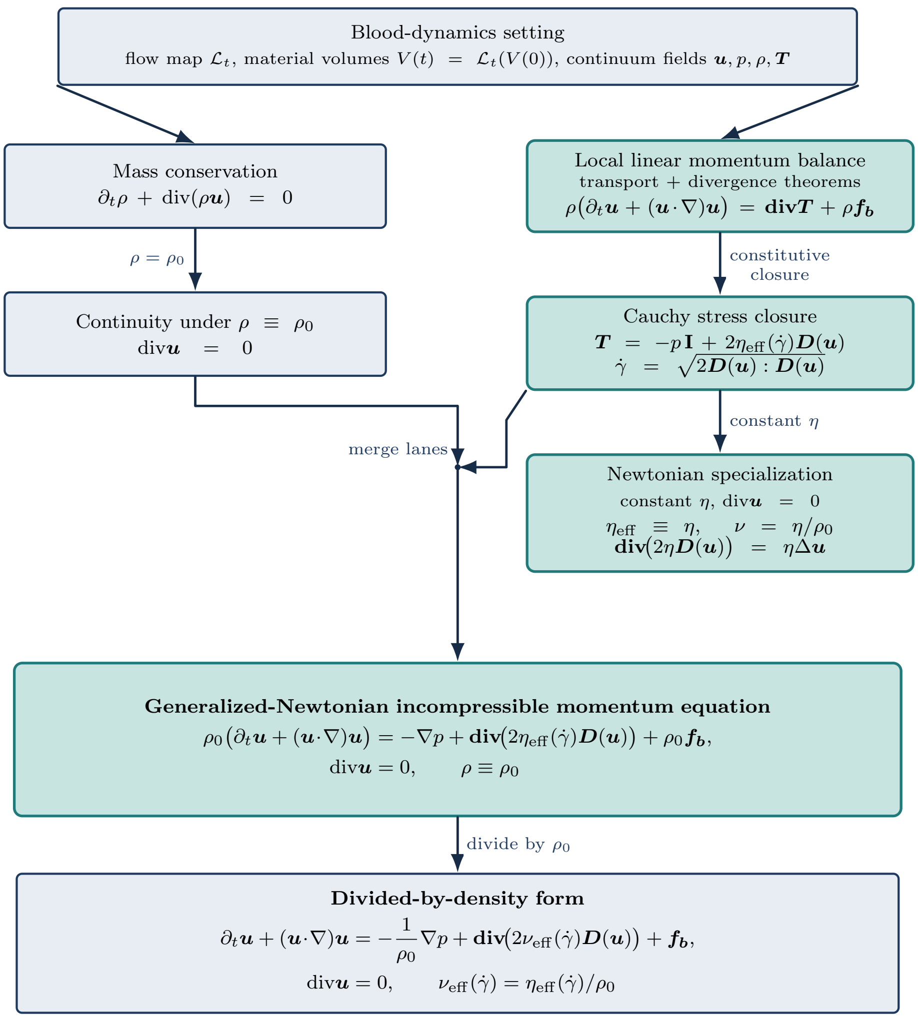

Figure 5: From the blood-dynamics setting to incompressible Navier–Stokes. The flow-map and material-volume assumptions support the conservation statements. The mass-conservation/incompressibility lane and the stress-closure lane merge into the generalized-Newtonian incompressible momentum form. The Newtonian box is a constant-viscosity specialization, and the divided-by-density form records the corresponding kinematic-viscosity scaling.

With this kinematic setting fixed, mass conservation for the constant-density incompressible model gives

$$\operatorname{div}\boldsymbol{u} = 0 \qquad \text{in } \Omega_B(t).$$

Linear momentum conservation balances inertia, pressure, viscous stress, and body forces:

$$\rho\,(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}) \;=\; \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}.$$

Throughout the Navier–Stokes statements below, $\boldsymbol{f_b}$ denotes body force per unit mass, so $\rho\boldsymbol{f_b}$ is the corresponding body force per unit volume.

Definition 1.17 (Conservative incompressible Navier–Stokes)**.** On $\Omega_B(t)$, the constant-density Newtonian system may be written in conservative tensor-divergence form as

$$\begin{cases} \partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = -\nabla p + \mathbf{div}(2\eta\boldsymbol{D}(\boldsymbol{u})) + \rho\boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \qquad \rho \equiv \rho_0 > 0. \end{cases}$$

For constant density and divergence-free velocity, $\mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \rho(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$. Thus the same Newtonian momentum equation can be read in advective form.

Definition 1.18 (Generalized-Newtonian incompressible Navier–Stokes)**.** For an incompressible generalized-Newtonian fluid, use the shear-rate convention

$$\dot\gamma \;:=\; \sqrt{2\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{u})}.$$

The velocity $\boldsymbol{u}$, pressure p, body force per unit mass $\boldsymbol{f_b}$, and constant density $\rho\equiv\rho_0>0$ satisfy

$$\begin{cases} \rho\big(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\big) = -\nabla p + \mathbf{div}\Big(2\eta_{\mathrm{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})\Big) + \rho\boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0. \end{cases}$$

Definition 1.19 (Density-scaled generalized Navier–Stokes)**.** With $\nu_{\mathrm{eff}}(\dot\gamma) \;:=\; \eta_{\mathrm{eff}}(\dot\gamma)/\rho$, the divided-by-density form is

$$\begin{cases} \partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \mathbf{div}\Big(2\nu_{\mathrm{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})\Big) + \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0. \end{cases}$$

The density-scaled form is particularly useful for numerical simulations, because it separates the convective $((\boldsymbol{u}\cdot\nabla)\boldsymbol{u})$ and diffusive / viscous-stress $(\mathbf{div}(2\nu_{\mathrm{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})))$ contributions in the momentum equation, allowing for more straightforward numerical treatment of each term independently.

Definition 1.20 (Newtonian incompressible Navier–Stokes)**.** If $\eta_{\mathrm{eff}}\equiv\eta$ is constant, then sufficiently smooth divergence-free velocity fields satisfy $\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \Delta \boldsymbol{u}$. The Newtonian equations are

$$\begin{cases} \rho\big(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\big) = -\nabla p + \eta\Delta\boldsymbol{u} + \rho \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \end{cases} \qquad \begin{cases} \partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \nu\Delta\boldsymbol{u} + \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \end{cases}$$

where $\nu := \eta/\rho$.

Thus the 3D PDE tier is not introduced independently: flow maps and material volumes support Reynolds transport, Reynolds transport gives mass and momentum balance, the Cauchy stress law closes momentum, and incompressibility selects the Navier–Stokes systems above.

Remark 1.21 (Navier–Stokes system classification). Each Navier–Stokes form above is a coupled time-dependent PDE system for the three velocity components and the pressure on a three-dimensional domain. The momentum equations evolve velocity, while pressure acts as a Lagrange multiplier enforcing the first-order incompressibility constraint $\mathrm{div}\boldsymbol{u} = 0$. Thus the system is not a scalar parabolic equation: it is a mixed incompressible evolution system with parabolic viscous smoothing coupled to an elliptic pressure constraint.

System	**Type and order**	**Classification detail**
Definition 1.17	Conservative incompressible Newtonian form: first order in time; first-order conservative transport and pressure terms; first-order incompressibility constraint; second-order viscous stress operator.	With constant ρ and η, smooth divergence-free fields convert this form to the advective Newtonian form. It is nonlinear through momentum transport and is a mixed parabolic–elliptic system after the pressure constraint is imposed.
Definition 1.18	Generalized-Newtonian incompressible form: first order in time; first-order advective and pressure terms; second-order stress-divergence operator with coefficients depending on $\boldsymbol{D}(\boldsymbol{u})$ through $\eta_{\mathrm{eff}}(\dot\gamma)$.	The highest-order stress operator is generally quasilinear. Convection and strain-rate-dependent viscosity both contribute nonlinearity. Positive regular viscosity laws give the usual incompressible parabolic–elliptic reading; degenerate or singular laws require separate assumptions.
Definition 1.19	Density-scaled generalized form: same differential order and unknowns as Definition 1.18; division by constant density rescales the pressure gradient and viscous coefficient.	This is not a new analytic class. It is the same mixed incompressible quasilinear system written with kinematic effective viscosity ν_{eff} and body force per unit mass.
Definition 1.20	Newtonian incompressible advective form: first order in time; first-order quadratic convection and pressure terms; first-order divergence constraint; second-order linear Laplacian diffusion.	Constant viscosity removes the quasilinear viscosity dependence, leaving a semilinear incompressible parabolic–elliptic system with nonlinear lower-order transport and an elliptic pressure constraint.

1.3.2 Boundary Data and Well-Posedness Context

For a single-vessel three-dimensional domain, write the boundary decomposition

$$\partial\Omega_B(t) = \Gamma_{\mathrm{in}}(t) \cup \Gamma_{\mathrm{out}}(t) \cup \Gamma_w(t).$$

Here $\Gamma_{\mathrm{in}}(t)$ denotes an inlet portion where a velocity, flow-rate, or profile condition may be prescribed. The outlet portion $\Gamma_{\mathrm{out}}(t)$ carries pressure, traction, or reduced downstream network data. For a fixed-wall no-slip continuum model, the wall portion $\Gamma_w(t) \subset \partial\Omega_B(t)$ carries $\boldsymbol{u} = 0$; this condition is not imposed on the inlet or outlet portions of $\partial\Omega_B(t)$. Other continuum or FSI tiers may use no-penetration, wall-law, or structure-coupling data instead. A complete initial-boundary value problem also states $\boldsymbol{u}(0,\cdot) = \boldsymbol{u}_0$, the domain, forcing, compatibility assumptions, and the solution class.

Remark 1.22 (Global regularity context). For smooth, spatially localized initial data in three dimensions, global existence and smoothness for Navier–Stokes remains an open Clay Millennium Prize problem [16, Problem statement]. This is mathematical context for the 3D continuum model, not a direct limitation of the reduced models discussed below.

Well-posedness results for a three-dimensional Navier–Stokes initial-boundary value problem are problem-specific: the statement must fix the domain class, boundary conditions, forcing, compatibility assumptions, initial-data space, solution class, and time horizon. This report uses that vocabulary only as PDE background for the model hierarchy. It does not use the Clay global-regularity problem as support for a local theorem, and it does not prove a well-posedness result for the moving arterial domain or for the reduced numerical solver. The domain and function-space notation relevant to such statements is summarized in Appendix D; coordinate and energy background is collected in Appendix F.

The result is a three-dimensional continuum problem with declared field unknowns, stress closure, boundary partitions, and interpretation limits. Section 2.2 now uses this 3D setting as the top tier of the model hierarchy before introducing coupled, reduced, and lumped descriptions.

1.4 Research Problem and Questions

The research problem is to define and check an auditable reduced-order model for an idealized stenotic vessel without overstating what the current evidence can show. The report is organized around three answerable questions:

1. Which variables, wall law, rheology/profile closures, source terms, and boundary-state approximation define the closed stenosis-aware 1D balance law?
2. Under the recorded finite-volume solver, time integrators, descriptor checks, and benchmark norms, what implementation properties are supported by the reproducibility and self-convergence evidence?

3. Under the declared `CrossSectionQuadratureOperator`, what section-wise velocity and physical-flow discrepancies separate the 1D solution from the available resolved 3D velocity data?

1.5 Objectives and Contributions

The contribution is a consistent model-and-output specification for an idealized stenotic vessel, an internally reproducible implementation and benchmark suite for the selected extended 1D model, a targeted numerical audit that exposes a geometry-rest preservation failure, and a bounded cross-model velocity comparison against two resolved 3D computational datasets. The comparison is diagnostic rather than clinical, experimental, or production-accuracy validation.

1.6 Scope and Limits of the Study

The smooth geometry is a deliberate numerical device: one analytic radius profile supplies controlled derivatives, wall normals, severity parameters, and repeatable mesh families for the 1D, fixed-wall 3D Stokes, and later 3D/FSI model tiers. The completed numerical record is correspondingly bounded: pressure-ratio quantities are reduced model-output vocabulary, with no clinical FFR or CT-FFR measurement relationship.

Claim	**Evidence in this report**	**Permitted wording**
Model formulation	Derivation, closure tables, and implementation map	Defines a selected 1D model.
Smooth idealized geometry	Analytic C^∞ stenosis profile and shared finite-volume / finite-element geometry views	Supports controlled numerical studies, not patient-specific morphology.
Code behavior	Tests, self-convergence, descriptor checks, cross-integrator discrepancies, rest-state drift rows, and manifests	Supports reported implementation properties and exposes the non-well-balanced rest-state defect.
Resolved velocity comparison	Tetrahedral cross-section quadrature at $t = 1.0$ s, with production sensitivity still incomplete	Reports single-realization diagnostic cross-model descriptors.
Pressure accuracy	No computed pressure comparison	Future work only.
Clinical validity	No experimental or clinical measurement comparison	No clinical validation claim.

Convention C.3 records the admissibility requirements for any later pressure-ratio output; the completed numerical record here reports verification diagnostics and velocity-focused cross-model comparison only.

1.7 Report Organization

- Section 2 reviews the hemodynamic modeling requirements, full-order and reduced-order model classes, stenosis-specific closure issues, and the distinction between verification, cross-model comparison, and validation.
- Section 3 specifies the selected 1D model, solver-coordinate convention, stenosis geometry, boundary contract, finite-volume/time-integration realization, comparison operator, metrics, and provenance record.
- Section 4 reports the implementation and numerical-verification evidence recorded by the benchmark suite, including the failed geometry-rest preservation check that limits the later comparison.
- Section 5 reports the resolved-3D velocity comparison and localizes the observed discrepancies.
- Sections 6 and 7 interpret the evidence, state the limitations, and summarize the bounded contribution.

2 Background and Literature Review

This chapter reviews the state-of-the-art modeling and numerical-method context needed for the reported study. The continuum blood-flow setting and governing-equation baseline have already been fixed in Section 1.2 and Section 1.3; proof-level transport identities remain in the appendices. The review here therefore compares model tiers, closure choices, numerical methods, verification practice, and one-dimensional to three-dimensional observation operators. It positions the selected solver in that landscape, but it does not claim clinical validation, pressure-ratio accuracy, or a state-of-the-art solver result.

2.1 Stenotic Hemodynamics and Relevant Observables

Stenosis is a geometric and hemodynamic problem: a local lumen reduction can alter axial velocity, pressure drop, wall loading, recirculation, and the relationship between flow and downstream perfusion. Clinical and computational literatures therefore use different observables depending on the question. Pressure-ratio work emphasizes pressure drops and fractional-flow style indices [5, 6, 17], whereas computational fluid-dynamics studies can expose local velocity, traction, and wall-shear structures that are not section-averaged quantities [8, 10]. The present report uses velocity observables only. It does not report pressure drop, FFR, CT-FFR, wall shear stress, or clinical validation.

Reduced models are attractive because stenosis studies often require repeated parameter sweeps, reproducible solver comparisons, or network-level coupling. The cost is that the relevant output must be compatible with the reduced state. For a one-dimensional area–flow model, section mean velocity and physical flow are natural; resolved radial profile, secondary flow, skewness, and wall traction are not native outputs unless an additional reconstruction or observation map is declared.

2.2 State-of-the-Art Model Families

The current modeling landscape is best read as a hierarchy of model roles, not as a single ladder of accuracy. Three-dimensional CFD and FSI models retain geometry-resolved velocity and stress fields and can represent local stenotic acceleration, separation, and wall interaction more directly than reduced models [18, 19]. They also require detailed geometry, material assumptions, boundary histories, mesh generation, and substantially higher computational cost. Coupled 3D–1D or geometrical multiscale formulations address domain-size and boundary limitations by joining local resolved regions with reduced vascular networks [9, 20, 21].

One-dimensional distributed models retain axial propagation and pressure–area–flow dynamics while replacing the transverse velocity field by closure terms [15, 22, 23]. Zero-dimensional networks and terminal models are still cheaper and are useful for lumped impedance and perfusion effects, but they do not resolve axial stenosis localization. Intermediate 2D or axisymmetric models can retain more local structure than 1D systems with less cost than full 3D, but they still require their own symmetry and boundary assumptions.

Recent surrogate, PINN, and operator-learning methods add another axis: trained maps can reduce evaluation cost or assimilate data, but their evidence depends on training data, uncertainty treatment, boundary-condition coverage, and comparison with mechanistic solvers or measurements [24, 25, 26, 27]. They are therefore useful state-of-the-art context and future-work vocabulary for stenosed-vessel modeling, but they are not used as forward generators in the present report.

Table 1: State-of-the-art model and method families relevant to this report. The table maps field context to the evidence standard needed before a result can support a claim in this thesis.

Family	**Representative role**	**Evidence requirement**	**Reading in this report**
3D CFD / FSI and multiscale 3D–1D–0D	Resolved local fields, wall interaction, and coupled circulation [19, 20].	Matched geometry, wall and material data, boundary histories, mesh/time studies, and output operators.	Reference vocabulary and comparison target, not the selected repeated forward generator.
Distributed 1D stenosis-aware models	Low-cost area–flow dynamics with closure-indexed wall, profile, friction, and stenosis terms [22, 28].	Explicit closure contract, boundary rule, coordinate map, and verification of the computed realization.	Selected model class; the implementation is an $R_{\max}$-normalized Canic-derived variant.
Open and reproducible 1D solver workflows	Published solver surfaces, benchmark cases, and reusable finite-volume infrastructure [29, 30].	Runnable commands, persisted descriptors, benchmark outputs, hashes, and documented optional inputs.	Workflow comparator and reproducibility standard, not an independent-solver parity claim.
Well-balanced and high-order 1D numerical schemes	Equilibrium-preserving finite-volume or high-order discretizations for elastic-vessel networks and source terms [31, 32].	Rest-state/equilibrium preservation, manufactured solutions, refinement, and boundary-regime diagnostics.	Context for why the present rest-state drift is a material numerical limitation.
Surrogates, PINNs, and operator learning	Fast learned maps or physics-constrained reconstructions for hemodynamic quantities [24, 26, 27].	Training-data provenance, uncertainty, out-of-distribution tests, and comparison with accepted reference outputs.	Bounded outlook only; no learned model is part of the completed numerical study.

Table 2: Model-tier roles relevant to the present report. The listed observables are admissible outputs only after the corresponding model and observation map have been declared.

Tier	**Retained physics**	**Data and cost**	**Typical admissible outputs**
3D CFD / FSI	Resolved velocity, local gradients, geometry, and possibly wall motion.	Detailed mesh, material and boundary histories; high cost.	Local velocity, pressure, traction, wall shear stress, section averages.
Axisymmetric / 2D	More local structure than 1D under symmetry or planar assumptions.	Intermediate geometry and mesh burden.	Axial/radial fields consistent with the symmetry assumption.
1D distributed	Axial wave propagation, area–flow dynamics, wall and friction closures.	Vessel centerline, radius, wall law, terminal and inlet data; low cost.	Area, flow, section mean velocity, reduced pressure variables.
0D network	Lumped resistance, compliance, inertance, and terminal coupling.	Lowest spatial data burden; low cost.	Compartment pressures and flows, not local stenotic fields.
Geometrical multiscale	Local resolved region coupled to reduced upstream/downstream networks.	Requires interface conditions and compatible observables.	Mixed local and network outputs with interface consistency checks.

2.3 One-Dimensional Models for Nonuniform and Stenotic Vessels

The defining step in a 1D vessel model is cross-sectional averaging. The state typically stores a section area and volumetric flow, while the discarded transverse structure re-enters through closure choices: momentum-flux factor, velocity profile, wall law, friction, rheology, and source terms for nonuniform reference geometry [22, 23]. These choices are not interchangeable. A profile factor changes the convective flux; a wall law changes the pressure–area relation and wave speed; a friction or rheology law changes dissipative closure; and a stenosis-loss or variable-radius term changes how geometry enters the reduced momentum balance.

Classical stenosis literature also shows that reduced pressure-loss formulas and unsteady stenosis corrections can be useful but model a different closure object than a distributed 1D finite-volume system [33, 34, 35]. Recent stenosis-focused reduced models, including the Canic et al. extended 1D formulation, add variable-radius contributions intended to better represent stenotic geometry in a distributed balance [28]. Coronary and arterial-tree models make the same point at network scale: useful predictions depend on the declared wall, terminal, and closure assumptions [36, 37].

Table 3: Closure dimensions that should be separated when comparing stenosis-aware 1D models.

Closure dimension	**Common choices**	**Effect on evidence**
State variables	Area–flow, radius–flow, pressure–flow, or network compartment variables.	Determines which outputs are native and which require reconstruction.
Wall model	Fixed geometry, algebraic pressure–area relation, FSI coupling, terminal compliance.	Controls wave speed and pressure interpretation.
Profile / momentum flux	Flat, parabolic, power-law, empirical correction factors.	Controls convective flux and any reconstructed radial profile.
Friction / rheology	Newtonian, Carreau, Carreau–Yasuda, Casson, power-law, tube-flow laws.	Controls dissipation and shear-rate proxy interpretation [11, 13, 14].
Stenosis treatment	Reference-radius variation, empirical loss terms, variable-radius distributed terms.	Determines whether geometry appears only in area or also as a momentum source.
Numerical method	Finite volume, discontinuous Galerkin, WENO, Lax–Friedrichs-type, well-balanced, and method-specific boundaries.	Determines conservation, equilibrium preservation, stability evidence, and boundary evidence needs [31, 38, 39].
Validation target	Benchmarks, manufactured solutions, 3D fields, pressure ratios, clinical measurements.	Determines whether the result is verification, cross-model comparison, or validation.

2.4 Numerical Methods and Verification of 1D Arterial Solvers

The numerical-method state of the art for 1D arterial flow is shaped by the hyperbolic balance-law structure of the area–flow equations. Finite-volume methods remain attractive because they track conservation and discontinuity-like features through numerical fluxes, while benchmark studies have also compared finite element, discontinuous Galerkin, finite difference, and simplified network-oriented schemes [29, 37, 38]. In recent open implementations, MUSCL finite-volume discretizations and Lax–Friedrichs-type fluxes are common because they give robust low-cost baselines with simple boundary and coupling treatments [30, 39].

High-order and equilibrium-aware methods address different limitations of such baselines. DG and WENO reconstructions can improve smooth-region resolution, but their evidence depends on source quadrature, boundary treatment, positivity handling, and whether limiter fallback or projections alter the formal order. Well-balanced schemes target a separate requirement: a physically meaningful steady or rest state should remain steady under the discrete operator. Recent work on well-balanced explicit, semi-implicit, and high-order blood-flow schemes makes this a central numerical property for elastic vessels and network applications [31, 32]. That literature is directly relevant to this report because the selected Rusanov/MUSCL realization is deliberately reported as not exactly well balanced for the stenotic geometry-rest state.

Verification asks whether the implementation solves the declared equations and produces reproducible numerical records. For hyperbolic finite-volume solvers, relevant evidence includes conservation diagnostics, CFL reporting, positivity/limiting records, grid and timestep refinement, boundary-regime checks, exact equilibria, and manufactured solutions [38]. Arterial solver benchmarks provide a useful cross-code reference for wave propagation and network behavior, but they do not replace problem-specific verification of a selected stenosis realization [29, 30].

This distinction matters for the present report. A benchmark-stage completion row, an output-consistency check, or a cross-integrator discrepancy can be useful evidence about the computational realization, but it is weaker than a manufactured-solution convergence table or a demonstrated discrete equilibrium. A fixed-area characteristic boundary rule also requires evidence that the sign condition under which it is applied holds in the sampled run. Verification therefore has to persist diagnostic quantities, not only final plots.

The literature also separates method order from study validity. A high-order scheme can still be the wrong evidence for a pressure-ratio claim if the pressure observation maps, wall model, inlet/outlet histories, and comparison data are not matched. Conversely, a lower-order method can still support a bounded implementation claim if its equations, numerical record, diagnostics, and limitations are explicit. The present report follows the second path: it states the selected method, exposes its rest-state defect, and limits the resolved comparison to velocity-output discrepancies.

2.5 Cross-Dimensional Comparison and Validation

A 1D–3D comparison is meaningful only after the compared observable has been mapped to a common quantity. For velocity, this can mean reducing a resolved 3D velocity field to section averages, radial profiles, or point/slab samples. For pressure and FFR-style quantities, the required maps are different and must include proximal/distal pressure observations and time-averaging conventions [3, 31, 40]. Agreement under one operator is not agreement under another.

Validation is still a separate evidential category. A comparison with another simulation can identify cross-model discrepancies, but it does not establish physical accuracy for the target question unless the reference model and its boundary, wall, material, and history data are themselves accepted for that purpose. The present work therefore uses cross-model velocity discrepancy language and reserves *error* for exact solutions, manufactured solutions, or accepted reference data under matched conditions.

2.6 Literature Synthesis and Research Gap

The reviewed literature supports a calibrated gap. State-of-the-art stenosis modeling spans resolved 3D/FSI simulations, geometrical multiscale coupling, distributed 1D reductions, open solver workflows, well-balanced and high-order schemes, and emerging surrogate or physics-informed learning methods. These families answer different questions. Resolved models expose local velocity, pressure, and traction fields at high setup cost. Reduced models support repeated studies only after their closure and observation maps are declared. Well-balanced and high-order schemes improve numerical evidence only for the equilibria and outputs they actually test. Learned surrogates reduce evaluation cost only after training data, uncertainty, and admissible operating regimes are made part of the evidence record.

The present work addresses a practical reproducibility and observation-operator gap for this implementation of an extended 1D stenosis model. It specifies the solver-coordinate map, the Canic-derived model variant, targeted verification evidence, and a declared plane–tetrahedron velocity comparison operator. Its claim is therefore intentionally narrower than a field-wide state-of-the-art claim. A stronger novelty or validation claim would require broader evidence that comparable auditable implementations and matched 1D–3D operators do not already exist, together with matched pressure, experimental, or clinical data. That broader claim is outside this report.

3 Mathematical and Computational Methodology

This section states the actual model and computational protocol used in the report. The preceding review explains why a reduced model is useful and where its assumptions enter; the present section fixes the selected variables, closure choices, solver-coordinate convention, boundary contract, numerical realization, comparison operator, metrics, and provenance record before any numerical outcomes are interpreted.

3.1 Problem Definition and Modeling Assumptions

The computational problem is a single idealized stenotic vessel represented by a selected one-dimensional area–flow model. The literature review has already separated full-order, reduced-order, and multiscale roles; the methods chapter therefore fixes the particular assumptions used in the numerical study. The 1D model assumes a distinguished axial direction and cross-sectionally averaged pressure and velocity. This keeps distributed pulse-wave and pressure–flow structure while discarding secondary flow, separation, recirculation, and resolved wall traction. The reduction follows standard compliant-artery and dimension-reduced modeling arguments [22, Sec. 2, pp. 253–257] [9, Ch. 3, Secs. 3.2 and 3.6] [23, Sec. 3, pp. 24–26].

3.2 Cross-Sectional Variables and Governing Equations

The reduced variables A, Q, $\bar{u}$, and p_{g} should be read as cross-sectional outputs of the continuum fields under additional averaging, geometry, wall-law, and profile assumptions, not as independent replacements for the blood-dynamics setting.

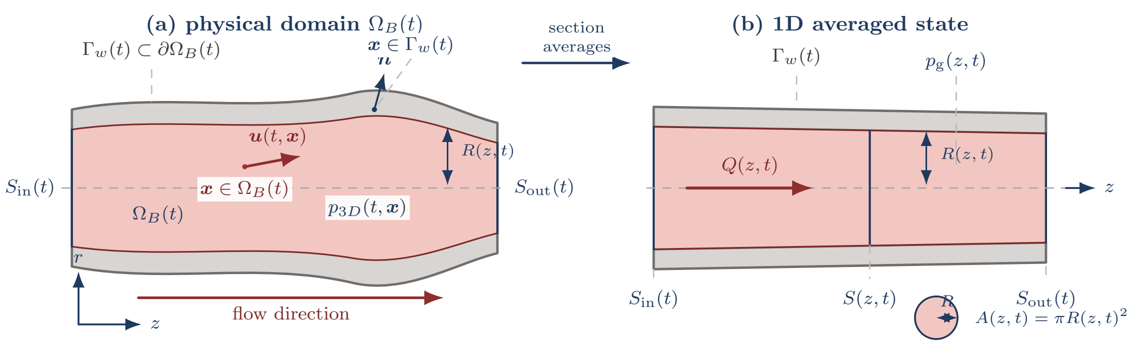

Figure 6: Compliant-vessel notation for the 1D reduction. The current configuration $\Omega_B(t)$ is represented by a straightened averaging domain: section averages over $S(z, t)$ define $A(z, t)$, $Q(z, t)$, $\bar{u}(z, t)$, and $p_{\mathrm{g}}(z, t)$, while $R(z, t)$ records the radius entering the wall and geometry closures. The figure is a bookkeeping reduction from continuum fields to retained 1D variables, not a resolved wall-traction or secondary-flow model.

Assumption 3.1 (Straight single-vessel geometry)**.** The selected 1D model assumes a vessel of length L aligned with the z-axis and a circular cross-section with radius $R(z, t)$. For $0 < z < L$, $S(z, t)$ is the interior cross-section used for averaging, not a boundary component of the open lumen.

Definition 3.2 (Section-averaged 1D fields)**.** The retained reduced variables are

$$
\begin{aligned}
A(z,t) &:= \int_{S(z,t)} 1\, dS, && \text{cross-sectional area,}\\
Q(z,t) &:= \int_{S(z,t)} u_z\, dS, && \text{volumetric flow rate,}\\
\bar{u}(z,t) &:= \frac{Q(z,t)}{A(z,t)}, && \text{mean axial velocity,}\\
p_{\mathrm{g}}(z,t) &:= \frac{1}{A(z,t)} \int_{S(z,t)} p_{3D,\mathrm{g}}\, dS, && \text{mean gauge pressure.}
\end{aligned}
$$

Here $p_{3D,\mathrm{g}}$ is continuum pressure in the wall-law gauge frame. The pressure-reference convention for reduced ratios is fixed separately in Convention C.3.

3.3 Selected Wall Law and Closure Model

Definition 3.3 (Classical parabolic-profile 1D baseline)**.** The classical baseline used as a model-family member is a parabolic-profile reduced model. A no-slip wall trace is compatible with a parabolic profile for fully developed Newtonian flow in a straight circular tube, but no-slip alone does not imply that profile in an unsteady stenotic vessel. The parabolic profile is therefore an independent cross-sectional closure with $\alpha_{\mathcal{P}} = 4/3$ and $g_{\mathcal{P}} = 4$. In the implementation manifests this member is still recorded under the historical token `classical-1d-no-slip`; in the mathematical discussion it is the classical parabolic-profile baseline. It keeps the same reduced state variables and selected Canic-derived pressure–area wall law as the extended 1D framework, but omits the Canic effective-alpha variable-radius correction. The descriptor `canic-extended-1d` is the historical manifest token for the current $R_{\max}$-normalized Canic-derived extended-model member.

Definition 3.4 (Closed 1D area-flow model)**.** For $\mathcal{W} = \mathcal{W}_{\mathrm{elastic1D}}$ and an incompressible fluid with density ρ, the working compliant single-vessel model is written with the composite axial pressure derivative

$$
\frac{d}{dz} p_{\mathrm{g}}(A(z,t), z, t) = \partial_A p_{\mathrm{g}}(A, z, t)\partial_z A + \partial_z p_{\mathrm{g}}(A, z, t)\big|_A .
$$

The reduced balance law is

$$
\partial_t A + \partial_z Q = 0, \tag{3.1}
$$

$$
\partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A} \right) + \frac{A}{\rho} \frac{d}{dz} p_{\mathrm{g}}(A(z,t), z, t) = -C_f(z, A, Q) \frac{Q}{A}. \tag{3.2}
$$

The parameter α is the momentum-flux correction factor induced by the selected cross-sectional velocity profile, C_f is the wall-friction closure, and $p_{\mathrm{g}}(A, z, t)$ is the wall-law gauge pressure closing the reduced system. This is the reduced $\mathcal{W}_{\mathrm{elastic1D}}$ member of the wall-model axis: wall response is represented through the pressure–area law rather than through a separate structural PDE [22, Sec. 2, pp. 253–255] [9, Ch. 3, Sec. 3.2, pp. 40–47].

Assumption 3.5 (Velocity-profile and rheology closure)**.** A closed 1D area-flow model must declare the momentum-flux correction factor α, the wall-friction closure C_f, any rheology descriptor $\mathcal{R}$, and the pressure–area wall closure $\mathcal{W}$. These choices determine how cross-sectional profile effects, wall friction, viscosity or effective viscosity, and wall compliance enter the reduced equations. Numerical-record details such as solver coordinates, shear-rate proxies, and discrete source terms are deferred to Appendix G.

Definition 3.6 (Selected Canic–Koiter pressure–area wall law)**.** The selected elastic wall closure is the Canic–Koiter thin-wall pressure–area law. In conventional area notation it may be written

$$p_{\mathrm{g}}(A,z,t) = p_{\mathrm{ext}}(z,t) + \frac{\beta(z)}{A_0(z)}\left(\sqrt{A(z,t)} - \sqrt{A_0(z)}\right). \tag{3.3}$$

Here $A_0(z)$ is the reference area, p_{ext} is the external pressure, and $\beta(z)$ is a wall-stiffness parameter. Some implementations absorb A_0^{-1} into β; the essential closure is that p_{g} is known once A_0, β, and p_{ext} are supplied. In the radius-squared coordinate $a = R^2$ used by the implemented finite-volume evolution, the selected continuous source-balanced member uses the Canic conservative-form reference radius $R_0^* = R_{\max}$:

$$p_{1,\mathrm{g}}(a,z,t) = p_{\mathrm{ext}}(z,t) + \frac{K}{R_{\max}^2}\left(\sqrt{a} - R_0(z)\right), \qquad K = \frac{Eh}{1-\sigma^2}.$$

With $A = \pi a$ and $A_0 = \pi R_0^2$, the conventional law in Eq. 3.3 reduces to this selected member only when

$$\beta(z) = \sqrt{\pi}\, K\, \frac{R_0(z)^2}{R_{\max}^2}.$$

Thus the implemented evolution uses a constant reference-radius denominator, not the local R_0^{-2} denominator. In this report $R_{\max}$ is the healthy reference-radius parameter used by the solver; for the default stenosis profile it equals $\max_z R_0(z)$. The selected comparison runs use the cgs numerical-record values $\rho = 1.055\,\mathrm{g/cm^3}$, $E = 5.02\times10^6\,\mathrm{dyn/cm^2}$, $h = 0.06\,\mathrm{cm}$, $\sigma = 0.5$, $K = 4.016\times10^5\,\mathrm{dyn/cm}$, $R_{\max} = 0.18\,\mathrm{cm}$, and $p_{\mathrm{ext}} \equiv 0$. This is the wall law paired with the implemented elastic potential and wall/geometry source below. Legacy pressure-output helpers in the implementation still report a local-denominator pressure diagnostic through `pressure(result,params)`; benchmark pressure extrema and pressure-diagnostic bars refer to that diagnostic, not to the selected evolution wall law. Those pressure values are not used here as pressure-drop, pressure-ratio, or FFR evidence. The diagnostic extended pressure correction is

$$p_{2,\mathrm{g}}(a,q,z,t) = g_{\mathcal{P}}\rho\nu_{\mathrm{eff}}^{\mathcal{R}}\frac{q}{a}\frac{R_0'(z)}{R_0(z)}.$$

This p_2 term is a variable-radius viscous correction, not a replacement wall law. In generalized-Newtonian descriptor runs, the implemented analytic p_2 derivative uses the local value of $\nu_{\mathrm{eff}}^{\mathcal{R}}$ and does not differentiate that effective-viscosity factor; it is therefore a locally frozen-viscosity computational closure for this term. Resolved FSI would replace the algebraic pressure–area closure with structural unknowns and fluid–structure interface conditions.

3.4 Stenosis Geometry and Implemented Coordinates

Definition 3.7 (Physical-to-solver map for the implemented 1D state)**.** The continuum 1D model uses physical area A_{phys} and physical flow Q_{phys}. The implemented radius-squared solver instead stores

$$a = R^2, \qquad q = \frac{Q_{\mathrm{phys}}}{\pi}.$$

Thus

$$A_{\mathrm{phys}} = \pi a, \qquad Q_{\mathrm{phys}} = \pi q, \qquad \bar{u} = \frac{Q_{\mathrm{phys}}}{A_{\mathrm{phys}}} = \frac{q}{a}.$$

With $K = Eh/(1-\sigma^2)$ and the Canic conservative-form convention $R_0^* = R_{\max}$, the selected evolution wall law gives the solver elastic flux contribution recorded in Appendix G.3:

$$\Psi_h(a) = \frac{K}{3\rho R_{\max}^2} a^{3/2}, \qquad \partial_a \Psi_h(a) = \frac{K}{2\rho R_{\max}^2}\sqrt{a}.$$

This map is the convention used by the Section 5 velocity and flow comparisons.

Definition 3.8 (Selected implemented balance law)**.** The selected one-dimensional evolution model advanced by the finite-volume solver is stated in solver coordinates:

$$\begin{aligned}
\partial_t a + \partial_z q &= 0,\\
\partial_t q + \partial_z \left[\alpha_{\mathrm{eff}}(z)\frac{q^2}{a} + \Psi_h(a)\right] &= S_{\mathrm{wall}}(a,z) + S_{\mathrm{fric}}(a,q,z) + S_\alpha(a,q,z) + S_{p_2}(a,q,\partial_z a,\partial_z q,z).
\end{aligned}$$

The terms are

$$\begin{aligned}
\alpha_{\mathrm{eff}}(z) &= \alpha_{\mathcal{P}} + \iota_{\mathcal{M}}\alpha_c(R_0'(z)), & \alpha_c(R_0') &= -\frac{2}{35}(R_0')^2,\\
\Psi_h(a) &= \frac{K}{3\rho R_{\max}^2}a^{3/2}, & S_{\mathrm{wall}}(a,z) &= \frac{K}{\rho R_{\max}^2} a R_0'(z),\\
S_{\mathrm{fric}}(a,q,z) &= -2\nu_{\mathrm{eff}}^{\mathcal{R}} g_{\mathcal{P}}\frac{q}{a}, & S_\alpha(a,q,z) &= \iota_{\mathcal{M}}\frac{q^2}{a}\alpha_c'(z),\\
S_{p_2}(a,q,\partial_z a,\partial_z q,z) &= -\iota_{\mathcal{M}} a\,\partial_z^{\mathrm{frz}} p_{2,\mathrm{g}}(a,q,z).
\end{aligned}$$

The indicator $\iota_{\mathcal{M}} = 1$ gives the `canic-extended-1d` member and $\iota_{\mathcal{M}} = 0$ gives the classical parabolic-profile baseline. The symbol $\partial_z^{\mathrm{frz}} p_{2,\mathrm{g}}$ denotes the density-scaled derivative of the Canic p_2 pressure correction with

$\nu_{\text{eff}}^{\mathcal{R}}$ held locally fixed during the derivative,

$$\partial_z^{\text{frz}} p_{2,\text{g}} := \frac{1}{\rho} \frac{d}{dz} p_{2,\text{g}} \Big|_{\nu_{\text{eff}}^{\mathcal{R}} \text{ held fixed}} .$$

The selected model therefore contains a derivative-dependent nonconservative correction, realized discretely through $D_z^h a$ and $D_z^h q$ in Appendix G.3, rather than a purely local source $S(U, z)$. Thus the classical baseline is recovered by setting $\alpha_{\text{eff}} = \alpha_{\mathcal{P}}$, $S_\alpha = 0$, and $S_{p_2} = 0$. The selected implemented comparison also assumes $p_{\text{ext}} \equiv 0$; if p_{ext} varies with z, the wall/geometry source would need the additional term $-(a/\rho)\partial_z p_{\text{ext}}$ in solver coordinates.

Proposition 3.9 (Rest-state compatibility of the selected wall source)**.** *Suppose p_{ext} is constant in the gauge convention and the boundary data are compatible with $q = 0$. Then $a(z) = R_0(z)^2$ and $q(z) = 0$ are an exact steady state of the selected continuous source-balanced evolution law.*

Proof. The mass equation gives $\partial_z q = 0$. In the momentum equation, all terms proportional to q vanish, so $S_{\text{fric}} = S_\alpha = S_{p_2} = 0$. At $a = R_0^2$,

$$\partial_z \Psi_h(R_0^2) = \partial_z \left(\frac{K}{3\rho R_{\max}^2} R_0^3 \right) = \frac{K}{\rho R_{\max}^2} R_0^2 R_0' = S_{\text{wall}}(R_0^2, z).$$

Thus the elastic flux derivative is exactly canceled by the wall/geometry source, and $\partial_t q = 0$. □

Definition 3.10 (Stenosis reference geometry)**.** A stenosis is encoded through the reference radius or reference area. Let $R_{\text{base}}(z)$ be the nominal healthy radius and $s(z) \in [0, 1)$ a fractional radius-reduction profile. Then

$$R_0(z) = R_{\text{base}}(z)\big(1 - s(z)\big), \qquad A_0(z) = \pi R_0(z)^2. \tag{3.4}$$

The baseline idealized stenotic-vessel profile is the smooth asymmetric kernel

$$g(z) = z - z_s + \kappa \exp\left(-\frac{(z - z_a)^2}{2\sigma_a^2} \right), \qquad \psi(z) = \exp\left(-\lambda g(z)^4\right), \tag{3.5}$$

$$s(z) = s_{\max}\psi(z), \qquad 0 \le s_{\max} < 1. \tag{3.6}$$

In the default centimeter-scale simulation record, $z_a = 2.5\,\text{cm}$, $z_s = 3.4\,\text{cm}$, $\sigma_a = 1\,\text{cm}$, $\kappa = 0.95\,\text{cm}$, and $\lambda = 50\,\text{cm}^{-4}$. This geometry is the baseline idealized simulation geometry in this report, not a patient-specific plaque morphology or clinical calibration.

Remark 3.11 (Stenosis-profile and variable-radius boundary). The profile in Eq. 3.6 is C^∞ and rapidly localized rather than compactly supported. This smooth idealized geometry is the baseline for the reduced simulations in this report and is also useful for fixed-wall 3D studies: the same analytic radius profile supplies controlled derivatives, wall normals, severity variation, and repeatable mesh-refinement families without introducing segmentation noise or patient-specific morphology. Canic et al. show that standard 1D variable-geometry models may miss stenotic pressure and velocity trends unless additional variable-radius terms are included [28, Abstract, Secs. 1–2]. Those corrections are included only by the `canic-extended-1d` forward-model descriptor; the classical parabolic-profile baseline currently recorded by the historical token `classical-1d-no-slip` retains the same selected wall law but omits the Canic variable-radius corrections.

3.5 Conservative Form and Source Terms

Definition 3.12 (Conservative flux/source notation for the reduced wall law)**.** For a reduced wall closure $\mathcal{W}$ admitting an elastic potential, define

$$\mathbf{U} = (A, Q)^\top, \qquad \mathbf{F}_{\mathcal{W}}(\mathbf{U}, z, t) = \begin{pmatrix} Q \\ \alpha Q^2/A + \Psi_{\mathcal{W}}(A, z, t) \end{pmatrix},$$

$$\mathbf{S}_{\mathcal{W},\mathcal{R}}(\mathbf{U}, z, t) = \begin{pmatrix} 0 \\ S^{\mathcal{W}}_{\text{geom}}(A, z, t) + S^{\mathcal{R}}_f(A, Q, z) \end{pmatrix}.$$

The potential and sources are tied to the selected closure descriptors by

$$\begin{aligned} \partial_A \Psi_{\mathcal{W}}(A, z, t) &= \frac{A}{\rho} \partial_A p^{\mathcal{W}}_{\mathrm{g}}(A, z, t), \\ S^{\mathcal{W}}_{\text{geom}}(A, z, t) &= \partial_z \Psi_{\mathcal{W}}(A, z, t)\big|_A - \frac{A}{\rho}\, \partial_z p^{\mathcal{W}}_{\mathrm{g}}(A, z, t)\big|_A, \\ S^{\mathcal{R}}_f(A, Q, z) &= -C^{\mathcal{R}}_f(z, A, Q) \frac{Q}{A}. \end{aligned}$$

For $\mathcal{W}_{\text{elastic1D}}$, $p^{\mathcal{W}}_{\mathrm{g}}$ is the wall law in Definition 3.6. In component form,

$$\partial_t A + \partial_z Q = 0, \tag{3.7}$$

$$\partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A} + \Psi_{\mathcal{W}}(A, z, t) \right) = S^{\mathcal{W}}_{\text{geom}}(A, z, t) + S^{\mathcal{R}}_f(A, Q, z), \tag{3.8}$$

The wall descriptor $\mathcal{W}$ determines the pressure–area law, elastic potential, wave speed, and geometry source; the rheology descriptor $\mathcal{R}$ determines the viscosity or friction closure. A resolved FSI model is therefore not obtained by merely relabeling S_{geom}: it introduces structural state and interface laws. The displayed split records the balance-law objects needed by finite-volume schemes, while a computed realization may still use a particular well-balanced or source-split discretization [38] [29, Summary, Secs. 2.1–2.3, Secs. 5–6] [20].

3.6 Initial and Boundary Conditions

The comparison runs start from the geometry-rest state and then impose the finite-volume boundary approximation stated below. The generic alternatives are included to identify the model-family contract, while the final paragraph records the literal boundary rule used in the reported runs.

Definition 3.13 (Single-vessel boundary data)**.** For a generic single-vessel solve on $z \in [0, L]$, one declares one inlet condition and one outlet condition, for example

$$\begin{aligned} &Q(0,t) = Q_{\mathrm{in}}(t) \quad \text{or} \quad \bar{u}(0,t) = \bar{u}_{\mathrm{in}}(t), \\ &p_{\mathrm{g}}(L,t) = p_{\mathrm{out,g}}(t) \quad \text{or} \quad p_{\mathrm{g}}(L,t) = p_{\mathrm{WK,g}}(Q(L,\cdot),t). \end{aligned}$$

The second outlet option denotes a terminal Windkessel-style relation. These are model-family possibilities, not the literal boundary rule used by the reported comparison. The realized finite-volume run prescribes an inlet solver flow derived from a nominal reference-area mean velocity,

$$q_{\mathrm{in}} = R_0(0)^2 \bar{u}_{\mathrm{in}},$$

solves the inlet ghost area from the approximate outgoing characteristic invariant, fixes the outlet ghost coordinate $a_{\mathrm{out}} = R_0(L)^2$, and obtains the outlet ghost flow from the corresponding approximate invariant. Hence the imposed inlet condition is a prescribed reference-area flow, not a fixed realized mean velocity $q_{\mathrm{in}}/a_{\mathrm{in}}$, and the outlet condition fixes the elastic reference pressure only for the selected wall component.

Definition 3.14 (Characteristic speeds for the reduced elastic subsystem)**.** Let $\lambda_{\pm}$ denote the two characteristic speeds of the positive-area elastic 1D subsystem obtained from the homogeneous (A, Q) balance law. Their formula and assumptions are given in Appendix G.2.

Assumption 3.15 (Subcritical boundary regime)**.** The baseline boundary treatment is used only in the subcritical regime $\lambda_- < 0 < \lambda_+$, where one characteristic enters through the inlet and one characteristic enters through the outlet.

Under this regime, a finite-volume scheme constrains the incoming information at each boundary and leaves the outgoing state determined by the interior solve. This is the reduced-model analogue of declaring enough boundary data for the selected equation class; it is not, by itself, a validation of a particular boundary discretization.

3.7 Source-to-Implementation Model Variant

The selected implementation is a documented Canic-derived extended model, not a literal reproduction of every source-paper parameter and normalization in [28]. The report therefore uses the phrase $R_{\max}$-

normalized Canic-derived extended model when source fidelity matters. Table 4 records the relevant source-to-implementation choices.

Table 4: Source-to-implementation map for the selected 1D model.

Feature	**Canic et al. source form**	**Implemented report realization**
Pressure denominator	Elastic pressure written with a local $R_0(z)^{-2}$ denominator.	Evolution wall potential and source use the constant $R_{\max}^{-2}$ normalization specified in Appendix G.3.
Profile parameters	Source simulations state $\gamma = 9$ and $\alpha = 1.1$.	Principal runs use the parabolic profile $\alpha_{\mathcal{P}} = 4/3$, $g_{\mathcal{P}} = 4$; power-profile runs are sensitivity/diagnostic records.
Friction coefficient	Profile-dependent wall/friction closure in the reduced model.	Implemented through the declared velocity-profile shear factor and rheology descriptor.
p_2 differentiation	Variable-radius pressure correction differentiated in the model.	Locally frozen-viscosity realization for the p_2 derivative.
External pressure	External pressure term may be included in the formulation.	$p_{\text{ext}} \equiv 0$ in the reported runs.
Coordinates	Physical area and flow notation in the source paper.	Solver state is (a, q) with $A_{\text{phys}} = \pi a$, $Q_{\text{phys}} = \pi q$.
Boundary rule	Boundary conditions are model/problem dependent.	Fixed-area characteristic approximation applied with persisted subcritical-regime diagnostics.
Discrete well balancing	Continuous rest equilibrium is a model-level property.	Rusanov/MUSCL is not exactly well balanced for spatially varying R_0; Chapter 4 reports zero-forcing rest-state drift.

3.8 Selected and Supported Discretizations

The reported C23/C40 comparison uses the finite-volume realization specified in Appendix G.3. In summary, the radius-squared state (a, q) is advanced on a one-dimensional grid with a MUSCL reconstruction, minmod limiting, source splitting for the wall/geometry and variable-radius terms, and native SSPRK3 time integration. The numerical comparison runs use $N = 400$ cells on $0 \le z \le 6$ cm, a timestep cap of 10^{-5} s, and a CFL cap of 0.45. The boundary-state approximation is the fixed-area characteristic contract recorded in Appendix G.5.

The local Julia project also exposes the implemented spatial and time-integration surface in Table 5. These options are implementation-check and sensitivity context unless a result table explicitly identifies them as the active method. The project dependencies include `SciMLBase` and `OrdinaryDiffEq`; adaptive `vern7` and `vern9` policies therefore use the existing SciML backend. The characteristic WENO3 finite-volume reconstruction is implemented in-repo and is available as an opt-in higher-order sensitivity method. WENO5 and ENO remain future method-development options in this report.

3.9 Cross-Sectional Velocity Reconstruction

The 1D state directly supplies a section mean, $\bar{u}_{1D}(z,t) = q(z,t)/a(z,t)$, and physical flow $Q_{1D}(z,t) = \pi q(z,t)$. Principal radial profile plots use the current-radius coordinate $r/\sqrt{a(z,t)}$ and the parabolic-profile

Table 5: Implemented discretization surface used by the report. The principal comparison in this report uses MUSCL finite volume with minmod limiting and native SSPRK3; the other entries are benchmark, sensitivity, or backend parity context supported by the local Julia project.

Surface	**Implemented option**	**Report role**
Spatial	First-order finite-volume Rusanov	Low-order cell-average baseline and DG $p = 0$ bridge.
Spatial	MUSCL finite volume with minmod-limited interface states and Rusanov flux	Principal C23/C40 comparison method and main empirical refinement target.
Spatial	Characteristic WENO3 finite volume with Rusanov flux	Opt-in native finite-volume implementation check and smooth-regime sensitivity method; not the principal comparison method.
Spatial	Richtmyer/Lax–Wendroff finite volume with minmod-limited interface states	Native fixed-step sensitivity method; not exposed through the SciML `ODEProblem` backend because its flux predictor depends on $\Delta t/\Delta z$.
Spatial	Modal DG with Legendre degree $p = 0, 1, 2$	Native DG benchmark rows; $p = 0$ reduces to the cell-average FV path, while $p = 1, 2$ are high-order spatial context rather than the principal comparison method.
Time	Forward Euler, SSPRK2, SSPRK3, and SSPRK54	Native fixed-step options; SSPRK3 is the selected reported integrator, while Euler, SSPRK2, and SSPRK54 are diagnostic or higher-order sensitivity rows.
Time	SciML/OrdinaryDiffEq `auto`, `tsit5`, `vern7`, `vern9`, and `rodas5p`	Adaptive backend-parity checks for semi-discrete operators that the current `ODEProblem` path supports.

reconstruction

$$u_{1D}(r; z, t) = 2\,\frac{q(z,t)}{a(z,t)}\left(1 - \left(\frac{r}{\sqrt{a(z,t)}}\right)^2\right)$$

with the supplemental reference-radius coordinate $r/R_0(z)$ retained as an operator-control output. That reconstruction is a profile diagnostic, not a resolved secondary-flow, separation, skewness, or wall-traction model.

3.10 Plane–Tetrahedron Comparison Operator

For a target plane $S_j = \Omega_{3\mathrm{D}} \cap \{z = z_j\}$, the default `CrossSectionQuadratureOperator` intersects every tetrahedron with S_j, linearly interpolates axial velocity on the cut edges, sorts the cut polygon vertices in the plane, and triangulates the polygon by a centroid fan. Let $\mathcal{Q}_j$ denote the resulting cut triangles. The reported 3D area, flow, and mean velocity are

$$A_{3D,j} = \sum_{\Delta\in\mathcal{Q}_j} |\Delta|, \qquad Q_{3D,j} = \sum_{\Delta\in\mathcal{Q}_j} |\Delta|\,\bar{u}_{z,\Delta}, \qquad \bar{u}_{3D,j} = Q_{3D,j}/A_{3D,j}.$$

For a cut triangle Δ whose vertices carry reconstructed axial velocities $u_{z,1}$, $u_{z,2}$, and $u_{z,3}$,

$$\bar{u}_{z,\Delta} = \frac{u_{z,1} + u_{z,2} + u_{z,3}}{3}.$$

The corresponding 1D quantities use physical flow, not the raw solver flow coordinate:

$$Q_{1D,j} = \pi q(z_j, t), \qquad \bar{u}_{1D,j} = q(z_j, t)/a(z_j, t).$$

Radial profile rows use the same cut triangles, binned by area-conservative three-point triangle quadrature in either $r/\sqrt{a(z,t)}$ or $r/R_0(z)$. The recorded controls include 10/20/40-bin sensitivity, area-weighted bin means, within-bin azimuthal variance, and the current-to-reference area mismatch. Empty bins are reported as invalid rows rather than zero-velocity observations.

3.11 Numerical Experiments, Discrepancy Metrics, and Provenance

The resolved-velocity comparison evaluates the selected 1D solution against two available resolved 3D velocity comparison datasets at nominally aligned final-time samples: the 1D output is sampled at 1.0 s and the XDMF velocity field records 0.9995 s. The cases are labelled C23 and C40 by stenosis severity. They are computational cross-model diagnostics, not pressure-flow results and not clinical validation.

Solver-coordinate convention. In the comparison record, the stored solver state `A` is $a = R^2$, not physical circular area. The physical map is Definition 3.7: $A_{\mathrm{phys}} = \pi a$, $Q_{\mathrm{phys}} = \pi q$, and $\bar{u} = q/a = Q_{\mathrm{phys}}/A_{\mathrm{phys}}$.

The manuscript tables and figures use the tracked static report assets under `figures/static/static/data/stenosis-comparison/`, interpreted under Convention G.1. The baseline regeneration command, source-path convention, and file hashes are listed in Appendix H. Scratch sensitivity reruns live under `simulations/output/revision-*` or `tmp/revision-evidence/` and are not the manuscript assets used for the current tables and figures. The raw XDMF/HDF5 inputs remain local ignored files under `simulations/data/3d/canic_case3/`; the report archives the generated CSV and PGFPlots-ready static data, not the raw fields.

Table 6: Case matching for the resolved-velocity comparison. The case labels C23 and C40 are used in the main text and figures; the source ID records the upstream local data directory.

Label	Source ID	Velocity metadata	Operator	3D time (s)	Time offset (s)
C23	77	`77/velocity.xdmf`	`CrossSectionQuadratureOperator`	0.9994999999999453	5.0×10^{-4}
C40	60	`60/velocity.xdmf`	`CrossSectionQuadratureOperator`	0.9994999999999453	5.0×10^{-4}

Table 7: Model-matching matrix for the 1D–3D velocity diagnostic. Status labels record what the archived report artifact supports, not what a future matched study might supply.

Feature	**1D model**	**3D comparison dataset**	**Status**	**Implication**
Reference-profile geometry	Same analytic stenosis profile and declared severity.	Source mesh metadata is interpreted against the same nominal profile.	approximately matched	The intended stenosis shape is shared.
Cut-operator area	Static reference area $A_{\mathrm{ref}} = \pi R_0^2$ is available.	Tetrahedral cuts audit against the same static reference area.	approximately matched	Cut areas show typical sub-percent agreement with localized multi-percent outliers.
Current lumen area	Dynamic 1D area is $A_{1D} = \pi a(z,t)$.	Fixed 3D cut area is $A_{3D}(z)$.	algebraically reconstructed	The tracked section asset supports the reported flow/area split, but the split is bookkeeping rather than a wall-motion attribution.
Wall model	Compliant reduced Canic–Koiter wall law.	Velocity and displacement XDMF files expose one local snapshot; wall-history and area-rate data are not persisted here.	unmatched	Wall effects and flow conservation cannot be isolated from the velocity mismatch.
Density	ρ recorded by the 1D parameters.	Not persisted in the tracked comparison tables.	unknown	Density-sensitive pressure or wave-speed claims are out of scope.
Viscosity/rheology	Newtonian $\nu = 0.04\ \mathrm{cm}^2/\mathrm{s}$.	Rheology and viscosity metadata are not persisted in the report artifact.	unknown	The velocity comparison is not a material-parameter validation.
Initial condition	Geometry-rest state, then forced by the inlet.	Prior transient history is not persisted.	unmatched	Final-time differences may include start-up history mismatch.
Inlet condition	Reference-area flow $q_{\mathrm{in}} = R_0(0)^2 \bar{u}_{\mathrm{in}}$ from nominal $\bar{u}_{\mathrm{in}} = 22.5\ \mathrm{cm/s}$.	Boundary condition used for the 3D field is not persisted here.	unknown	Flow mismatch cannot be attributed uniquely to closure discrepancy.
Outlet condition	Fixed-area characteristic approximation.	Outlet treatment is not persisted here.	unknown	Reflections and downstream constraints remain unquantified.
Time / transient history	Evaluated at $T = 1.0$ s from a start-up transient.	XDMF time is 0.9995 s; earlier history is not recorded.	unknown	The final timestamps are close, but phase/history matching is not established.
Numerical resolution	$N = 400$ finite-volume cells.	Node-centered velocity fields with loaded node counts reported.	unresolved	Scratch reruns provide diagnostic resolution checks, but the manuscript tables remain single-realization descriptors unless a dedicated output-sensitivity study is completed.
Velocity observable	Section mean $\bar{u} = q/a$ and physical flow πq.	Plane–tetrahedron cross-section quadrature of axial velocity.	matched	The compared observable is well defined.
Pressure availability	Pressure-output vocabulary exists but is not used here.	No pressure comparison is reported.	unmatched	This is not a pressure-drop, FFR, or CT-FFR study.

Table 8: Shared 1D realization and output-operator parameters for the $t = 1.0$ comparison. The generated comparison records characteristic-speed diagnostics, while realized CFL, mass, and positivity diagnostics are exposed by the solver and verification records.

Quantity	**Recorded value**
Forward model	`canic-extended-1d`.
Wall model	$\mathcal{W}_{\mathrm{elastic1D}}$ with the selected Canic–Koiter pressure–area law `canic-koiter-thin-membrane`.
Rheology	Newtonian rheology with $\nu = 0.04$ cm^2/s.
Velocity profile	Parabolic profile with $\alpha_{\mathcal{P}} = 4/3$.
Variable-radius terms	Canic effective-alpha and geometry-source corrections, including the locally frozen p_2 derivative in the extended member.
Physical parameters	$\rho = 1.055$ g/cm^3, $E = 5.02 \times 10^6$ dyn/cm^2, $h = 0.06$ cm, $\sigma = 0.5$, $K = 4.016 \times 10^5$ dyn/cm, $R_{\max} = 0.18$ cm, and $p_{\mathrm{ext}} \equiv 0$.
Discretization	$N = 400$, $L = 6$ cm, MUSCL finite volume with minmod limiter, native SSPRK3 time integration.
Time controls	$T = 1.0$ s from a start-up transient, timestep cap 10^{-5} s, CFL cap 0.45.
Initial and boundary state	Geometry-rest initial state $a = R_0^2$, $q = 0$; inlet $\bar{u}_{\mathrm{in}} = 22.5$ cm/s; outlet $a_{\mathrm{out}} = R_0(L)^2$ with the implemented characteristic approximation in Appendix G.5.
Section targets	200 equally spaced axial planes on $0 \leq z \leq 6$ cm.
Radial targets	Slices $z = 1.951$, 2.451, and 2.951 cm with principal 20-bin plots in $r/\sqrt{a(z,t)}$ plus supplemental $r/R_0(z)$ and 10/20/40-bin sensitivity rows.
Supplemental sensitivity	Node-slab arithmetic means are stored only to `node-slab-sensitivity.csv` for half-widths 0.0075, 0.0150753769, and 0.0301507538 cm.

Velocity and flow differences are reported as empirical sample L^1, L^2, and L^3 errors over valid section targets. With $d_{u,j} = \bar{u}_{1D,j} - \bar{u}_{3D,j}$ and $d_{Q,j} = Q_{1D,j} - Q_{3D,j}$, the normalized sample errors are

$$E_{u,p} = \left(\frac{1}{n} \sum_j |d_{u,j}|^p \right)^{1/p}, \qquad E_{Q,p} = \left(\frac{1}{n} \sum_j |d_{Q,j}|^p \right)^{1/p}, \qquad p \in \{1,2,3\},$$

$$E_{u,p}^{\mathrm{rel}} = \frac{\left(\sum_j |d_{u,j}|^p \right)^{1/p}}{\left(\sum_j |\bar{u}_{3D,j}|^p \right)^{1/p}}, \qquad p \in \{1,2,3\}.$$

4 Implementation Checks and Targeted Numerical Verification

This chapter reports the implementation checks and targeted numerical verification now available for the selected solver. The title is intentionally narrower than "full numerical verification." The evidence below supports the finite-volume implementation, diagnostics, reported output records, and selected verification problems used in this report. It does not establish clinical validation, pressure or FFR accuracy, or exhaustive verification of every possible model variant.

4.1 Scope of Convergence and Stability Evidence

The numerical evidence in this chapter is implementation verification rather than a formal convergence proof for the full nonlinear stenosis solver. The standard finite-volume context is supplied by a conservative hyperbolic discretization with consistent numerical fluxes and sources, CFL-controlled time stepping, limiter and Rusanov dissipation choices, and SSP Runge–Kutta time integration [38]. In 1D blood-flow practice, MUSCL/Rusanov and Lax–Friedrichs-type discretizations are commonly assessed through exact-solution, benchmark, grid-refinement, and equilibrium tests [29, 30, 31, 37, 39]. The report-level evidence then comes from manufactured-solution rows, zero-forcing rest-state drift checks, sampled refinement records, and admissibility diagnostics such as CFL, characteristic-sign, positivity, and mass-balance metadata. Discrete L^1, L^2, and L^∞ histories are therefore read as empirical error and boundedness diagnostics for the recorded runs. They are not, by themselves, stability proofs or convergence theorems for every admissible stenosis configuration.

Reading the diagnostics. The strongest positive evidence in this chapter is the manufactured-solution table because it compares the implemented operator against a known smooth solution. The rest-state drift table is an equilibrium and well-balancedness diagnostic, and here it records a negative result for the selected Rusanov/MUSCL flux–source realization. CFL, characteristic-sign, positivity, and mass-balance rows are admissibility and boundedness diagnostics for the accepted runs. Self-convergence and cross-integrator rows are empirical implementation-health checks rather than stability or convergence proofs. The WENO3, SSPRK54, nonselected DG, and rheology/profile rows are sensitivity and descriptor-context evidence unless a later claim explicitly names them as the active method.

4.2 Software and Descriptor Checks

The reproducible benchmark suite records implementation-check stages for the Julia solver: descriptor health, sampled self-convergence, native/SciML cross-integrator discrepancy, stationary-Stokes projection rows, rheology/profile sensitivity, OpenBF-style boundary handling, and resolved-velocity output diagnostics. These stages check that the selected implementation surface remains runnable and that documented output summaries retain their expected columns. Descriptor-health rows exercise the implemented spatial surface from Table 5: first-order finite volume, MUSCL finite volume, characteristic WENO3 finite volume, Richtmyer/Lax–Wendroff finite volume, and modal DG degrees $p = 0, 1, 2$. The smoke profile pairs those spatial descriptors with SSPRK3; the full profile also expands native descriptor-health rows across Euler, SSPRK2, SSPRK3, and SSPRK54. Refinement rows provide FV/DG spatial context, while backend-parity rows compare native SSPRK3 against the supported SciML/OrdinaryDiffEq policies for semi-discrete operators admitted by that backend, including `vern7` and `vern9` in the full profile.

```
./scripts/stenosis-hemodynamics benchmark \
  --profile overnight \
  --output-dir simulations/output/package_benchmark/<run_id> \
  --overwrite \
  --include-resolved3d \
  --publish-report-assets

pipenv run python scripts/render_package_benchmark_figures.py \
  --benchmark-dir simulations/output/package_benchmark/<run_id>
```

Table 9: Package benchmark stage summary.

Stage	Rows	OK	Skipped/error
case results	18	18	0
refinement	128	128	0
backend parity	18	18	0
stokes ic	12	12	0
rheology profile	60	60	0
boundary openbf	3	3	0
resolved3d	4	4	0

4.3 Manufactured-Solution Verification

The implementation now exposes an opt-in manufactured forcing term. The production equations use `NoForcing()` by default. For verification only, `ManufacturedForcing()` supplies a smooth exact state $(a^\star(z,t), q^\star(z,t))$, exact initial data, exact boundary states, and source terms inserted into the existing finite-volume right-hand side. In solver coordinates, let

$$s(z) = \sin\left(\frac{\pi z}{L}\right), \qquad a_0(z) = R_0(z)^2, \qquad \omega = \frac{2\pi}{T_{\text{mms}}}.$$

The default manufactured record uses

$$\epsilon_a = 0.02, \qquad U_{\text{mms}} = 2.0\,\text{cm/s}, \qquad T_{\text{mms}} = 0.02\,\text{s},$$

and defines

$$
\begin{aligned}
a^\star(z,t) &= a_0(z)\left[1+\epsilon_a s(z)\cos(\omega t)\right],\\
q^\star(z,t) &= a_0(z)U_{\mathrm{mms}}s(z)^2\sin(\omega t).
\end{aligned}
$$

The tabulated run uses $0 \le z \le L = 6\,\mathrm{cm}$ and $0 \le t \le 2\times 10^{-4}\,\mathrm{s}$ unless the CLI options below are changed. Initial data are sampled from $(a^\star, q^\star)$ at $t=0$, and the boundary state routine receives the exact traces at $z=0$ and $z=L$. The inserted forcing terms are the residuals required by the recorded flux and source realization,

$$
\begin{aligned}
f_a(z,t) &= \partial_t a^\star + \partial_z F_a(a^\star, q^\star; z),\\
f_q(z,t) &= \partial_t q^\star + \partial_z F_q(a^\star, q^\star; z) - S_q(a^\star, q^\star; z),
\end{aligned}
$$

where F_a, F_q and S_q are the selected discrete-model point flux and source functions. The spatial derivatives used to evaluate the forcing are generated by the verification helper, so the test checks the implemented operator against this declared manufactured record rather than a physical stenosis experiment. The runner

```
./scripts/stenosis-hemodynamics verify mms \
  --nxs 50,100,200,400 \
  --dt-values 2e-5,1e-5,5e-6,2.5e-6 \
  --output-dir figures/static/static/tables/verification \
  --overwrite
```

records cell-center discrete errors

$$
\|e\|_1 = \frac{1}{N}\sum_i |e_i|, \qquad \|e\|_2 = \left(\frac{1}{N}\sum_i e_i^2\right)^{1/2}, \qquad \|e\|_\infty = \max_i |e_i|,
$$

with observed orders computed from the L^2 rows as

$$
p_{\mathrm{obs}} = \frac{\log(E_{\mathrm{coarse}}/E_{\mathrm{fine}})}{\log(\Delta_{\mathrm{coarse}}/\Delta_{\mathrm{fine}})}.
$$

For spatial rows Δ is the grid spacing; for requested-timestep rows it is the requested timestep. The native solver also records realized CFL diagnostics because the accepted timestep is $\min(\Delta t_{\mathrm{request}}, \Delta t_{\mathrm{CFL}})$. These rows are method-of-manufactured-solutions evidence for the implemented operator; the temporal rows should not be read as a clean temporal-order estimate when the spatial error floor or CFL cap dominates. The same runner can be repeated with `--space fv-weno3` to exercise the characteristic WENO3 operator, but WENO3 order claims are kept cautious because source quadrature, boundary treatment, limiter fallback, and positive-area projection can cap the full-system observed rate. They are not physical validation cases.

The spatial rows support an empirical convergence reading for the implemented forced operator, but the

Table 10: Manufactured-solution verification errors. The order columns use the L^2 errors.

Study	N	Δt	$\|e_a\|_1$	$\|e_a\|_2$	$\|e_a\|_\infty$	p_a	$\|e_q\|_1$	$\|e_q\|_2$	$\|e_q\|_\infty$	p_q
spatial	50	2.500e-06	4.154e-07	9.503e-07	3.138e-06	1.567	0.00106	0.00351	0.01693	1.255
spatial	100	2.500e-06	1.172e-07	3.208e-07	1.368e-06	1.613	3.172e-04	0.001471	0.01014	1.466
spatial	200	2.500e-06	3.066e-08	1.049e-07	5.830e-07	1.639	8.195e-05	5.323e-04	0.005214	1.502
spatial	400	2.500e-06	7.732e-09	3.368e-08	2.410e-07	–	2.057e-05	1.879e-04	0.002612	–
temporal	400	2.000e-05	7.741e-09	3.369e-08	2.406e-07	0.000e+00	2.058e-05	1.879e-04	0.002612	0.000e+00
temporal	400	1.000e-05	7.741e-09	3.369e-08	2.406e-07	2.672e-04	2.058e-05	1.879e-04	0.002612	-7.372e-06
temporal	400	5.000e-06	7.736e-09	3.369e-08	2.408e-07	1.815e-04	2.058e-05	1.879e-04	0.002612	2.217e-05
temporal	400	2.500e-06	7.732e-09	3.368e-08	2.410e-07	–	2.057e-05	1.879e-04	0.002612	–

observed rates are below the formal smooth-region MUSCL/SSPRK3 expectation. In this table, the reported observed order is taken from the L^2 rows; the L^1 history records integrated error behavior, while L^∞ highlights localized boundary, limiter, or source-discretization effects. The subformal observed rates are consistent with limiter activation, source discretization, boundary treatment, or an unresolved spatial/temporal error balance.

Illustrative p- and h-refinement split. The same manufactured-solution setting is used for a deliberately narrow p-versus-h demonstration. In the h-refinement family, the DG polynomial degree is fixed and the element count is increased. In the p-refinement family, the mesh is fixed and the modal polynomial degree is increased through the implemented DG range. The shared horizontal cost measure is the number of state degrees of freedom. The runner and renderer are

```
./scripts/stenosis-hemodynamics verify ph-refinement \
  --h-nxs 20,40,80,160 \
  --h-degree 2 \
  --degrees 0,1,2,3,4 \
  --p-nx 40 \
  --output-dir figures/static/static/data/verification \
  --summary-tex figures/static/static/tables/verification/p_h_refinement_demo.tex \
  --overwrite

pipenv run python scripts/render_ph_refinement_demo.py
```

The resulting rows are read as an illustrative smooth-solution comparison. Boundary treatment, positivity limiting, source quadrature, and the fixed timestep can cap the ideal p-refinement trend, so flat or regressing high-order rows are reported as implementation sensitivity rather than hidden spectral convergence.

4.4 Zero-Forcing Geometry-Rest Drift

The continuous model has a geometry-rest equilibrium at $a_i = R_0(z_i)^2$, $q_i = 0$ under zero forcing. The discrete Rusanov/MUSCL operator is not exactly well balanced for spatially varying R_0 because the mass

Table 11: Manufactured-solution p- and h-refinement demonstration. The h-refinement rows report observed order from adjacent grid spacings; the p-refinement rows report error reduction relative to the previous polynomial degree.

Sweep	p	N	DOFs	$\|e_a\|_2$	h-order	p-reduction	$\|e_q\|_2$	h-order	p-reduction
h-refinement	2	20	120	9.797e-06	1.457	–	0.009629	1.052	–
h-refinement	2	40	240	3.569e-06	1.488	–	0.004643	1.209	–
h-refinement	2	80	480	1.272e-06	1.51	–	0.002009	1.424	–
h-refinement	2	160	960	4.467e-07	–	–	7.489e-04	–	–
p-refinement	0	40	80	1.078e-05	–	–	0.002585	–	–
p-refinement	1	40	160	3.569e-06	–	3.022	0.004645	–	0.5565
p-refinement	2	40	240	3.569e-06	–	1	0.004643	–	1
p-refinement	3	40	320	3.569e-06	–	0.9999	0.004643	–	1
p-refinement	4	40	400	3.569e-06	–	1	0.004642	–	1

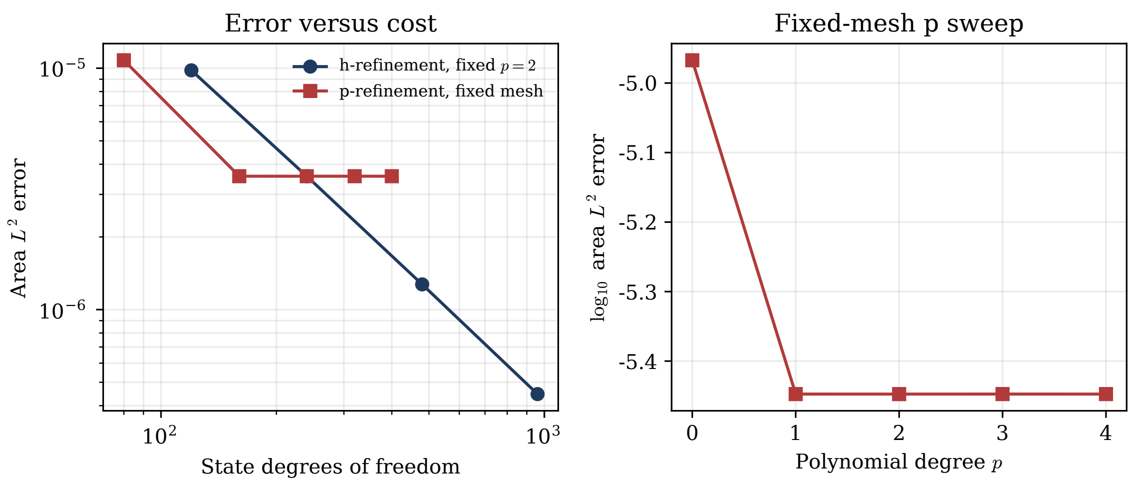

Figure 7: Manufactured-solution p- and h-refinement demonstration for the modal DG surface. The comparison isolates fixed-degree h-refinement from a fixed-mesh p sweep and should be interpreted as smooth-case implementation evidence, not physical validation.

flux includes numerical viscosity. The verification runner therefore starts the selected C23 and C40 geometries from the rest state with a zero-velocity inlet and reports

$$\max_i |q_i|, \qquad \max_i |a_i - R_0(z_i)^2|, \qquad \left|\sum_i a_i \Delta z - \sum_i a_i(0)\Delta z\right|$$

as functions of grid size and elapsed time. This test is required context for any comparison run initialized from the same geometry-rest state.

```
./scripts/stenosis-hemodynamics verify rest \
  --severities 23,40 \
  --nxs 100,200,400,800 \
  --elapsed-times 0,0.001,0.01,0.1,1.0 \
  --output-dir figures/static/static/tables/verification \
```

```
--overwrite
```

Table 12: Zero-forcing, zero-inlet geometry-rest drift summary. Values are maxima over the reported positive elapsed times for the two finest grids. The normalization uses the comparison-run flow scale $q_{\mathrm{comp}} = 2.288/\pi = 0.7283\,\mathrm{cm}^3/\mathrm{s}$.

Severity	N	$t_{\max q}$	$\max\lvert q_i\rvert$	$\max\lvert q_i\rvert/q_{\mathrm{comp}}$	$\max\lvert a_i - R_{0,i}^2\rvert$	$\max\lvert\Delta M\rvert$	Subcrit. min	$\max\lvert\Delta t_{\mathrm{term}}\rvert$
23	400	0.01	1.559	2.141	7.422e-04	0.001577	800.9	0.000e+00
23	800	0.01	1.566	2.151	7.401e-04	0.001574	800.4	0.000e+00
40	400	0.01	1.64	2.252	7.984e-04	0.001944	626.1	0.000e+00
40	800	0.01	1.658	2.276	7.941e-04	0.001925	624.9	0.000e+00

Interpretation of rest-state drift. The corrected rest-state run has zero body forcing and zero inlet flow, so the reported motion is not an imposed boundary through-flow. The comparison-flow scale is $q_{\mathrm{comp}} \approx 2.288/\pi = 0.7283\,\mathrm{cm}^3/\mathrm{s}$, close to the production through-flow scale in Table 16. The largest corrected rest-state values on the two finest grids are about 2.14–2.28 q_{comp} by $t = 0.01$ s and remain about one q_{comp} at $t = 1.0$ s on the finest grid. This is a material non-well-balanced artifact of the present Rusanov/MUSCL flux–source realization, possibly amplified by the characteristic boundary response; it is not small relative to the C23/C40 comparison flow scale. The C23/C40 comparison runs are initialized from the same geometry-rest state and use the same operator, so their start-up transient may include artificial rest-state motion before and during the imposed through-flow. This limits the C23/C40 comparison to a velocity-output and reproducibility diagnostic rather than a clean discretization-accuracy result. The contribution claimed here is therefore an auditable, diagnostically exposed implementation, not a well-balanced stenosis discretization.

4.5 Boundary and Conservation Diagnostics

The native solver records per-run diagnostics in `SimulationDiagnostics`: accepted timestep extrema, realized CFL extrema, λ_- and λ_+ extrema, the minimum subcriticality margin $\min(-\lambda_-, \lambda_+)$, mass extrema and mass defect, positivity projection count, and total area correction from the positivity floor. These diagnostics are reported by direct simulation summaries, benchmark case rows, rest-state drift rows, and resolved-velocity comparison summaries.

The fixed-area characteristic boundary rule remains a method applied under the subcritical sign condition $\lambda_- < 0 < \lambda_+$. A positive characteristic radicand is not enough. The persisted sign extrema and subcriticality margin are therefore the relevant evidence for the boundary regime in each accepted native run.

4.6 Self-Convergence and Cross-Integrator Checks

Self-convergence and backend-parity rows are empirical implementation checks. They can reveal grid sensitivity, boundedness problems, or integration-policy discrepancies, but they do not replace the exact-solution

MMS rows and do not establish stability of the nonlinear semi-discrete operator.

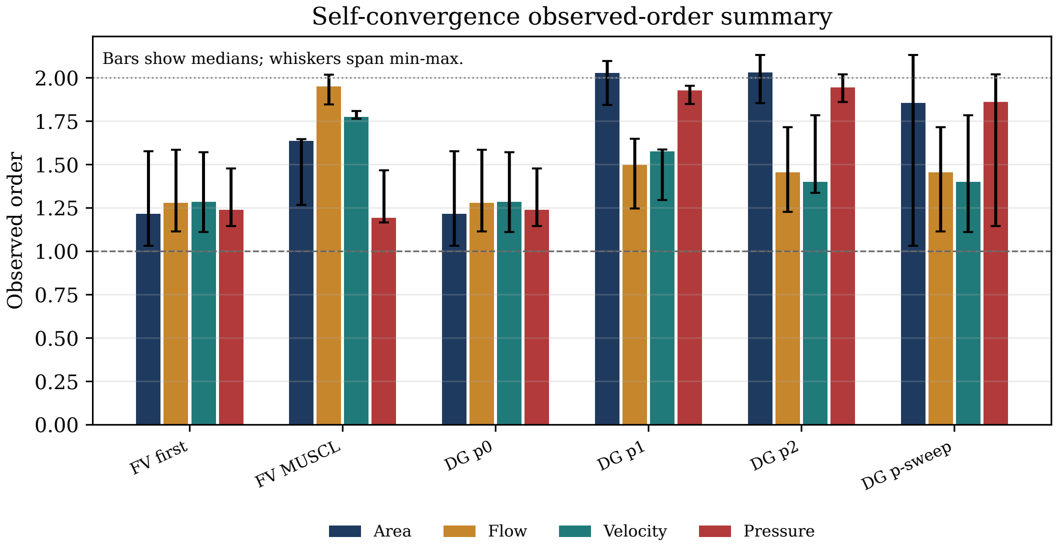

Figure 8: Self-convergence observed-order summary reported by the benchmark renderer. This is a sampled implementation-check diagnostic; the MMS table in Section 4.3 is the stronger exact-solution verification evidence.

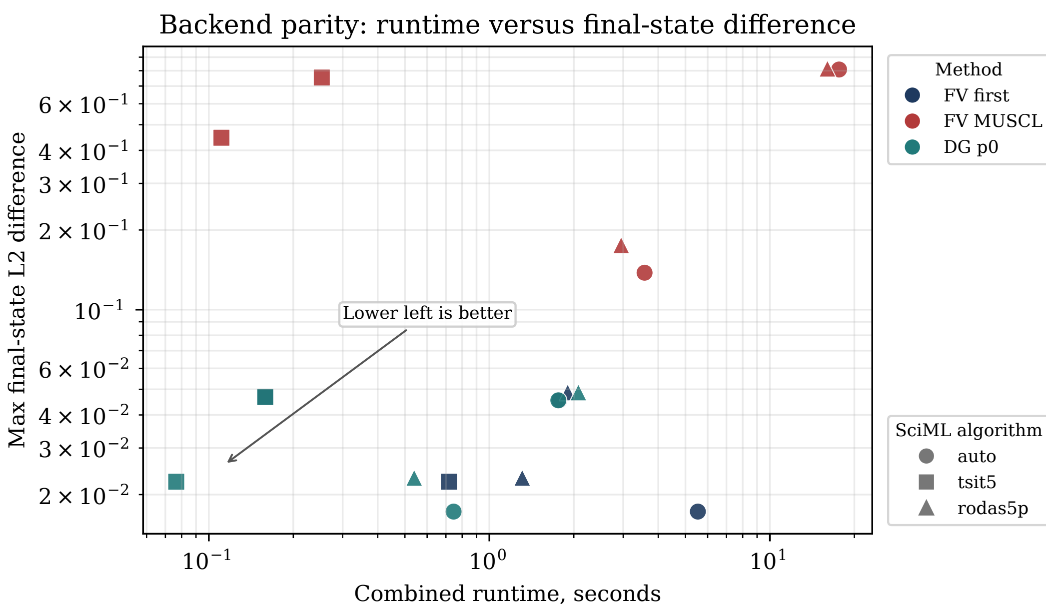

Figure 9: Native/SciML cross-integrator discrepancy versus runtime for compatible methods. The native path uses the stagewise positivity limiter; the SciML path integrates the unprojected semi-discrete system. The figure is therefore a cross-integrator diagnostic, not a proof of solver equivalence.

4.7 Stationary-Stokes Sensitivity

The stationary-Stokes module records a two-level mesh sensitivity summary rather than only a projection-existence check. It compares projected one-dimensional section averages with finite-element section samples across generated stationary Stokes meshes and reports finest-mesh-relative discrepancies. This remains an initializer/projection diagnostic; it is not a convergence theorem or transient 3D validation of the 1D solver.

Table 13: Stationary-Stokes projection and mesh-refinement diagnostics. The finest-mesh relative discrepancy is zero on the finest row by construction; it is included to show the refinement reference used.

Case	Nodes	Cells	$\langle Q \rangle_{\mathrm{FE}}$	rel. diff.$(u_{\mathrm{FE}}, u_{\mathrm{proj}})$	rel. diff.$(u_{\mathrm{FE}}, u_{\mathrm{finest}})$	max $\lvert \tau_w \rvert$
stokes-s23-8x2x8	153	576	0.1743	0.2012	0.3643	5.644
stokes-s23-16x4x16	1105	5376	0.1743	0.1928	0.000e+00	8.446
stokes-s40-8x2x8	153	576	0.1286	0.378	0.5212	6.034
stokes-s40-16x4x16	1105	5376	0.1286	0.2201	0.000e+00	11.06

4.8 Resolved-Velocity Benchmark Diagnostic

The package benchmark also keeps a resolved-velocity output-operator stage. Its role in this chapter is limited: it checks that the `CrossSectionQuadratureOperator` report path remains executable and records the same kind of mean absolute velocity discrepancy used in Section 5. The detailed interpretation of the resolved comparison belongs to Chapter 5.

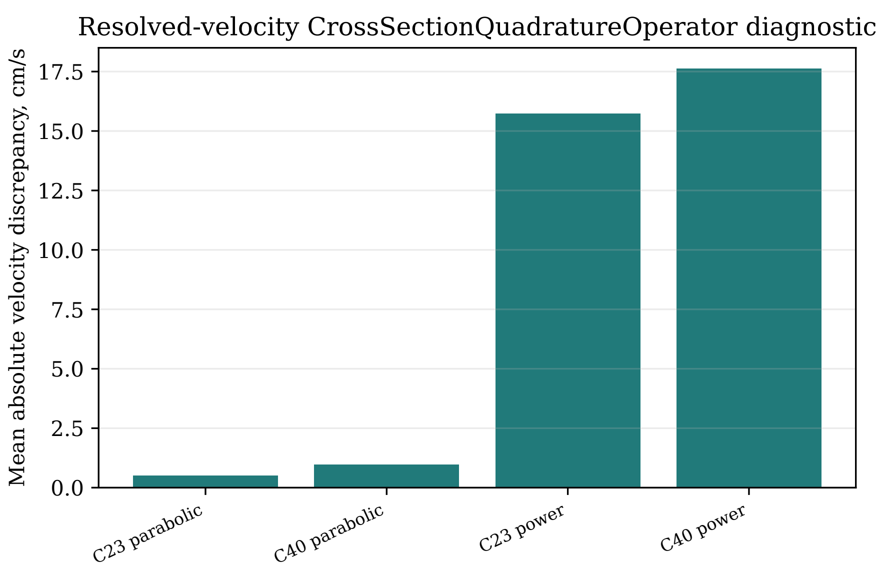

Figure 10: Resolved-velocity output-operator diagnostic summary for the available local cases. The rendered value is the section-wise mean absolute velocity discrepancy for the declared `CrossSectionQuadratureOperator`.

4.9 Descriptor Sensitivity Checks

The remaining benchmark figure is a descriptor-level sensitivity check. It is included here to show that the rheology/profile options remain connected to the benchmark record, but it is not used to support the principal C23/C40 comparison claim.

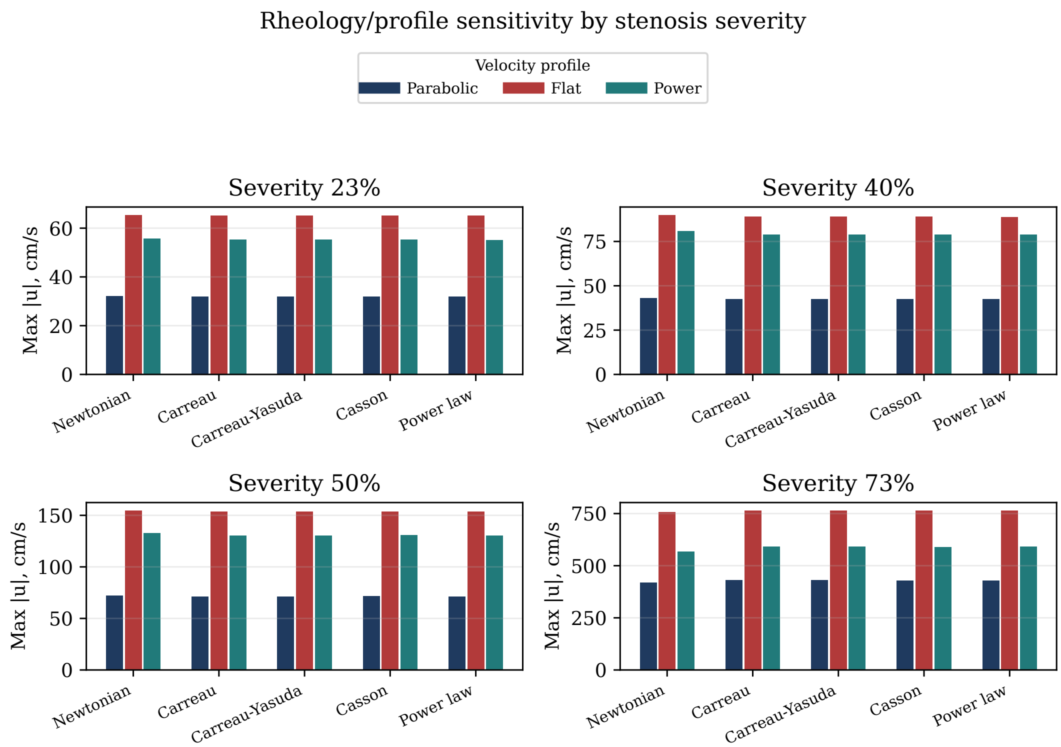

Figure 11: Rheology/profile sensitivity summary for declared descriptor combinations. This stage checks descriptor response direction; it is not part of the principal C23/C40 comparison evidence.

4.10 Verification Summary

The chapter supports a bounded numerical-verification claim, not clinical validation or a convergence theorem for the full nonlinear stenosis problem. Its strongest positive evidence is the exact-solution MMS table for the implemented forced operator. Its main negative evidence is the zero-forcing rest-state drift table, which shows that the selected Rusanov/MUSCL flux–source realization is not well balanced for the recorded geometries. CFL and characteristic-speed metadata, mass-balance and positivity metadata, sampled self-convergence, cross-integrator checks, and stationary-Stokes projection sensitivity provide additional implementation-health and admissibility evidence. Rows for runtime, platform, broad rheology/profile sensitivity, WENO3, SSPRK54, Verner adaptive policies, Lax–Wendroff, and nonselected DG remain sensitivity or appendix-level context unless a principal research claim explicitly names them.

5 Cross-Model Velocity Comparison

5.1 Three-Dimensional Comparison Datasets

This section reports the final-time resolved-velocity comparison whose protocol is defined in Section 3.11. The comparison is velocity-focused: pressure-drop, pressure-ratio, FFR, and clinical validation claims remain outside the completed numerical evidence. Because the zero-input rest-state diagnostic is not small at the comparison time scale and a production-level grid/time sensitivity study is not complete in the tracked report assets, the numerical values in this section are single-realization descriptors under the declared output operator.

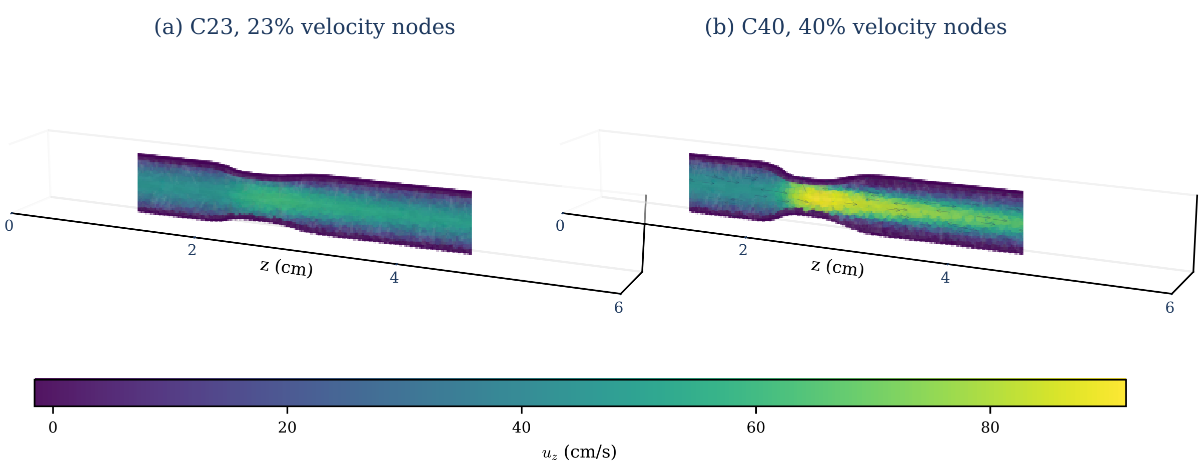

Figure 12: Resolved 3D node-centered axial velocity fields for the two $T = 1$ comparison cases used in Section 5. Panel (a) shows C23 with 22,691 loaded nodes; panel (b) shows C40 with 21,975 loaded nodes. The shared color scale makes the stronger axial-velocity concentration in C40 visible. Sparse arrows indicate local vector direction, and the transparent surface frames the 3D point cloud. These are renderings of the resolved velocity comparison data, not the stationary-Stokes pathline proxy or a pressure-drop, FFR, or patient-specific clinical calibration measurement.

5.2 Cross-Sectional Sampling Protocol

The same quadrature record audits whether the cut operator is sampling the expected static cross-section. With the reference area $A_{\text{ref}}(z_j) = \pi R_0(z_j)^2$, define

$$\epsilon_A(z_j) = \frac{|A_{3D}(z_j) - A_{\text{ref}}(z_j)|}{A_{\text{ref}}(z_j)}.$$

Table 14 reports this geometry/operator audit for the same 200 section targets. It shows that the tetrahedral cut areas agree with the expected fixed stenosis cross-sections to sub-percent typical discrepancy, with the largest differences confined to a few section targets. It does not measure the dynamic area discrepancy entering the 1D velocity denominator. The relevant unpersisted quantity for velocity attribution is

$$\epsilon_A^{\text{model}}(z_j, t) = \frac{|\pi a(z_j, t) - A_{3D}(z_j)|}{A_{3D}(z_j)}.$$

The algebraic split used here for flow/area bookkeeping is

$$\frac{Q_{1D}}{A_{1D}} - \frac{Q_{3D}}{A_{3D}} = \frac{Q_{1D} - Q_{3D}}{A_{1D}} + Q_{3D}\left(\frac{1}{A_{1D}} - \frac{1}{A_{3D}}\right), \qquad A_{1D} = \pi a.$$

The tracked section-quadrature data store enough section-wise values to reconstruct this split on retained sections with nonzero mean-velocity denominators: $A_{1D} = Q_{1D}/\bar{u}_{1D}$ and $A_{3D} = Q_{3D}/\bar{u}_{3D}$. Table 15 summarizes the resulting contribution magnitudes. This decomposition is an accounting identity for the reported samples, not a causal attribution of the flow mismatch.

5.3 Section-Averaged Velocity Results

Table 14: Quadrature cut-area audit against the static reference $A_{\text{ref}}(z) = \pi R_0(z)^2$. Discrepancies are percentages over valid section targets and are stored in the tracked `area-audit.dat` report asset.

Case	Sections	min ϵ_A	median ϵ_A	mean ϵ_A	max ϵ_A
C23	200	0.045	0.281	0.389	2.94
C40	200	0.005	0.291	0.397	4.46

The largest C40 velocity discrepancy occurs at the same section as the largest static cut-area error in Table 14. Recomputing the summary after excluding the two C40 sections with static area error above 4% moves the maximum discrepancy from 9.92 cm/s at $z = 2.261$ cm to 8.74 cm/s at $z = 2.291$ cm, where the static area error is below 1%. Thus the localized C40 peak is not solely an area-audit artifact, but the exact maximum value and location remain operator-sensitive.

Table 15: Section-wise decomposition of the mean-velocity discrepancy into physical-flow and area-denominator contributions. Values are in cm/s and are reconstructed from the tracked section-quadrature report asset.

Case	$\langle\lvert(Q_{1D} - Q_{3D})/A_{1D}\rvert\rangle$	$\langle\lvert Q_{3D}(A_{1D}^{-1} - A_{3D}^{-1})\rvert\rangle$	$z_{\max\lvert d_u\rvert}$	$\lvert d_u\rvert_{\max}$	Dominant term at max
C23	0.529	0.112	2.291	2.35	Flow contribution
C40	1.012	0.133	2.261	9.92	Flow contribution

The comparison records now report both the requested and completed 1D times. In the tracked C23/C40 sample, the 1D run completes at 1.0 s and the XDMF velocity field is sampled at 0.9995 s, so the cross-model sampling offset is 5×10^{-4} s. That offset is numerically small, but it does not establish matched transient histories. The reported characteristic-radicand scales correspond to one-way traversal times on the order of milliseconds for a 6 cm vessel, so boundary and start-up histories can influence the state many times before the sampled final time.

Table 16: Quadrature-backed section velocity and physical-flow summary at nominally aligned final-time samples. These are single-realization descriptors, not converged production-accuracy estimates. Mean velocities and velocity errors are in cm/s; physical flows and flow errors are in cm^3/s. Angle brackets in this table denote axial sample means over the 200 retained comparison sections, not temporal averages or inner products. The 1D physical flow is $Q_{1D} = \pi q$.

Case	Sections	$\langle \bar{u}_{1D} \rangle$	$\langle \bar{u}_{3D} \rangle$	$\langle Q_{1D} \rangle$	$\langle Q_{3D} \rangle$	$E_{u,1}$	$E_{u,2}$	$E_{u,3}$	rel. $E_{u,1}$	rel. $E_{u,2}$	rel. $E_{u,3}$	$E_{Q,1}$	$E_{Q,2}$	$E_{Q,3}$
C23	200	24.11	23.63	2.288	2.241	0.505	0.664	0.853	0.0214	0.0277	0.0351	0.0482	0.0509	0.0537
C40	200	26.43	25.52	2.287	2.219	0.966	1.87	2.82	0.0379	0.0688	0.0961	0.0709	0.0893	0.112

Table 17: Recorded solver diagnostics for the same runs. The positive characteristic-radicand values are in solver units of cm^2/s^2 and support the reported characteristic-speed evaluation. The generated comparison summaries now also persist λ_-, λ_+, and the subcriticality margin required by Assumption 3.15.

Case	$\min \alpha_{\mathrm{eff}}$	$\max \alpha_{\mathrm{eff}}$	$\min \chi^2$	$\min \lambda_-$	$\max \lambda_-$	$\min \lambda_+$	$\max \lambda_+$	$\min\{\lambda_+, -\lambda_-\}$
C23	1.33058	1.33333	8.15×10^5	-998.85	-852.17	953.33	1058.76	852.17
C40	1.32500	1.33333	6.36×10^5	-999.08	-713.79	880.69	1058.94	713.79

The stenosis throat is the minimizer $z^\star = \operatorname{argmin}_z R_0(z)$ of the smooth reference-radius profile in Eq. 3.6. For the comparison geometry this gives $z^\star \approx 2.451\,\mathrm{cm}$ for both severities, with $R_{\min} = 0.1386\,\mathrm{cm}$ for C23 and $R_{\min} = 0.1080\,\mathrm{cm}$ for C40. The shaded "throat zone" in Figure 13 is the displayed interval $2.4 \le z \le 2.8\,\mathrm{cm}$; the radial-profile slices use $z^\star - 0.5\,\mathrm{cm}$, $z^\star$, and $z^\star + 0.5\,\mathrm{cm}$. The largest section discrepancies listed in Table 18 sit upstream of that displayed band, in the acceleration/shoulder region.

Figure 13 shows the area-mean axial velocity comparison. The 1D and quadrature-averaged 3D means are closest away from the largest acceleration zone, while the largest mismatch in this two-case pair is localized in the upstream C40 shoulder. This is a descriptive comparison under the declared output operator, not evidence of model accuracy. The C40 maximum section discrepancy is 9.92 cm/s at $z = 2.261$ cm; C23 reaches 2.35 cm/s at $z = 2.291$ cm.

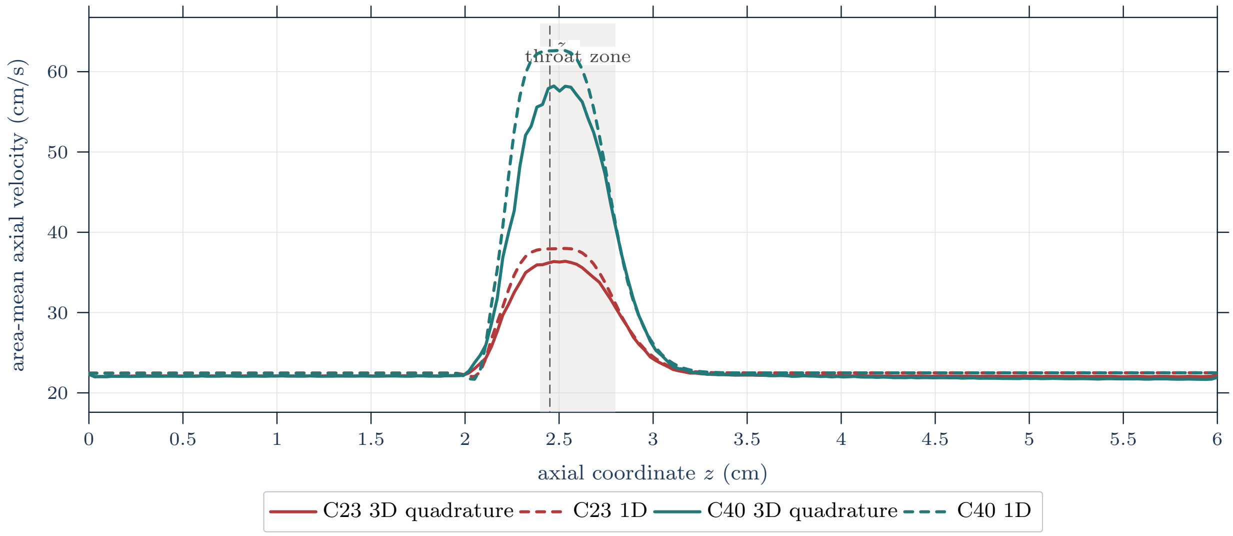

Figure 13: Area-mean axial velocity comparison at $t = 1.0$ s. Solid curves are tetrahedral cross-section quadrature means from the resolved 3D velocity field; dashed curves are interpolated 1D means $\bar{u} = q/a = Q_{\mathrm{phys}}/A_{\mathrm{phys}}$. The shaded band marks the displayed throat-centered stenosis region; the largest listed section discrepancies occur just upstream of this band.

Table 18: Five largest section-wise velocity discrepancies across the two included cases. Velocities and velocity discrepancies are in cm/s; physical flows and flow discrepancies are in cm^3/s, with $Q_{1D} = \pi q$.

Rank	Case	z (cm)	$\bar{u}_{1D}$	$\bar{u}_{3D}$	$\lvert d_u \rvert$	Q_{1D}	Q_{3D}	$\lvert d_Q \rvert$	Tri.
1	C40	2.261	52.58	42.66	9.92	2.201	1.862	0.339	759
2	C40	2.291	56.94	48.20	8.74	2.230	1.903	0.327	749
3	C40	2.352	61.43	53.22	8.21	2.270	1.980	0.290	743
4	C40	2.322	59.81	52.09	7.72	2.253	1.970	0.283	771
5	C40	2.231	46.95	39.94	7.01	2.174	1.918	0.256	833

5.4 Reconstructed Radial-Profile Results

The radial-profile reconstruction in Figure 14 is a secondary diagnostic. It asks whether the 1D velocity-profile closure reconstructed from the section mean resembles area-weighted radial bins of the resolved 3D cut. The throat and downstream slices show the largest C40 discrepancy. This is suggestive of profile-shape effects omitted by the 1D closure, but it does not uniquely diagnose secondary flow, skewness, separation, or any single omitted mechanism. The reconstructed parabolic profile used for the dashed 1D curve is

$$u_{1D}(r; z, t) = 2\,\frac{q(z,t)}{a(z,t)}\left(1 - \left(\frac{r}{\sqrt{a(z,t)}}\right)^2\right)$$

on the current-radius coordinate. Supplemental rows retain the static reference-radius coordinate $r/R_0(z)$ and 10/20/40-bin sensitivity, with area-weighted radial means, area mismatch, and within-bin azimuthal variance. The MAE/RMSE radial summaries weight valid radial bins equally after those area-weighted bin means are formed.

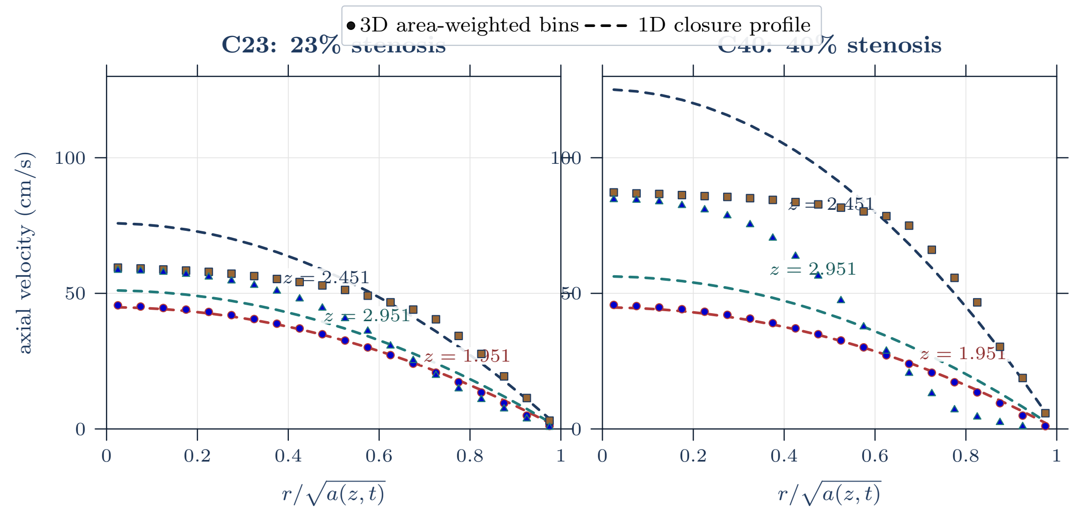

Figure 14: Radial velocity profile comparison at the declared throat-centered slices. Markers are resolved 3D area-weighted radial-bin means from the quadrature triangles on the current-radius coordinate; dashed curves are the 1D closure profile reconstructed from q/a. Colored in-panel labels identify the axial stations.

Table 19: Radial-profile discrepancy by case and slice. Bin means are area-weighted over the quadrature triangles in each bin; the displayed MAE and RMSE summaries weight the valid radial-bin means equally after binning. Velocity discrepancies are in cm/s; radial-bin physical-flow discrepancies are in cm^3/s and use $Q_{1D,\mathrm{bin}} = \bar{u}_{1D,\mathrm{bin}} A_{\mathrm{bin}}$. All listed slices have 20 valid radial bins.

Case	z (cm)	MAE(u_r)	RMSE(u_r)	max $\lvert d_{u_r}\rvert$	MAE(Q_r)	RMSE(Q_r)
C23	1.951	0.493	0.614	1.34	0.00290	0.00498
C23	2.451	7.47	9.47	16.35	0.0130	0.0152
C23	2.951	5.15	5.62	7.62	0.0191	0.0229
C40	1.951	0.561	0.649	1.41	0.00303	0.00478
C40	2.451	16.79	21.78	37.73	0.0163	0.0220
C40	2.951	16.47	18.89	28.36	0.0495	0.0568

5.5 Spatial Localization of Discrepancies

These discrepancies should be read in light of the cross-dimensional comparison scope in Section 2.5 and the operator definition in Section 3.10. The quadrature operator gives a physical cross-sectional average of the resolved velocity field at the nominally aligned final-time samples, but the comparison still changes model tier, wall treatment, boundary realization, and transient history. The result therefore localizes velocity mismatch; it does not attribute that mismatch uniquely to the 1D closure and does not supply pressure, FFR, or clinical accuracy evidence. The next defensible pressure-focused study would need declared pressure observations and a matched evidence record, not only this velocity diagnostic.

6 Discussion

6.1 Interpretation of the Verification Evidence

The verification evidence supports a bounded implementation claim: the reported solver configuration, benchmark records, and output summaries reproduce the finite-volume and comparison records used in this report. The implementation checks exercise descriptor health, sampled self-convergence, native/SciML cross-integrator discrepancies, stationary-Stokes projection rows, boundary handling, rheology/profile sensitivity, and resolved-velocity output diagnostics. These checks make the numerical record traceable rather than anecdotal.

The same evidence should not be read as a stronger theorem or validation claim. The benchmark rows do not prove convergence of every model component, do not establish pressure accuracy, and do not show agreement with clinical or experimental measurements. For the current comparison cases, the generated summaries persist characteristic-speed diagnostics; CFL, mass, and positivity diagnostics are exposed by the solver and verification records rather than as tracked C23/C40 comparison time histories. The remaining limitations are that the zero-forcing rest-state test reveals a material non-well-balanced start-up artifact, and production grid, timestep, start-up, and stationarity sensitivity remain incomplete for the C23/C40 comparison runs. The present result is therefore best described as a checked implementation record with documented limitations, not as complete numerical verification of every possible output.

6.2 Interpretation of the Cross-Model Discrepancies

The largest resolved-velocity disagreement in this pair occurs in C40, the stronger stenosis case. The section-mean comparison in Table 16 records the descriptive widening: the velocity L^1 error increases from 0.505 cm/s in C23 to 0.966 cm/s in C40, while the velocity L^3 error increases from 0.853 cm/s to 2.82 cm/s. The largest section discrepancy increases from 2.35 cm/s to 9.92 cm/s. Table 18 localizes the largest section discrepancies just upstream of the displayed throat band, in the C40 acceleration/shoulder region. Because the numerical and data-matching gates remain unresolved, this two-case pattern should not be read as a causal stenosis-severity result.

The radial-profile comparison gives a complementary reading. The throat and downstream C40 slices show the largest profile discrepancies, especially in Table 19. That is consistent with the expectation that a one-dimensional parabolic reconstruction will be most strained where the resolved field contains strong transverse structure. It is not, by itself, a diagnosis of secondary flow, skewness, separation, or any one omitted 3D mechanism. The radial plots show where the 1D profile closure fails to reproduce the resolved radial bins; they do not identify a unique physical cause.

6.3 Effects of Unmatched Wall Models, Boundary Conditions, and Histories

The present comparison changes more than spatial dimension. The 1D computation uses a compliant reduced Canic–Koiter wall law and a fixed-area characteristic boundary approximation, while the local resolved files expose velocity, pressure, and displacement as single final-time snapshots rather than a matched wall-history record. These differences matter because velocity discrepancy is not solely a profile-closure discrepancy. Area mismatch can change $\bar{u} = Q/A$ even when physical flow is close; flow mismatch can arise from different inlet, outlet, wall, or start-up histories; and the final-time difference between 1.0 s and 0.9995 s does not establish matched phases or matched histories.

The cut-area audit in Table 14 is therefore an operator check, not an attribution result. It shows typical sub-percent agreement between the tetrahedral cuts and the static reference areas, with localized multi-percent outliers. Table 15 adds a section-wise split between physical-flow and area-denominator contributions for the reported samples; that algebraic split is useful bookkeeping, but it does not by itself establish whether the flow mismatch is physical, numerical, or data-history driven. The comparison should therefore be read as a coupled velocity, flow, geometry, and modeling diagnostic rather than as a pure test of the cross-sectional profile closure.

6.4 Methodological Limitations

The most important limitation is evidential scope. The report compares a selected 1D realization with two resolved computational velocity datasets under a declared plane–tetrahedron output operator. It does not compare pressures, pressure drops, clinical FFR, CT-FFR, wall shear stress, structural deformation, or experimental measurements. The pressure-ratio notation in the report therefore remains an output convention for later studies, not a completed pressure-ratio result.

A second limitation is that the comparison is not a fully matched 1D–3D validation study. Matching would require, at minimum, a declared common geometry, wall model, rheology, material parameters, inlet and outlet histories, initialization, time alignment, numerical-resolution study, and output operator. Some of those ingredients are available here, especially the analytic reference geometry and the velocity output operator. Others are absent or only partially persisted, so the current comparison can localize discrepancies but cannot uniquely attribute them. The section-flow audit is another unresolved gate: the available 3D cuts vary by about 7.7% of the C23 mean section flow and 20.2% of the C40 mean section flow, and the comparison files do not by themselves establish whether this is fixed-wall numerical/operator error, single-snapshot moving-wall geometry, or a moving-wall effect requiring area-rate and wall-history data.

6.5 Implications and Future Work

The immediate implication is practical: the next useful step is a more controlled comparison record. That record should persist matched inlet and outlet histories, wall and material parameters, positivity/CFL/mass diagnostics, subcritical characteristic-speed signs, grid and operator sensitivity for the actual C23/C40 outputs, high-area-error section sensitivity, and an explicit classical-versus-extended ablation. It should also preserve the existing physical-flow and area-denominator decomposition as a standard output rather than as an after-the-fact table.

A pressure-focused extension would require a separate design. It would need proximal and distal pressure observation maps, reference-pressure and averaging conventions, pressure data from the comparison model, and a clear statement of whether the target is a reduced pressure diagnostic, a pressure-drop comparison, or a clinical pressure-ratio comparison. Until such a study is added, the defensible contribution remains the audited extended-1D model specification, the reproducible numerical workflow, and the declared operator for comparing 1D and resolved-3D velocity outputs.

7 Conclusions and Limitations

This report has separated three claims that are often blurred in reduced-order hemodynamics work: model specification, numerical verification, and cross-model comparison. The selected model is now stated in both physical variables $(A_{\text{phys}}, Q_{\text{phys}})$ and solver variables $(a, q) = (A_{\text{phys}}/\pi, Q_{\text{phys}}/\pi)$. It is an $R_{\max}$-normalized Canic-derived, radius-squared finite-volume model with $R_0^* = R_{\max}$ in the continuous source-balanced elastic wall law, a parabolic-profile baseline closure, optional Canic variable-radius terms, and a fixed-area characteristic boundary approximation.

The verification record documents an internally reproducible numerical record for the selected solver and recorded artifacts. Descriptor checks, self-convergence rows, cross-integrator discrepancies, stationary-Stokes projection rows, boundary diagnostics, rheology/profile sensitivity, and resolved-velocity output diagnostics support the reported finite-volume/time-integration data contracts. They remain summary diagnostics rather than full time histories, and the corrected rest-state test shows that the selected discretization is not well balanced for the stenotic geometry-rest state. At the finest checked grid, the zero-input artificial flow at $t = 1.0$ s is still of the same order as the comparison-flow scale, so the later C23/C40 velocity comparison cannot be interpreted as a production-accuracy study.

The 1D–3D velocity comparison shows localized discrepancy under the declared cross-section quadrature operator. For C23 and C40 at $t = 1.0$ s, the section-wise area-mean velocity discrepancies have L^1 errors of 0.505 and 0.966 cm/s, L^2 errors of 0.664 and 1.87 cm/s, and L^3 errors of 0.853 and 2.82 cm/s. The largest listed section discrepancies are 2.35 and 9.92 cm/s. The larger discrepancies in this two-case pair occur in the upstream C40 shoulder and in the reconstructed radial profiles; they are single-realization descriptive comparison outputs, not a validated stenosis-dependence or model-accuracy claim.

The bounded contribution is therefore precise rather than clinical: a selected extended 1D stenosis model, an auditable finite-volume/time-integration implementation, and a cross-sectional velocity comparison operator that identifies where the current 1D and resolved-3D computational outputs differ. The report does not establish pressure-drop accuracy, pressure-ratio or FFR behavior, clinical validity, full transient 3D parity, structural deformation, or a fully matched 1D–3D validation. Those remain future studies with different evidence requirements.

A Acronyms and Abbreviations

General abbreviations.

a.e.	almost everywhere	bpm	beats per minute
e.g.	*exempli gratia* (for example)	i.e.	*id est* (that is)

Clinical, modeling, and numerical acronyms.

CAS	coronary artery stenosis	FSI	fluid–structure interaction
CCTA	coronary computed tomography angiography	RBC	red blood cell
CT-FFR	computed tomography-derived fractional flow reserve	FFR	fractional flow reserve
IVUS	intravascular ultrasound	WSS	wall shear stress
OCT	optical coherence tomography	CFD	computational fluid dynamics
ODE	ordinary differential equation	PDE	partial differential equation
IC	initial condition	BC	boundary condition
FDM	finite difference method	FEM	finite element method
FVM	finite volume method	ROM	reduced-order model
0D	zero-dimensional	1D	one-dimensional
2D	two-dimensional	3D	three-dimensional

B Mathematical Notation

Notation B.1 (Report mathematical notation)**.** This appendix collects the mathematical notation used throughout this report.

Symbol	**Meaning**
$:=$	defines
$\equiv$	notational equivalence / identified with
$\mathbb{R}$	real numbers
$\mathbb{R}^+$	strictly positive real numbers
$\mathbb{R}^-$	strictly negative real numbers
$\mathbb{R}^n$	n-dimensional real vector space
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	nonnegative integers $\{0, 1, 2, \dots\}$
$\{\boldsymbol{v_1}, \dots, \boldsymbol{v_n}\}$	general basis of $\mathbb{R}^n$
$\{\boldsymbol{e_1}, \dots, \boldsymbol{e_n}\}$	standard basis of $\mathbb{R}^n$
$[n]$	index set $\{1, 2, \dots, n\} \subset \mathbb{N}$ for $n \in \mathbb{N}$
I_T	open time interval $(0, T_{\text{final}})$ for transient interior-time arguments
$\overline{I}_T$	closed time interval $[0, T_{\text{final}}]$ for initial-time data and flow maps
K	compact space-time rectangle, such as $[0, \ell] \times [0, T_{\text{final}}]$, used for output-field norms when locally stated
$\Omega \subset \mathbb{R}^n$	bounded domain
$\overline{\Omega}$	the closure of Ω
$\partial\Omega$	the boundary of Ω
$\Omega_B(t)$	spatial blood domain at time t
$\mathcal{Q}_T$	spatial-temporal domain $\{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}$
$C^k(\Omega)$	functions with continuous classical derivatives through order k in Ω
$C^k(\overline{\Omega})$	functions whose derivatives through order k extend continuously to $\overline{\Omega}$
$C^k_c(\Omega)$	k times continuously differentiable functions with compact support in Ω
$C^\infty_c(\Omega)$	smooth functions with compact support in Ω

Symbol	Meaning
$C(K)$	continuous functions on a stated compact set K
$L^p(\Omega)$	Lebesgue space of p-integrable functions modulo equality a.e.
$d\boldsymbol{x}$	Lebesgue volume element on $\mathbb{R}^n$
$dS_{\boldsymbol{x}}$	surface measure element on $\partial\Omega \subset \mathbb{R}^n$
dV	3D volume element on domain $\Omega \subset \mathbb{R}^3$
dS	3D surface element on boundary $\partial\Omega \subset \mathbb{R}^3$
∇	gradient operator
$\partial_{x_i}\phi$	partial derivative of ϕ with respect to coordinate x_i
$\partial_t\phi$, $\partial_z f$, $\partial_A p$	compact partial derivatives with respect to named coordinates t, z, and A
$\partial_r u$, $\partial_\theta u$	compact partial derivatives in cylindrical coordinates
$\Delta\phi = \nabla \cdot \nabla\phi$	Laplacian of scalar field ϕ
div	divergence of a vector field
div	row-wise divergence of a tensor field
$\boldsymbol{v}_i$	i-th component of vector $\boldsymbol{v}$
$\boldsymbol{S} : \boldsymbol{T}$	Frobenius contraction $\sum_{i,j} S_{ij}T_{ij}$ for second-order tensors
$\langle \cdot,\ \cdot \rangle_X$	inner product on vector space X
$\langle \boldsymbol{u},\ \boldsymbol{v} \rangle \ \equiv\ \langle \boldsymbol{u},\ \boldsymbol{v} \rangle_{\mathbb{R}^n}$	inner product of vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$
$\partial_{\hat{\boldsymbol{n}}}\phi$	normal derivative $\langle \nabla\phi,\ \hat{\boldsymbol{n}} \rangle$ of scalar field ϕ on $\partial\Omega$
$\lVert\cdot\rVert_X$	norm on a specified normed space X; bare $\lVert\cdot\rVert$ is local shorthand
$\lVert\cdot\rVert_{\mathrm{F}}$	Frobenius norm of a matrix or second-order tensor

C Domain Specific Notation

Definition C.1 (SI unit convention)**.** Unless otherwise stated, physical units are SI: length in meters, time in seconds, density in $\mathrm{kg/m^3}$, pressure in Pa, dynamic viscosity in $\mathrm{Pa \cdot s}$, and kinematic viscosity in $\mathrm{m^2/s}$.

Convention C.2 (Centimeter–second numerical-record convention)**.** The SI convention above governs continuum notation and unqualified analytical parameters. The Julia solver outputs and the Section 5 tables instead use the centimeter–gram–second convention of the numerical realization: lengths are in cm, time in s, density in $\mathrm{g/cm^3}$, velocity in cm/s, pressure and stress in $\mathrm{dyn/cm^2}$, dynamic viscosity in $\mathrm{g/(cm\,s)} = \mathrm{dyn\,s/cm^2}$, and kinematic viscosity in $\mathrm{cm^2/s}$. Thus the default solver geometry parameters L, $R_{\max}$, z_a, z_s, σ_a, and κ are centimeter quantities, while the stenosis localization λ in $\exp(-\lambda g^4)$ has units $\mathrm{cm^{-4}}$. The radius-squared solver state coordinate $a = R^2$ has units $\mathrm{cm^2}$; when a physical circular area is meant it is $A_{\mathrm{phys}} = \pi a$, also reported in square centimeters.

Output fields carry their display units in their names, but some existing CSV headers retain legacy solver names. In particular, `z_cm` and `R0_cm` are geometry fields. The code-level headers `A_cm2` and `Q_cm3_s` should be read in this manuscript as the solver-coordinate fields a_{cm^2} and $q_{\mathrm{cm}^3/\mathrm{s}}$, not as physical circular area and flow. The preferred report labels are `a_cm2`, `q_cm3_s`, `Aphys_cm2`, `Qphys_cm3_s`, and `uavg_cm_s`. For the radius-squared solver convention, the corresponding physical circular-area flow is $Q_{\mathrm{phys}} = \pi q$, while `uavg_cm_s` denotes $q/a = Q_{\mathrm{phys}}/A_{\mathrm{phys}}$. The Section 5 velocity/discrepancy columns are in cm/s, `pressure_dyn_cm2` and initial-condition pressure-drop fields are in $\mathrm{dyn/cm^2}$, and `nu_eff_cm2_s` is in $\mathrm{cm^2/s}$. A pressure supplied through a Pa-valued command-line option is converted before storage using $1\,\mathrm{Pa} = 10\,\mathrm{dyn/cm^2}$. Any numerical-output value used in an SI formula must first be converted; for example $1\,\mathrm{cm} = 10^{-2}\,\mathrm{m}$, $1\,\mathrm{cm^2/s} = 10^{-4}\,\mathrm{m^2/s}$, $1\,\mathrm{g/(cm\,s)} = 10^{-1}\,\mathrm{Pa \cdot s}$, and $1\,\mathrm{dyn/cm^2} = 10^{-1}\,\mathrm{Pa}$.

Convention C.3 (Pressure-output convention)**.** Pressure-drop and pressure-ratio quantities are reduced output vocabulary. A pressure-drop trace requires proximal and distal scalar pressure observations, and a pressure-ratio output is reported only with a ratio-output specification that declares proximal and distal scalar pressure observations, a time-averaging window, a pressure reference, and the denominator admissibility condition below. The general ratio-output specification is

$$\pi_{\mathrm{PR}} \;:=\; (\mathcal{O}_{\mathrm{prox}}, \mathcal{O}_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}}).$$

Here $\mathcal{O}_{\mathrm{prox}}$ and $\mathcal{O}_{\mathrm{dist}}$ are the declared scalar observation maps applied to the gauge-pressure field p_{g}, and p_{ref} is the fixed scalar pressure reference added to observed gauge-pressure traces before forming the ratio. In the 1D axial-tap or cross-sectional pressure-average specialization, this report may use the shorthand

$$\pi_{\mathrm{PR}} = (z_{\mathrm{prox}}, z_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}}),$$

with the pressure observation maps understood from the declared tap or cross-sectional averaging convention. Given a model gauge-pressure field, the observed pressure traces are

$$P_{\mathrm{prox}}(t) \;:=\; \mathcal{O}_{\mathrm{prox}}[p_{\mathrm{g}}(\cdot,t)], \qquad\qquad P_{\mathrm{dist}}(t) \;:=\; \mathcal{O}_{\mathrm{dist}}[p_{\mathrm{g}}(\cdot,t)].$$

The corresponding pressure-drop trace is

$$\Delta p(t) \;:=\; P_{\mathrm{prox}}(t) - P_{\mathrm{dist}}(t).$$

A pressure ratio is not invariant under common additive pressure shifts. For tap and cross-sectional-average observations, admissible observation maps must preserve constant shifts,

$$\mathcal{O}[p+c] = \mathcal{O}[p] + c \qquad \text{for scalar constants } c.$$

The reference-consistent ratio traces are therefore defined after observation:

$$P^{\mathrm{ratio}}_{\mathrm{prox}}(t) \;:=\; p_{\mathrm{ref}} + P_{\mathrm{prox}}(t), \qquad\qquad P^{\mathrm{ratio}}_{\mathrm{dist}}(t) \;:=\; p_{\mathrm{ref}} + P_{\mathrm{dist}}(t).$$

The subscript HF labels a declared high-flow averaging window in the study record; it is not a clinical hyperemia model unless an external measurement comparison is separately specified [2, Ch. 5, Sec. 5.5, pp. 72–73] [5, p. 1703]. The ratio-output specification is admissible for a computed pressure trace when $\mathcal{T}_{\mathrm{HF}} \subset I_T$ is Lebesgue-measurable with $0 < |\mathcal{T}_{\mathrm{HF}}| < \infty$, the proximal and distal ratio-pressure traces are integrable on that window, and

$$\langle P^{\mathrm{ratio}}_{\mathrm{prox}}(t)\rangle_{\mathcal{T}_{\mathrm{HF}}} \geq P_{\min} > 0.$$

The threshold $P_{\min}$ must be declared with the study record. No such pressure-ratio record is reported in the present numerical comparison. For an admissible π_{PR}, the corresponding reduced pressure-ratio output is

$$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}}) \;:=\; \frac{\langle P^{\mathrm{ratio}}_{\mathrm{dist}}(t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}{\langle P^{\mathrm{ratio}}_{\mathrm{prox}}(t)\rangle_{\mathcal{T}_{\mathrm{HF}}}}.$$

For an integrable scalar trace,

$$\langle f\rangle_{\mathcal{T}_{\mathrm{HF}}} \;:=\; \frac{1}{|\mathcal{T}_{\mathrm{HF}}|}\int_{\mathcal{T}_{\mathrm{HF}}} f(t)\,dt.$$

This bracket denotes time averaging, not the inner product notation in Appendix B. An FFR-like shorthand refers to this reduced ratio only for records carrying an admissible π_{PR}; the shorthand does not add a clinical measurement model [2, Ch. 5, Sec. 5.5, pp. 72–73] [5, p. 1703]. If π_{PR} is absent or inadmissible, the pressure ratio is outside the reported output space.

Convention C.4 (Shear-output convention)**.** Shear terminology is model-tier dependent. A resolved wall shear stress output requires a resolved stress tensor, wall normal, wall location, output operator, and, for a signed component, a chosen tangent direction [8, Abstract, Sec. 2, pp. 8–10]. A reduced shear proxy such as $\tau_{w,z}^{1D}$ requires the model tier, wall closure, rheology descriptor, velocity-profile or friction closure, evaluation location or spatial averaging operator, and sign or time-window convention. Reduced shear proxies are closure diagnostics; they are not interchangeable with resolved traction-based wall shear stress unless a separate comparison map is declared.

Symbol	Meaning	Units
R	vessel radius; vessel diameter is $d_v := 2R$	m
d_v	vessel diameter	m
η	dynamic viscosity	$\mathrm{Pa} \cdot \mathrm{s}$
$\nu := \eta/\rho$	kinematic viscosity	m^2/s
$\eta_{\mathrm{eff}}(\dot{\gamma})$	effective dynamic viscosity for generalized Newtonian models	$\mathrm{Pa} \cdot \mathrm{s}$
$\nu_{\mathrm{eff}}(\dot{\gamma}) := \eta_{\mathrm{eff}}(\dot{\gamma})/\rho$	effective kinematic viscosity after density scaling	m^2/s
τ	shear stress	Pa
$\dot{\gamma}$	shear rate	s^{-1}
ρ	density field	kg/m^3
p	pressure field	Pa
$\boldsymbol{u}$	velocity field	m/s
$D(\boldsymbol{u})$	rate-of-deformation tensor, $D(\boldsymbol{u}) := \frac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\top}\right)$	s^{-1}
$W_0 := d_v\sqrt{\frac{\omega\rho}{\eta}}$	Womersley number, diameter-based convention	$-$
Re	Reynolds number	$-$
t	time	s
T or T_{final}	terminal time; use T_{final} where it could be confused with stress tensors	s
ω	angular frequency	rad/s
(r, θ, z)	cylindrical coordinates used only when 3D coordinate forms are written	$\mathrm{m}, -, \mathrm{m}$
$(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$	cylindrical orthonormal basis vectors	$-$
τ_{ij}	cylindrical viscous-stress component	Pa
$\boldsymbol{f_b}$	body force per unit mass	m/s^2
$\rho\boldsymbol{f_b}$	body force per unit volume	N/m^3

Reduced-model and forward-map notation.

Symbol	Meaning	Units
$A(z,t)$	cross-sectional lumen area in the 1D model	m^2
$Q(z,t)$	volumetric flow rate through a cross-section	m^3/s
$\bar{u}(z,t) := Q/A$	section-mean axial velocity	m/s
$I_z = (0,L)$	axial single-vessel interval for a 1D formulation	m
$\mathbf{U} = (A,Q)^\top$	conservative area-flow state vector	$(\mathrm{m}^2, \mathrm{m}^3/\mathrm{s})$
$\mathcal{P}$	cross-sectional velocity-profile descriptor used by the 1D closure; admitted values include flat, parabolic, and power-family profiles	mixed
$\alpha_{\mathcal{P}}$	momentum-flux correction induced by the declared profile $\mathcal{P}$; the parabolic baseline uses $\alpha_{\mathcal{P}} = 4/3$	–
γ	power-profile exponent in $u(r) = ((\gamma+2)/\gamma)\bar{u}(1-(r/R)^\gamma)$	–
$g_{\mathcal{P}}$	profile shear-rate factor used by the 1D rheology proxy; for power profiles $g_{\mathcal{P}} = \gamma + 2$, while flat profiles assign this value explicitly	–
$R(A(z,t)) = \sqrt{A(z,t)/\pi}$	current reduced lumen radius derived from the 1D area	m
$R_0(z)$	reference stenosed radius used by the geometry and wall source	m
$R_{\max}$	healthy reference-radius parameter used by the selected Canic–Koiter solver wall law; equals $\max_z R_0(z)$ for the default stenosis profile	m
$R_{\text{base}}(z)$	nominal healthy baseline radius before applying $s(z)$	m
$\mathcal{R}$	rheology descriptor in the 1D model; distinct from radius symbols R, R_0, and R_{base}; admitted values are `newtonian`, `carreau`, `carreau-yasuda`, `casson`, and `power-law`	mixed
$\mathcal{W}$	wall descriptor; the selected 1D report model uses the Canic–Koiter member of $\mathcal{W}_{\text{elastic1D}}$, while fixed-wall and resolved-FSI wall models require separate declarations	mixed
$\mathcal{C}$	coupling/interface descriptor; the selected single-vessel study uses $\mathcal{C}_{\text{isolated}}$	mixed
$s(z)$	dimensionless stenosis profile, with $0 \le s(z) < 1$	–
$A_0(z) = \pi R_0(z)^2$	wall-law reference area after applying the stenosis profile	m^2
$s_{\max}$	maximum fractional radius-reduction parameter in $s(z)$	–
$g(z)$	smooth asymmetric stenosis-kernel coordinate	m
$\psi(z) = \exp(-\lambda g(z)^4)$	smooth rapidly localized stenosis kernel used by the baseline profile $s(z) = s_{\max}\psi(z)$	–
z_a	center of the Gaussian asymmetry term in $g(z)$	m
z_s	axial shift parameter in $g(z)$	m

Symbol	Meaning	Units
σ_a	width parameter for the Gaussian asymmetry term in $g(z)$	m
κ	amplitude of the Gaussian asymmetry term in $g(z)$	m
λ	localization strength in $\psi(z) = \exp(-\lambda g(z)^4)$	m^{-4}
$\beta(z)$	wall-stiffness parameter in $p_{\mathrm{g}} = p_{\mathrm{ext}} + \beta A_0^{-1}(\sqrt{A} - \sqrt{A_0})$; for the selected $R_{\max}$ law, $\beta(z) = \sqrt{\pi} K R_0(z)^2/R_{\max}^2$	$\mathrm{Pa} \cdot \mathrm{m}$
$p_{\mathrm{ext}}(z,t)$	external-pressure value in the wall-law gauge convention; baseline value $p_{\mathrm{ext}} \equiv 0$ unless declared otherwise	Pa
$\Psi(A,z,t)$ or $\Psi_{\mathcal{W}}(A,z,t)$	elastic flux potential satisfying $\partial_A \Psi_{\mathcal{W}} = (A/\rho)\partial_A p_{\mathrm{g}}^{\mathcal{W}}$ and normalized by $\Psi_{\mathcal{W}}(A_0(z),z,t) = 0$ only when that normalization is declared; the selected solver potential $\Psi_h(a)$ is unnormalized and balanced at the continuous source level	$\mathrm{m}^4/\mathrm{s}^2$
$\mathbf{F}$	area-flow flux map; not a residual or objective functional	problem-dependent
$\mathbf{S}$	source-term map in the 1D balance-law template	problem-dependent
$S_{\mathrm{geom}}^{\mathcal{W}}$	fixed-area wall/geometry source associated with the selected reduced wall closure $\mathcal{W}$	$\mathrm{m}^3/\mathrm{s}^2$
$S_f^{\mathcal{R}}$	wall-friction source associated with the selected rheology or effective-viscosity closure $\mathcal{R}$	$\mathrm{m}^3/\mathrm{s}^2$
$\widehat{\mathbf{F}}_{i+1/2}^n$	numerical flux at a finite-volume cell interface and time level	problem-dependent
$\widehat{\mathbf{S}}_i^n$	cell source quadrature used by a computed finite-volume scheme	problem-dependent
$\Delta z_i, \Delta t^n$	finite-volume cell length and timestep in a computed solve	m, s
$\dot{\gamma}_{1D}$	computed shear-rate proxy $g_{\mathcal{P}}\lvert q\rvert/(a\sqrt{a}) = g_{\mathcal{P}}\lvert Q_{\mathrm{phys}}\rvert/(\pi a\sqrt{a})$, where $a = R^2$ and $Q_{\mathrm{phys}} = \pi q$	s^{-1}
$\nu_{\mathrm{eff}}^{\mathcal{R}}$	closure-indexed effective kinematic viscosity used in the reduced viscous/source and pressure-correction terms	m^2/s
C_f or $C_f(z,A)$	wall-friction coefficient in the 1D source $-C_f Q/A$; constant-viscosity value $C_f = 8\pi\eta/\rho$	m^2/s
$\eta_{\mathrm{fric}}(z,t)$	dynamic viscosity used by the reduced wall-friction and signed-shear-proxy formula; $\eta_{\mathrm{fric}} = \eta$ for constant-viscosity realizations and $\eta_{\mathrm{fric}} = \eta_{\mathrm{eff}}^{\mathcal{R}}(\dot{\gamma}_{1D})$ for admitted non-Newtonian descriptor realizations	$\mathrm{Pa} \cdot \mathrm{s}$
c_{wave} or c	1D pulse-wave speed, $c^2 = (A/\rho)\partial_A p_{\mathrm{g}}$ when no concentration field is in scope	m/s

Symbol	Meaning	Units
$e \in \mathcal{E}$	expert/closure record for a closure-indexed 1D forward-map family; suppress e only for single-expert baseline records	mixed
$\mathcal{G}_{1D,h}^{r_h,e}$	computed fixed-grid 1D map indexed by numerical record r_h and expert record e	mixed

Symbol	Meaning	Units
p_{3D}	continuum pressure in the 3D/NS setting	Pa
$p_{3D,\mathrm{g}}$	continuum pressure after subtracting the wall-law gauge reference	Pa
p_{g}	1D gauge pressure produced by the wall law	Pa
$p_{\mathrm{ratio}} := p_{\mathrm{ref}} + p_{\mathrm{g}}$	field-level reference-consistent pressure shorthand; ratio traces are defined after scalar observation by Convention C.3	Pa
p_{ref}	fixed pressure reference recorded by the ratio-output specification	Pa
$\mathcal{O}_{\mathrm{prox}}, \mathcal{O}_{\mathrm{dist}}$	proximal and distal scalar pressure observation maps declared for pressure-drop or pressure-ratio outputs	input/output Pa
$P_{\mathrm{prox}}, P_{\mathrm{dist}}$	observed proximal and distal gauge-pressure traces after applying the declared pressure observation maps	Pa
$P^{\mathrm{ratio}}_{\mathrm{prox}}, P^{\mathrm{ratio}}_{\mathrm{dist}}$	observed proximal and distal pressure traces after adding p_{ref}	Pa
$\pi_{\Delta p} = (z_{\mathrm{prox}}, z_{\mathrm{dist}})$	1D shorthand pressure-drop tap specification; the observation maps define the actual scalar traces	m, m
π_{PR}	ratio-output specification $(\mathcal{O}_{\mathrm{prox}}, \mathcal{O}_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}})$; axial taps may be specified by shorthand	mixed
$\langle f \rangle_{\mathcal{T}_{\mathrm{HF}}}$	time average of an integrable trace over the declared high-flow window; not the Appendix B inner product	same as f
$\langle P^{\mathrm{ratio}}_{\mathrm{prox}} \rangle_{\mathcal{T}_{\mathrm{HF}}} \geq P_{\min} > 0$	denominator admissibility condition required before PR_{1D} is reported	Pa
$P_{\min}$	declared positive lower bound for the proximal ratio-pressure average	Pa
$p_{\mathrm{out,g}}$	prescribed outlet gauge pressure in the same reference frame as p_{g}	Pa
$\Delta p(t)$	proximal-minus-distal scalar pressure trace after the declared observations	Pa
$\overline{\Delta p}_{\mathcal{T}_{\mathrm{HF}}}$	scalar pressure-drop output averaged over the declared high-flow window $\mathcal{T}_{\mathrm{HF}}$	Pa
FFR-like shorthand	optional shorthand for PR_{1D} only when Convention C.3 and admissibility requirements are met	–
$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$	reduced pressure-ratio output under the declared observation maps, pressure reference, and averaging window	–
τ_w or $\lvert\tau_w\rvert$	resolved traction-based wall-shear-stress output after declaring the stress tensor, wall normal, location, and output operator	Pa
$\tau^{1D}_{w,z}$	signed 1D axial shear proxy after declaring the model tier, closures, location, and sign or window convention	Pa
`shear_rate_1_s`	CSV output field for the computed shear-rate proxy $\dot{\gamma}_{1D}$	s^{-1}
`nu_eff_cm2_s`	CSV output field for the effective kinematic viscosity used by the declared rheology closure	cm^2/s
`rheology`	CSV and summary output field naming the declared rheology descriptor	–

Remark C.5 (Viscosity notation). We write η for dynamic viscosity and

$$\nu := \frac{\eta}{\rho}$$

for kinematic viscosity. The distinction is dimensional: this report follows the convention that the coefficient in the Cauchy stress tensor is a dynamic viscosity: $D(\boldsymbol{u})$ has units of inverse time, so $\eta D(\boldsymbol{u})$ has pressure units. For generalized Newtonian models, the scalar viscosity function multiplying $2D(\boldsymbol{u})$ in the extra stress is likewise an effective dynamic viscosity, denoted

$$\eta_{\text{eff}}(\dot{\gamma}), \qquad \dot{\gamma} \;:=\; \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}.$$

Here "this quantity" means the stress-level viscosity coefficient, not the divided-by-density coefficient. Several sources write the generalized viscosity function as $\mu(|D|^2)$, while related hemodynamics sources also use μ for viscosity coefficients in reduced and continuum settings [9, Sec. 3.1, p. 35, Sec. 3.6, p. 215] [8, Sec. 1.1, pp. 6–7, Eqs. (1.2)–(1.4), Fig. 4.4, p. 18]. To keep this report unambiguous, ν is always kinematic viscosity, while η and η_{eff} are dynamic viscosities.

D Mathematical Foundations

This appendix supplies citation anchors for the analytic notation used in the main chapters and in Appendix B. It separates domain and boundary language, classical function spaces, field notation, differential operators, and norm conventions so later sections can cite only the convention they need.

Convention D.1 (Standing regularity convention)**.** Unless a section states a stronger or weaker hypothesis, classical identities are used only under the regularity needed for their terms to be defined. In particular, fields have the displayed classical derivatives in the indicated domains, boundary traces are defined when boundary terms are written, and integration-by-parts or adjoint identities are invoked only when the required boundary terms are meaningful. When a weak formulation is discussed, the relevant Sobolev space, trace theorem, and boundary regularity assumptions are stated explicitly; Sobolev traces are not inferred from this classical convention. At minimum, boundaries are assumed piecewise C^1 when classical boundary integrals or integration-by-parts identities are used. Outward unit normals are then interpreted on the regular boundary pieces, hence surface-a.e. When a pointwise normal derivative is asserted on all of $\partial\Omega$, the boundary is assumed at least C^1.

Definition D.2 (Domain and boundary notation)**.** Let $\Omega \subset \mathbb{R}^n$ be a bounded open connected set, with $n \in \mathbb{N}$, and denote its closure by $\overline{\Omega}$. In the hemodynamics settings considered in this report, only $n \in [3]$ is used. The standing regularity convention in Convention D.1 applies whenever traces, outward unit normals, or integration by parts are used. For open Ω,

$$\overline{\Omega} = \Omega \cup \partial\Omega, \qquad \partial\Omega = \overline{\Omega} \setminus \Omega.$$

Since Ω is bounded, $\overline{\Omega}$ is closed and bounded and is therefore compact by the Heine–Borel theorem. The open set Ω itself need not be compact. Since Ω is bounded, $\partial\Omega$ is also compact. When the standing regularity convention supplies a piecewise C^1 boundary, the outward unit normal $\hat{\boldsymbol{n}}$ is interpreted on the regular boundary pieces, hence $dS_{\boldsymbol{x}}$-a.e. on $\partial\Omega$.

Definition D.3 (Time intervals and compact space-time sets)**.** For a final time $T_{\text{final}} > 0$, this report uses

$$I_T \;:=\; (0, T_{\text{final}}), \qquad \overline{I}_T \;:=\; [0, T_{\text{final}}].$$

The open interval I_T is used for interior-time differential identities and space-time PDE regions, while $\overline{I}_T$ is used when initial-time data, endpoint data, sampled time grids, or flow maps are included. When a local final time such as T_* is stated, the same convention applies with T_{final} replaced by that local final time.

For one-dimensional output fields, a compact space-time rectangle is written as

$$K \;:=\; [0, \ell] \times [0, T_{\text{final}}]$$

or by an explicitly stated local variant. Spaces such as $C(K)$ and $C^1([0,\ell])$ mean continuous functions on the named compact set, with the supremum norm when a norm is written. Spaces such as $L^p(K)$ use the product Lebesgue measure and the same almost-everywhere convention as Definition D.7.

Definition D.4 (Field and multi-index notation)**.** For $\Omega \subset \mathbb{R}^n$, we use the following field notation:

$$\begin{aligned}
&\phi : \Omega \to \mathbb{R} \ \text{ is a } \textit{scalar field}, \\
&\boldsymbol{f} : \Omega \to \mathbb{R}^n \ \text{ is a } \textit{vector field}, \\
&\boldsymbol{T} : \Omega \to \mathbb{R}^{n\times n} \ \text{ is a } \textit{(second-order) tensor field},
\end{aligned}$$

and we write $\phi(\boldsymbol{x})$, $\boldsymbol{f}(\boldsymbol{x})$, and $\boldsymbol{T}(\boldsymbol{x})$[1] for the values at a point $\boldsymbol{x} \in \Omega$. When referring to a physical quantity, its measurement units are indicated as $\left[\ \cdot\ \right]$, whereas dimensionless quantities are indicated as $\left[\ -\ \right]$. The k-th partial derivative of ϕ with respect to coordinate x_i is

$$\partial^k_{x_i}\phi \ \equiv \ \frac{\partial^k \phi}{\partial x_i^k}.$$

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$, set $|\alpha| = \alpha_1 + \cdots + \alpha_n$. When the derivative exists classically, the corresponding derivative of a scalar field ϕ is

$$D^\alpha \phi \ \equiv \ \frac{\partial^{|\alpha|}\phi}{\partial x_1^{\alpha_1}\,\partial x_2^{\alpha_2}\,\ldots\,\partial x_n^{\alpha_n}}.$$

For $|\alpha| = 1$, $D^\alpha\phi$ is a first-order partial derivative of ϕ, e.g. $\partial_{x_i}\phi$ where $\alpha_i = 1$ and $\alpha_j = 0$ for all $j \in [n]\setminus\{i\}$.

Definition D.5 (Classical function-space notation)**.** Given a domain Ω and a scalar field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, a scalar function space on Ω is an $\mathbb{F}$-vector subspace of the maps $f : \Omega \to \mathbb{F}$. Unless a complex-valued space is explicitly specified, this report uses real-valued spaces. Thus $C^0(\Omega)$ denotes the real-valued continuous functions on the open domain Ω. For $k \in \mathbb{N}_0$, $C^k(\Omega)$ denotes functions $\phi : \Omega \to \mathbb{R}$ such that $D^\alpha\phi$ exists and is continuous in Ω for every $|\alpha| \le k$. By contrast, $C^k(\overline{\Omega})$ denotes functions in $C^k(\Omega)$ whose derivatives through order k extend continuously to the closure $\overline{\Omega}$. The space $C^\infty(\Omega)$ is the intersection of all $C^k(\Omega)$ for $k \in \mathbb{N}_0$, i.e. the infinitely differentiable functions on Ω. The notation $C^k(\Omega; \mathbb{R}^m)$ denotes the maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$ with components in $C^k(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in C^k(\Omega)$ for all $i \in [m]$.

Definition D.6 (Differential-operator conventions)**.** For real-valued $\phi \in C^1(\Omega)$, the partial derivative with respect to coordinate x_i is

$$\partial_{x_i}\phi = \frac{\partial \phi}{\partial x_i}.$$

[1] A second-order tensor value $\boldsymbol{T}(\boldsymbol{x}) \in \mathbb{R}^{n\times n}$ can be identified with the matrix of a linear map $\mathbb{R}^n \to \mathbb{R}^n$ acting on vectors by matrix-vector multiplication.

Named-coordinate partial derivatives use the same compact convention; for example,

$$\partial_t \phi = \frac{\partial \phi}{\partial t}, \qquad \partial_z f = \frac{\partial f}{\partial z}, \qquad \partial_A p = \frac{\partial p}{\partial A}.$$

Cylindrical derivatives such as $\partial_r u$ and $\partial_\theta u$ are read analogously. Boundary notation retains ∂, as in $\partial V(t)$ and $\partial\Omega$. For $\boldsymbol{x} \in \Omega$ and a direction $\boldsymbol{v} \in \mathbb{R}^n$, the directional derivative is

$$D_{\boldsymbol{v}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \boldsymbol{v}\rangle_{\mathbb{R}^n}.$$

If $\phi \in C^1(\overline{\Omega})$, $\boldsymbol{x} \in \partial\Omega$, and $\hat{\boldsymbol{n}}$ is the outward unit normal at $\boldsymbol{x}$, then the boundary normal derivative is

$$\partial_{\hat{\boldsymbol{n}}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \hat{\boldsymbol{n}}\rangle_{\mathbb{R}^n}.$$

For Lebesgue measure $d\boldsymbol{x}$ on $\mathbb{R}^n$ the integral of ϕ in Ω is

$$\int_\Omega \phi(\boldsymbol{x})\, d\boldsymbol{x}.$$

If $\partial\Omega$ is of class C^1 and $\phi \in C^0(\overline{\Omega})$, then the classical boundary trace $\phi|_{\partial\Omega}$ is continuous on $\partial\Omega$. Since Ω is bounded and $\partial\Omega$ is a compact C^1 hypersurface, this trace belongs to $L^1(\partial\Omega; dS_{\boldsymbol{x}})$. The corresponding boundary integral is written

$$\int_{\partial\Omega} \phi(\boldsymbol{x})\; dS_{\boldsymbol{x}}.$$

We use the column-gradient convention for scalar fields. For $\phi \in C^1(\Omega)$, the gradient is

$$\nabla\phi \;:=\; (\partial_{x_i}\phi)_{i\in[n]} \in \mathbb{R}^n.$$

For $\phi \in C^2(\Omega)$, the Laplacian is

$$\Delta\phi \;:=\; \nabla\cdot\nabla\phi \;=\; \sum_{i\in[n]} \partial_{x_i}^2\phi.$$

For vector-valued $\boldsymbol{f} \in C^1(\Omega;\mathbb{R}^n)$ with components $\boldsymbol{f} = (f_i)_{i\in[n]}$, the vector gradient is the Jacobian with entries $(\nabla\boldsymbol{f})_{ij} \;:=\; \partial_{x_j} f_i$. The divergence is the trace of this Jacobian:

$$\operatorname{div}\boldsymbol{f} \;:=\; \nabla\cdot\boldsymbol{f} \;=\; \sum_{i\in[n]} \partial_{x_i} f_i \;=\; \operatorname{tr}(\nabla\boldsymbol{f}).$$

For $\boldsymbol{f} \in C^2(\Omega;\mathbb{R}^n)$, the Laplacian is defined componentwise. The domain integral is also componentwise whenever the components are integrable:

$$\begin{aligned}\Delta \boldsymbol{f} \;&:=\; (\Delta f_i)_{i\in[n]}\\ \int_\Omega \boldsymbol{f}(\boldsymbol{x})\;d\boldsymbol{x} \;&:=\; \left(\int_\Omega f_i(\boldsymbol{x})\;d\boldsymbol{x}\right)_{i\in[n]}.\end{aligned}$$

For tensor-valued $\boldsymbol{T} \in C^1(\Omega;\mathbb{R}^{n\times n})$ with entries $\boldsymbol{T} = (T_{ij})_{i,j\in[n]}$, we use the row-divergence convention $(\mathbf{div}\boldsymbol{T})_i = \sum_{j\in[n]} \partial_{x_j} T_{ij}$. Equivalently, if $\boldsymbol{t}_i = (T_{ij})_{j\in[n]}$ denotes the i-th row of $\boldsymbol{T}$, then

$$\begin{aligned}\mathbf{div}\,\boldsymbol{T} \;&:=\; (\operatorname{div}\boldsymbol{t}_i)_{i\in[n]}\\ &=\; (\nabla\cdot\boldsymbol{t}_i)_{i\in[n]}\\ &=\; \left(\sum_{j\in[n]} \partial_{x_j} T_{ij}\right)_{i\in[n]}.\end{aligned}$$

For $\boldsymbol{T} \in C^2(\Omega;\mathbb{R}^{n\times n})$, the Laplacian is defined componentwise. The domain integral is also defined componentwise whenever the entries are integrable:

$$\begin{aligned}\Delta \boldsymbol{T} \;&:=\; (\Delta T_{ij})_{i,j\in[n]}\\ \int_\Omega \boldsymbol{T}(\boldsymbol{x})\;d\boldsymbol{x} \;&:=\; \left(\int_\Omega T_{ij}(\boldsymbol{x})\;d\boldsymbol{x}\right)_{i,j\in[n]}.\end{aligned}$$

Definition D.7 (C^k and L^p norm conventions)**.** Let $\Omega \subset \mathbb{R}^n$ be bounded and open. We write $C^k(\overline{\Omega})$ for the functions $\phi \in C^k(\Omega)$ such that each derivative $D^\alpha\phi$, $|\alpha| \le k$, extends continuously to $\overline{\Omega}$. The norm is

$$\|\phi\|_{C^k(\overline{\Omega})} \;:=\; \sum_{|\alpha|\le k} \sup_{\boldsymbol{x}\in\overline{\Omega}} |D^\alpha \phi(\boldsymbol{x})|.$$

With this norm, $C^k(\overline{\Omega})$ is a Banach space: every Cauchy sequence in $C^k(\overline{\Omega})$ converges, with respect to $\|\cdot\|_{C^k(\overline{\Omega})}$, to a limit in $C^k(\overline{\Omega})$. In particular, for $k = 0$, we have

$$\|\phi\|_{C^0(\overline{\Omega})} \;:=\; \sup_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})| = \max_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})|\,.$$

For continuous functions on compact sets, the max norm agrees with the usual supremum norm.

For $1 \le p < \infty$, $L^p(\Omega)$ denotes the space of equivalence classes of real-valued measurable functions on Ω, identified up to equality almost everywhere, with finite norm

$$\|\phi\|_{L^p(\Omega)} \;:=\; \left(\int_\Omega |\phi(\boldsymbol{x})|^p\,d\boldsymbol{x}\right)^{1/p}.$$

For $p = \infty$,

$$\|\phi\|_{L^\infty(\Omega)} \;:=\; \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} |\phi(\boldsymbol{x})|.$$

With this almost-everywhere identification, $L^p(\Omega)$ is a Banach space for $1 \le p \le \infty$.

The notation $L^p(\Omega; \mathbb{R}^m)$ denotes equivalence classes of measurable maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$, identified up to equality a.e., whose components belong to $L^p(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in L^p(\Omega)$ for all $i \in [m]$. When a norm on $L^p(\Omega; \mathbb{R}^m)$ is needed, we use the Euclidean norm in the range. For $1 \le p < \infty$,

$$\|\boldsymbol{f}\|_{L^p(\Omega;\mathbb{R}^m)} \;:=\; \left(\int_\Omega \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}^p \, d\boldsymbol{x}\right)^{1/p}.$$

For $p = \infty$, the corresponding convention is

$$\|\boldsymbol{f}\|_{L^\infty(\Omega;\mathbb{R}^m)} \;:=\; \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}.$$

For matrix or second-order tensor values, the pointwise range norm is the Frobenius norm

$$\|\boldsymbol{T}\|_{\mathrm{F}} \;:=\; (\boldsymbol{T} : \boldsymbol{T})^{1/2} = \left(\sum_{i,j\in[n]} T_{ij}^2\right)^{1/2}.$$

Thus tensor-valued L^p norms are formed by replacing $|\phi(\boldsymbol{x})|$ in the scalar L^p formula with $\|\boldsymbol{T}(\boldsymbol{x})\|_{\mathrm{F}}$.

E Continuum Derivation Details

This appendix supplements the continuum mechanics details that support the main-body assumptions without interrupting the reduced-order forward-model narrative within this report. It is appendix background for notation and model hierarchy, not additional evidence for the numerical study. All identities below are classical identities under the regularity needed for the displayed derivatives, traces, and changes of variables. The standing regularity convention, boundary notation, and differential-operator convention are those of Convention D.1, Definition D.2, and Definition D.6. In particular, **div** is the row-divergence of a tensor field, and boundary forms assume the stated trace and normal hypotheses; no weak formulation is being asserted in this appendix.

E.1 Material Derivative and Trajectory Details

For a scalar field $\phi \in C^1(\mathcal{Q}_T)$ and a sufficiently regular flow map $\mathcal{L}_t$, define the Lagrangian pullback $\widehat{\phi}(t,\boldsymbol{\xi}) := \phi(t,\mathcal{L}_t(\boldsymbol{\xi}))$. Differentiating with respect to time along a trajectory and using $\partial_t\mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t,\mathcal{L}_t(\boldsymbol{\xi}))$ gives

$$\frac{d}{dt}\phi(t,\mathcal{L}_t(\boldsymbol{\xi})) = \partial_t\phi(t,\boldsymbol{x}) + \nabla\phi(t,\boldsymbol{x})\cdot\boldsymbol{u}(t,\boldsymbol{x}), \qquad \boldsymbol{x} = \mathcal{L}_t(\boldsymbol{\xi}).$$

The identity

$$\operatorname{div}(\phi\boldsymbol{u}) = \nabla\phi\cdot\boldsymbol{u} + \phi\operatorname{div}\boldsymbol{u}$$

also gives

$$D_t\phi = \partial_t\phi + \operatorname{div}(\phi\boldsymbol{u}) - \phi\operatorname{div}\boldsymbol{u}.$$

E.2 Jacobian Evolution and Reynolds Transport

Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}}\mathcal{L}_t$ and $\widehat{J}_t = \det\widehat{\boldsymbol{F}}_t$. Here $\mathcal{L}_t$ is assumed to be an orientation-preserving C^1 change of variables on the material volume being considered, with the additional time regularity needed to differentiate $\widehat{\boldsymbol{F}}_t$ and $\widehat{J}_t$. The standard Jacobi identity gives

$$\partial_t\widehat{J}_t = \widehat{J}_t\operatorname{tr}\Big(\partial_t\widehat{\boldsymbol{F}}_t\,\widehat{\boldsymbol{F}}_t^{-1}\Big) = \widehat{J}_t\operatorname{div}\boldsymbol{u}(t,\mathcal{L}_t(\boldsymbol{\xi})),$$

which is the Lagrangian form of $D_tJ_t = J_t\operatorname{div}\boldsymbol{u}$ [10, Part I, Ch. 1, Sec. 1.3].

For $V(t) = \mathcal{L}_t(V(0))$, change variables in $I(t) = \int_{V(t)} \phi(t, \boldsymbol{x})\, d\boldsymbol{x}$:

$$I(t) = \int_{V(0)} \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))\widehat{J}_t(\boldsymbol{\xi})\, d\boldsymbol{\xi}.$$

Differentiating under the integral sign and substituting the Jacobian-evolution identity gives

$$I'(t) = \int_{V(t)} \left(D_t\phi + \phi\mathrm{div}\boldsymbol{u}\right)\, d\boldsymbol{x} = \int_{V(t)} \left(\partial_t\phi + \mathrm{div}(\phi\boldsymbol{u})\right)\, d\boldsymbol{x}.$$

If $V(t)$ has Lipschitz boundary and the trace of $\phi\boldsymbol{u}$ is integrable on $\partial V(t)$, the divergence theorem gives the boundary form

$$I'(t) = \int_{V(t)} \partial_t\phi\ d\boldsymbol{x} + \int_{\partial V(t)} \phi\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dS.$$

E.3 Linear Momentum Balance Details

Starting from the material-volume balance

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS + \int_{V(t)} \rho\boldsymbol{f_b}\, dV,$$

Reynolds transport gives

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{V(t)} \left(D_t(\rho\boldsymbol{u}) + \rho\boldsymbol{u}\,\mathrm{div}\boldsymbol{u}\right) dV.$$

Using mass conservation, $D_t\rho + \rho\mathrm{div}\boldsymbol{u} = 0$, this reduces to

$$\int_{V(t)} \rho D_t\boldsymbol{u}\, dV = \int_{V(t)} \rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) dV.$$

The surface traction term satisfies

$$\int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS = \int_{V(t)} \mathbf{div}(\boldsymbol{T})\, dV.$$

This is the tensor row-divergence convention from Definition D.6, applied under the regularity needed for the stress trace $\boldsymbol{T}\hat{\boldsymbol{n}}$ and the volume integral of $\mathbf{div}\boldsymbol{T}$. Since $V(t)$ is arbitrary within the localization class, the differential form is

$$\rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}.$$

The conservative Eulerian form

$$\partial_t(\rho \boldsymbol{u}) + \mathbf{div}(\rho \boldsymbol{u} \otimes \boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}$$

is equivalent whenever the continuity equation holds, because

$$\partial_t(\rho \boldsymbol{u}) + \mathbf{div}(\rho \boldsymbol{u} \otimes \boldsymbol{u}) = \rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}\right) + \boldsymbol{u}\left(\partial_t \rho + \mathrm{div}(\rho \boldsymbol{u})\right).$$

F Navier–Stokes Coordinate and Energy Details

This appendix collects mathematical orientation material for the 3D Navier–Stokes model. It is background for the continuum hierarchy; it is not evidence for the 1D study specification or any downstream validation claim. All formulas are read under the standing regularity and differential-operator conventions in Appendix D. In particular, $\mathbf{div}$ is row-divergence for tensor fields, $\Delta \boldsymbol{u}$ is the componentwise vector Laplacian, tensor norms are Frobenius norms when written, and η is dynamic viscosity while $\nu = \eta/\rho$ is kinematic viscosity as in Remark C.5.

F.1 Viscous-Term Simplification and Energy Context

For a Newtonian fluid with constant dynamic viscosity η and sufficiently smooth velocity field,

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \mathbf{div}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top) = \eta \Delta \boldsymbol{u} + \eta \nabla(\mathrm{div} \boldsymbol{u}).$$

For incompressible flow, $\mathrm{div}\boldsymbol{u} = 0$, so

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \Delta \boldsymbol{u}, \qquad \rho^{-1} \mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \nu \Delta \boldsymbol{u}, \qquad \nu = \eta/\rho.$$

This term is the momentum-diffusion contribution; it does not disappear in the incompressible Newtonian limit.

Classical local well-posedness results for Navier–Stokes depend on the domain, boundary conditions, forcing, initial-data space, and solution class. Under sufficiently smooth compatible data one obtains a unique strong solution on a maximal time interval. If the maximal interval is finite, continuation fails through blow-up of the norm controlled by the chosen well-posedness theory.

F.2 Cylindrical Coordinate Form

For (r, θ, z) with orthonormal basis $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$,

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad \boldsymbol{u} = u_r \boldsymbol{e}_r + u_\theta \boldsymbol{e}_\theta + u_z \boldsymbol{e}_z.$$

The component formulas are local cylindrical-coordinate formulas on patches where $r > 0$. Axis regularity, wall traces, and boundary conditions require separate hypotheses and are not supplied by the coordinate notation alone. The differential operators used in cylindrical divided-by-density Navier–Stokes forms are

$$
\begin{aligned}
(\boldsymbol{u}\cdot\nabla) &= u_r\partial_r + \frac{u_\theta}{r}\partial_\theta + u_z\partial_z,\\
\nabla p &= \partial_r p\,\boldsymbol{e}_r + \frac{1}{r}\partial_\theta p\,\boldsymbol{e}_\theta + \partial_z p\,\boldsymbol{e}_z,\\
\mathrm{div}\boldsymbol{u} &= \frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z.
\end{aligned}
$$

The rate-of-deformation components in the cylindrical orthonormal basis are

$$
\begin{aligned}
&D_{rr} = \partial_r u_r, && D_{\theta\theta} = \frac{1}{r}\partial_\theta u_\theta + \frac{u_r}{r}, && D_{zz} = \partial_z u_z,\\
&D_{r\theta} = \frac{1}{2}\left(\partial_r u_\theta - \frac{u_\theta}{r} + \frac{1}{r}\partial_\theta u_r\right), && D_{rz} = \frac{1}{2}(\partial_r u_z + \partial_z u_r), && D_{\theta z} = \frac{1}{2}\left(\frac{1}{r}\partial_\theta u_z + \partial_z u_\theta\right).
\end{aligned}
$$

Use the scalar shear-rate convention $\dot\gamma = \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}$. Then

$$
\tau_{ij} = 2\eta_{\mathrm{eff}}(\dot\gamma)D_{ij}, \qquad i,j\in\{r,\theta,z\}.
$$

When a source writes the law as a function of $D : D$ or $|D|_{\mathrm{F}}^2$, this report translates it through this convention, consistent with Remark C.5. Writing $\boldsymbol{f} = f_r\boldsymbol{e}_r + f_\theta\boldsymbol{e}_\theta + f_z\boldsymbol{e}_z$, the momentum components are

$$
\begin{aligned}
\text{(radial)}\quad & \partial_t u_r + u_r\partial_r u_r + \frac{u_\theta}{r}\partial_\theta u_r + u_z\partial_z u_r - \frac{u_\theta^2}{r}\\
&\quad = -\frac{1}{\rho}\partial_r p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rr}) + \frac{1}{r}\partial_\theta\tau_{r\theta} + \partial_z\tau_{rz} - \frac{1}{r}\tau_{\theta\theta}\right] + f_r,\\
\text{(azimuthal)}\quad & \partial_t u_\theta + u_r\partial_r u_\theta + \frac{u_\theta}{r}\partial_\theta u_\theta + u_z\partial_z u_\theta + \frac{u_r u_\theta}{r}\\
&\quad = -\frac{1}{\rho r}\partial_\theta p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{r\theta}) + \frac{1}{r}\partial_\theta\tau_{\theta\theta} + \partial_z\tau_{\theta z} + \frac{1}{r}\tau_{r\theta}\right] + f_\theta,\\
\text{(axial)}\quad & \partial_t u_z + u_r\partial_r u_z + \frac{u_\theta}{r}\partial_\theta u_z + u_z\partial_z u_z\\
&\quad = -\frac{1}{\rho}\partial_z p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rz}) + \frac{1}{r}\partial_\theta\tau_{\theta z} + \partial_z\tau_{zz}\right] + f_z.
\end{aligned}
$$

The incompressibility condition is

$$
\frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z = 0.
$$

G Numerical Methods Details

This appendix supplies the numerical-method detail behind Section 3.2 and the comparison operator in Section 3.10. It is supporting method detail, not a claim of external-solver parity for the implemented solver. The continuous identities are read under Convention D.1; computed quantities are read through the following numerical-record convention.

Convention G.1 (Computed realization record)**.** A computed realization is interpreted only through the model, closure, discretization, grid, timestep policy, boundary-state rule, output operator, and diagnostic metadata recorded for that realization. Symbols such as D_t, D_z, numerical fluxes, residual arrays, and sampled norms denote recorded discrete objects unless a section explicitly switches back to a continuum identity.

The time interval, output rectangle, and norm notation follow Definition D.3 and Definition D.7; discrete residuals use the recorded D_t, D_z, grid, and quadrature weights rather than unnamed continuum derivatives.

G.1 Fixed-wall stationary-Stokes refinement record

The local three-dimensional refinement study is limited to a fixed-wall stationary-Stokes problem on the smooth idealized stenosis geometry. The public entrypoint `run_stationary_stokes_refinement` constructs generated Gridap tetrahedral meshes from the same C^∞ radius profile used by the reduced model, solves a pressure-drop-driven stationary Stokes problem, evaluates section-averaged velocity and pressure from the finite-element field, and compares those averages with the resistance/projection quantities used by the stationary-Stokes initializer. This keeps the initializer behavior unchanged: the study is an additional diagnostic record, not a replacement for the 1D initial-condition contract.

For a Newtonian fixed-wall Stokes field, the sampled stress and wall-traction diagnostics use

$$\begin{aligned} \boldsymbol{\sigma}(\boldsymbol{u}, p) &= -p\mathbf{I} + \mu\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\top\right), & \mu &= \rho\nu, \\ \mathbf{t}_w &= \boldsymbol{\sigma}\mathbf{n}, & \boldsymbol{\tau}_w &= \mathbf{t}_w - (\mathbf{t}_w \cdot \mathbf{n})\mathbf{n}. \end{aligned}$$

Here $\mathbf{n}$ is the analytic outward normal of the C^∞ wall profile. The reported means and maxima are fixed-wall fluid-stress diagnostics sampled from the finite-element solution. They are not structural displacement, wall velocity, structural stress, or resolved FSI interface-work outputs.

G.2 Elastic-potential balance-law form

The reduced state is $\mathbf{U} = (A, Q)^\top$. For a member of the $\mathcal{W}_{\text{elastic1D}}$ balance-law contract with prescribed momentum-flux correction α, the reference elastic-potential form is

$$\begin{aligned} \partial_t A + \partial_z Q &= 0, \\ \partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A} + \Psi(A, z, t) \right) &= S_{\text{geom}}(A, z, t) - C_f(z, A, Q) \frac{Q}{A}. \end{aligned}$$

The elastic potential satisfies

$$\partial_A \Psi(A, z, t) = \frac{A}{\rho} \partial_A p_{\text{g}}(A, z, t),$$

and the geometry source is

$$S_{\text{geom}}(A, z, t) = \left. \partial_z \Psi(A, z, t) \right|_A - \frac{A}{\rho} \left. \partial_z p_{\text{g}}(A, z, t) \right|_A .$$

For the selected Canic–Koiter wall-law member, assuming p_{ext} is independent of A,

$$c^2(A, z) = \frac{A}{\rho} \partial_A p_{\text{g}}(A, z, t) = \frac{\beta(z)\sqrt{A}}{2\rho A_0(z)}.$$

Under the selected solver-coordinate map $A = \pi a$ and $A_0 = \pi R_0^2$, the implemented member corresponds to $\beta(z) = \sqrt{\pi} K R_0(z)^2 / R_{\max}^2$. Therefore

$$c^2(a, z) = \frac{K}{2\rho R_{\max}^2} \sqrt{a}$$

for the evolution wall law used in the reported finite-volume runs. Freeze z, the wall-closure data, and the source terms, and consider the homogeneous (A, Q) flux Jacobian associated with $(Q, \alpha Q^2/A + \Psi(A, z, t))^\top$. On a positive-area state $A > 0$, with flow rate Q, mean velocity Q/A, prescribed momentum-flux correction α, and $c^2(A, z) := (A/\rho)\partial_A p_{\text{g}}(A, z, t)$, its eigenvalues are

$$\lambda_\pm(A, Q; z) = \alpha \frac{Q}{A} \pm \sqrt{c^2(A, z) + \alpha(\alpha - 1) \left(\frac{Q}{A} \right)^2}.$$

For the displayed elastic wall law, $c^2 > 0$ when $\beta > 0$, $A > 0$, $A_0 > 0$, and $\rho > 0$. Strict hyperbolicity of the homogeneous elastic subsystem further requires the characteristic radicand

$$c^2(A, z) + \alpha(\alpha - 1) \left(\frac{Q}{A} \right)^2 > 0$$

to be positive on the reported state set. These speeds reduce to $Q/A \pm c(A,z)$ in the special $\alpha = 1$ reference case. The subcritical case used in Assumption 3.15 is $\lambda_- < 0 < \lambda_+$, so one characteristic enters through the inlet, one characteristic enters through the outlet, and one scalar boundary condition is imposed at each end of the vessel. The displayed derivatives of A_0, β, p_{g}, and Ψ are continuum notation for the model contract. A computed record may realize them through explicitly recorded finite-difference or source-split operators; changing that realization changes the computed map.

G.3 Semi-discrete solver operator

The implemented solver stores a radius-squared area coordinate in the variable `A`. In this subsection write that coordinate as $a = R^2$. The physical circular cross-sectional area is $A_{\mathrm{phys}} = \pi a$. This is a solver-specific convention; the continuum and classical 1D model sections keep A for physical cross-sectional area.

For N finite-volume cells, let

$$z_i = \left(i - \frac{1}{2}\right)\Delta z, \qquad \Delta z = \frac{L}{N}, \qquad u_h(t) = (a_1(t), \ldots, a_N(t), q_1(t), \ldots, q_N(t))^\top .$$

The semi-discrete system exposed by `semidiscretize` and `rhs!` is

$$\dot{u}_h = \mathcal{L}_h(u_h, t; \theta),$$

where θ denotes the recorded physical, geometry, grid, and solver parameters. Componentwise,

$$\begin{aligned}
\dot{a}_i &= -\frac{\widehat{F}^a_{i+1/2} - \widehat{F}^a_{i-1/2}}{\Delta z}, \\
\dot{q}_i &= -\frac{\widehat{F}^q_{i+1/2} - \widehat{F}^q_{i-1/2}}{\Delta z} + \widehat{S}_i.
\end{aligned}$$

All divisions by a use the positivity-regularized value of the radius-squared area coordinate.

With $K = Eh/(1-\sigma^2)$, Canic conservative-form reference radius $R_0^* = R_{\max}$, velocity-profile momentum factor $\alpha_{\mathcal{P}}$, and $\alpha_c(R_0') = -2(R_0')^2/35$, define $\iota_{\mathcal{M}} = 1$ for the `canic-extended-1d` forward-model descriptor and $\iota_{\mathcal{M}} = 0$ for `classical-1d-no-slip`, the manifest token for the classical parabolic-profile baseline. The semi-discrete full-state flux is

$$\mathbf{F}(a,q,z) = \left(q,\, (\alpha_{\mathcal{P}} + \iota_{\mathcal{M}}\alpha_c(R_0'(z)))\,\frac{q^2}{a} + \frac{K}{3\rho R_{\max}^2} a^{3/2}\right)^\top .$$

At each interface,

$$\widehat{\mathbf{F}}_{i+1/2} = \frac{1}{2}\left[\mathbf{F}(\mathbf{U}^-_{i+1/2}, z_{i+1/2}) + \mathbf{F}(\mathbf{U}^+_{i+1/2}, z_{i+1/2})\right] - \frac{1}{2} s_{i+1/2}\left(\mathbf{U}^+_{i+1/2} - \mathbf{U}^-_{i+1/2}\right),$$

where $\mathbf{U} = (a, q)^\top$. The MUSCL finite-volume method limits the conservative solver variables component-wise. For an interior cell,

$$\delta_i^a = \operatorname{minmod}(a_i - a_{i-1},\, a_{i+1} - a_i), \qquad \delta_i^q = \operatorname{minmod}(q_i - q_{i-1},\, q_{i+1} - q_i),$$

$$\mathbf{U}^-_{i+1/2} = (a_i + \tfrac{1}{2}\delta_i^a,\, q_i + \tfrac{1}{2}\delta_i^q)^\top, \qquad \mathbf{U}^+_{i+1/2} = (a_{i+1} - \tfrac{1}{2}\delta_{i+1}^a,\, q_{i+1} - \tfrac{1}{2}\delta_{i+1}^q)^\top.$$

The active limiter is the two-argument minmod limiter, equivalent to the $\theta = 1$ member of the generalized minmod family. Boundary-adjacent cell slopes are set to zero and paired with the ghost states described in Appendix G.5. Positivity limiting is applied to reconstructed a states before flux evaluation. The cell source is not MUSCL-reconstructed; it is evaluated at cell centers using the recorded D_z^h differences below. The first-order finite-volume method is the separate path that passes neighboring cell and ghost states directly to the same Rusanov flux.

The implemented finite-volume surface also includes an opt-in characteristic WENO3 reconstruction. At an interior interface $i + 1/2$ with the four-cell stencil $(i - 1, i, i + 1, i + 2)$, the implementation freezes the homogeneous flux-Jacobian eigenvalues at the local reference state

$$\mathbf{U}_{\mathrm{ref}} = \frac{1}{2}(\mathbf{U}_i + \mathbf{U}_{i+1})$$

and projects conservative states into characteristic components

$$w^- = \frac{\lambda^+ a - q}{\lambda^+ - \lambda^-}, \qquad w^+ = \frac{-\lambda^- a + q}{\lambda^+ - \lambda^-}.$$

For a scalar characteristic component with stencil values (v_{i-1}, v_i, v_{i+1}), the left state uses

$$p_0^- = -\frac{1}{2}v_{i-1} + \frac{3}{2}v_i, \qquad p_1^- = \frac{1}{2}v_i + \frac{1}{2}v_{i+1},$$

$$\beta_0^- = (v_i - v_{i-1})^2, \qquad \beta_1^- = (v_{i+1} - v_i)^2,$$

$$\omega_k^- = \frac{d_k^-/(\varepsilon + \beta_k^-)^2}{\sum_{\ell=0}^{1} d_\ell^-/(\varepsilon + \beta_\ell^-)^2}, \qquad (d_0^-, d_1^-) = \left(\frac{1}{3}, \frac{2}{3}\right),$$

$$v^-_{i+1/2} = \omega_0^- p_0^- + \omega_1^- p_1^-.$$

The right state is formed symmetrically from (v_i, v_{i+1}, v_{i+2}) and then mapped back to (a, q) through $a = w^- + w^+$ and $q = \lambda^- w^- + \lambda^+ w^+$. The WENO3 path uses the same Rusanov flux and source discretization as MUSCL, applies the same positive-area protection to reconstructed a states, and falls back to the existing ghost-state/minmod reconstruction at the domain faces and boundary-adjacent interfaces where a full WENO3 stencil is not available.

The implemented spatial surface also includes a native Richtmyer/Lax–Wendroff finite-volume path. It uses the same minmod-limited interface states as above, then forms the half-step interface state

$$\mathbf{U}^{\mathrm{LW}}_{i+1/2} = \frac{1}{2}\left(\mathbf{U}^{-}_{i+1/2} + \mathbf{U}^{+}_{i+1/2}\right) - \frac{\Delta t}{2\Delta z}\left[\mathbf{F}(\mathbf{U}^{+}_{i+1/2}, z_{i+1/2}) - \mathbf{F}(\mathbf{U}^{-}_{i+1/2}, z_{i+1/2})\right],$$

with positivity limiting on its a component before the interface flux is evaluated as

$$\widehat{\mathbf{F}}^{\mathrm{LW}}_{i+1/2} = \mathbf{F}(\mathbf{U}^{\mathrm{LW}}_{i+1/2}, z_{i+1/2}).$$

Because this path depends explicitly on $\Delta t/\Delta z$, it is a native fixed-step alternative and is not the semi-discrete operator passed to the SciML `ODEProblem` backend.

The native modal DG option represents each cell state by Legendre coefficients,

$$\mathbf{U}_i(\xi, t) = \sum_{m=0}^{p} \widehat{\mathbf{U}}_{i,m}(t) P_m(\xi), \qquad -1 \le \xi \le 1, \qquad p \in \{0, 1, 2\}.$$

The implemented DG interface traces are evaluated at $\xi = \pm 1$ and passed to the same Rusanov flux form, while cell-volume flux and source contributions use the recorded quadrature rule. The degree-zero DG path is recorded as the cell-average finite-volume bridge; degrees one and two are native modal-DG benchmark rows, not the selected C23/C40 comparison method.

The local numerical speed for the Rusanov finite-volume and DG interface fluxes is

$$s_{i+1/2} = \max\left\{s_h(\mathbf{U}^{-}_{i+1/2}, z_{i+1/2}), s_h(\mathbf{U}^{+}_{i+1/2}, z_{i+1/2})\right\},$$

with

$$\begin{aligned} s_h(a, q, z) &= \max\left\{|\alpha_{\mathrm{eff}} u - \chi|, |\alpha_{\mathrm{eff}} u + \chi|\right\}, \\ \alpha_{\mathrm{eff}} &= \alpha_{\mathcal{P}} + \iota_{\mathcal{M}} \alpha_c(R_0'(z)), \qquad u = \frac{q}{a}, \\ \chi^2 &= (\alpha_{\mathrm{eff}} u)^2 - \alpha_{\mathrm{eff}} u^2 + \frac{K}{2\rho R_{\max}^2}\sqrt{a}. \end{aligned}$$

This standard Rusanov/MUSCL operator is not presented here as an exactly well-balanced discretization of the geometry-rest state. In particular, at $q_i = 0$ with spatially varying $a_i = R_0(z_i)^2$, the numerical viscosity in the mass flux generally contributes a nonzero discrete residual unless an equilibrium reconstruction or path-conservative flux is added. Thus "source-balanced" in the main text refers to the continuous flux/source pair; this report does not claim $\mathcal{L}_h(R_0(z_i)^2, 0) = 0$ to machine precision for the displayed operator.

The source term uses the specified centered interior and one-sided boundary difference D_z^h on the cell averages.

Let $g_{\mathcal{P}}$ denote the profile shear factor,

$$\dot{\gamma}_{1D,i} = g_{\mathcal{P}} \frac{|q_i|}{a_i \sqrt{a_i}}, \qquad \nu^{\mathcal{R}}_{\mathrm{eff},i} = \frac{\eta^{\mathcal{R}}_{\mathrm{eff}}(\dot{\gamma}_{1D,i})}{\rho},$$

and $\alpha'_c(z) = -4R'_0(z)R''_0(z)/35$. The implemented source is

$$\begin{aligned}
\widehat{S}_i &= -2\nu^{\mathcal{R}}_{\mathrm{eff},i} g_{\mathcal{P}} \frac{q_i}{a_i} + \frac{K}{\rho R^2_{\max}} a_i R'_0(z_i) - \iota_{\mathcal{M}} a_i \widehat{\partial_z p_2}_i + \iota_{\mathcal{M}} \frac{q_i^2}{a_i} \alpha'_c(z_i), \\
\widehat{\partial_z p_2}_i &= \nu^{\mathcal{R}}_{\mathrm{eff},i} g_{\mathcal{P}} \left[\frac{q_i R''_0(z_i)}{a_i R_0(z_i)} - \frac{q_i (R'_0(z_i))^2}{a_i R_0(z_i)^2} + \frac{(D^h_z q)_i R'_0(z_i)}{a_i R_0(z_i)} - \frac{q_i (D^h_z a)_i R'_0(z_i)}{a_i^2 R_0(z_i)} \right].
\end{aligned}$$

The $\widehat{\partial_z p_2}_i$ term is the density-scaled discrete derivative of the Canic variable-radius viscous pressure correction. It is tied to the `canic-extended-1d` realization and is identically inactive in the model-family specification when $\iota_{\mathcal{M}} = 0$, while the elastic stiffness terms above come from the selected Canic–Koiter wall law. For the `newtonian` descriptor, this reduces to the constant value $\nu^{\mathcal{R}}_{\mathrm{eff},i} = \nu$. For the non-Newtonian closures, the effective viscosity is evaluated with the shear-rate regularization and any declared interval projection before division by density, then held locally fixed in the analytic p_2 derivative. The realized generalized-Newtonian p_2 contribution is therefore a computational closure rather than a fully rederived variable-viscosity pressure correction. For the power-family profile, $g_{\mathcal{P}} = \gamma + 2 = \alpha_{\mathcal{P}}/(\alpha_{\mathcal{P}} - 1)$. The flat profile is not evaluated through that singular limit; it is assigned $\alpha_{\mathcal{P}} = 1$ and an explicit finite $g_{\mathcal{P}}$.

G.4 Time-integration realization

The native time integrator selected for the reported comparison advances the semi-discrete operator with the declared third-order strong-stability-preserving Runge–Kutta map

$$u_h^{n+1} = \Phi^{\mathrm{SSPRK3}}_{\Delta t_n} \left(u_h^n; \mathcal{L}_h \right).$$

In expanded form, using Π for the recorded positive-area projection on the a-components,

$$\begin{aligned}
u^{(1)} &= \Pi \left(u^n + \Delta t_n \mathcal{L}_h(u^n, t^n) \right), \\
u^{(2)} &= \Pi \left(\frac{3}{4} u^n + \frac{1}{4} \left[u^{(1)} + \Delta t_n \mathcal{L}_h(u^{(1)}, t^n + \Delta t_n) \right] \right), \\
u^{n+1} &= \Pi \left(\frac{1}{3} u^n + \frac{2}{3} \left[u^{(2)} + \Delta t_n \mathcal{L}_h(u^{(2)}, t^n + \Delta t_n) \right] \right).
\end{aligned}$$

The native code also exposes Forward Euler, SSPRK2, and the standard five-stage fourth-order SSPRK54 stepper for diagnostic and higher-order sensitivity rows; they are not the selected time integrators for the principal comparison. SSPRK54 reuses the same stage right-hand side, positive-area projection, and diagnostics accounting as the SSPRK3 path. The native step size is

$$\Delta t_n = \min\left\{\Delta t_{\max}, \frac{\mathrm{CFL}\,\Delta z}{\max_i s_h(a_i^n, q_i^n, z_i)}, T_{\text{final}} - t^n\right\}.$$

The SciML realization constructs an `ODEProblem` for the same $\dot{u}_h = \mathcal{L}_h(u_h, t; \theta)$ and applies the selected OrdinaryDiffEq policy recorded in the solve specification. The local policy surface exposes `auto`, `tsit5`, `vern7`, `vern9`, and `rodas5p`. Thus comparisons between these realizations change the time integrator, not the semi-discrete spatial operator, for the spatial methods supported by that backend. The native fully discrete path applies the stage projection Π, whereas the SciML path integrates the unprojected ODE system. If Π activates, the compared paths are no longer identical fully discrete problems; the archived comparison records do not persist projection-activation counts. The current SciML backend excludes the Richtmyer/Lax–Wendroff path and DG degrees greater than zero; those rows remain native fixed-step method checks.

G.5 Boundary-state realization

The finite-volume boundary realization imposes one incoming boundary condition at each boundary in the subcritical regime using an $\alpha = 1$, fixed-area characteristic approximation. This is the implemented boundary-state rule for the variable-α_{eff} solver, not an exact invariant derivation for the full variable-radius balance law. Let

$$c_0 = \left(\frac{K}{2\rho(R_0^*)^2}\right)^{1/2} = \left(\frac{K}{2\rho R_{\max}^2}\right)^{1/2}.$$

At the inlet, the incoming flow is $q_{\text{in}} = R_0(0)^2\bar{u}_{\text{in}}$. The outgoing invariant from the first interior cell is

$$w_- = \frac{q_1}{a_1} - 4c_0 a_1^{1/4},$$

and the ghost coordinate a_{in} solves

$$\frac{q_{\text{in}}}{a_{\text{in}}} - 4c_0 a_{\text{in}}^{1/4} = w_-.$$

At the outlet, the radius-squared coordinate is fixed to $a_{\text{out}} = R_0(L)^2$. The outgoing invariant from the last interior cell is

$$w_+ = \frac{q_N}{a_N} + 4c_0 a_N^{1/4},$$

and the ghost flow is

$$q_{\mathrm{out}} = a_{\mathrm{out}} w_+ - 4c_0 a_{\mathrm{out}}^{5/4}.$$

The boundary update therefore constrains the incoming information and leaves the outgoing information tied to the interior state. Boundary-quality plots and reflected-wave sidecars are diagnostics until explicit thresholds are admitted.

G.6 Residual diagnostics

Stored residuals are post-processing diagnostics on sampled fields, not proof of the continuous PDE residual. The symbols D_t and D_z in this subsection denote the recorded temporal and spatial finite-difference operators on the saved grid, not weak derivatives. For a stored field sequence, the semi-discrete defect is read as

$$\mathcal{E}_h^n = D_t u_h^n - \mathcal{L}_h(u_h^n, t^n; \theta).$$

Componentwise, this gives

$$\mathcal{E}_i^a(t^n) = D_t a_i^n - \mathcal{L}_{h,i}^a(u_h^n, t^n; \theta), \qquad \mathcal{E}_i^q(t^n) = D_t q_i^n - \mathcal{L}_{h,i}^q(u_h^n, t^n; \theta).$$

The E1 diagnostic summary reports these residuals on reviewed interior cells with recorded finite-field, positivity, CFL, and pressure-drop sign checks. The numbers support the dataset contract and local consistency reading; they do not establish a convergence theorem or external parity. Any residual norm or error norm reported from these arrays is a sampled norm tied to the numerical record, rather than an unqualified continuum $L^p(K)$ norm.

G.7 Full rest-state drift grid

Chapter 4 reports a compressed summary of the zero-forcing, zero-inlet geometry-rest drift test. The full generated grid and elapsed-time table is retained here so that the main-text interpretation can be checked against the underlying rows.

Table 20: Full zero-forcing, zero-inlet geometry-rest drift diagnostics.

Severity	N	t	$\lvert\Delta t_{\mathrm{term}}\rvert$	$\max\lvert q_i\rvert$	$\max\lvert a_i - R_{0,i}^2\rvert$	Mass defect	CFL max	Subcrit. margin
23	100	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
23	100	0.001	0.000e+00	0.7287	6.999e-04	7.293e-04	0.1772	894.0
23	100	0.01	2.394e-16	1.492	7.758e-04	0.001592	0.1819	814.2
23	100	0.1	6.078e-15	0.8474	4.064e-04	0.001091	0.1823	804.5
23	100	1.0	0.000e+00	0.8693	1.683e-04	1.603e-04	0.1823	804.5
23	200	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
23	200	0.001	0.000e+00	0.7288	7.002e-04	7.291e-04	0.3544	900.1
23	200	0.01	2.394e-16	1.534	7.559e-04	0.001588	0.364	814.9
23	200	0.1	6.078e-15	0.7431	4.249e-04	0.001215	0.3653	802.1
23	200	1.0	0.000e+00	0.7648	6.892e-05	1.437e-04	0.3653	802.1
23	400	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
23	400	0.001	0.000e+00	0.7289	7.003e-04	7.291e-04	0.45	901.5
23	400	0.01	0.000e+00	1.559	7.422e-04	0.001577	0.45	814.1
23	400	0.1	0.000e+00	0.7657	4.615e-04	0.001242	0.45	800.9
23	400	1.0	0.000e+00	0.738	5.022e-05	1.324e-04	0.45	800.9
23	800	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
23	800	0.001	0.000e+00	0.729	7.003e-04	7.290e-04	0.45	902.0
23	800	0.01	0.000e+00	1.566	7.401e-04	0.001574	0.45	814.0
23	800	0.1	0.000e+00	0.7852	4.824e-04	0.001249	0.45	800.4
23	800	1.0	0.000e+00	0.7313	4.902e-05	1.294e-04	0.45	800.4
40	100	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
40	100	0.001	0.000e+00	0.8534	6.999e-04	7.293e-04	0.1772	777.3
40	100	0.01	2.394e-16	1.524	9.094e-04	0.002057	0.1821	666.9
40	100	0.1	6.078e-15	0.8869	3.437e-04	6.292e-04	0.1828	640.4
40	100	1.0	0.000e+00	0.9366	3.367e-04	3.388e-04	0.1828	640.4
40	200	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
40	200	0.001	0.000e+00	0.7288	7.002e-04	7.291e-04	0.3544	791.6
40	200	0.01	2.394e-16	1.576	8.382e-04	0.001996	0.3648	666.7
40	200	0.1	6.078e-15	0.7281	3.155e-04	8.581e-04	0.3669	629.9
40	200	1.0	0.000e+00	0.7826	1.148e-04	2.240e-04	0.3669	629.9
40	400	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
40	400	0.001	0.000e+00	0.7289	7.003e-04	7.291e-04	0.45	794.7
40	400	0.01	0.000e+00	1.64	7.984e-04	0.001944	0.45	664.3
40	400	0.1	0.000e+00	0.7283	3.706e-04	9.607e-04	0.45	626.1
40	400	1.0	0.000e+00	0.7425	7.546e-05	1.731e-04	0.45	626.1
40	800	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	–	–
40	800	0.001	0.000e+00	0.729	7.003e-04	7.290e-04	0.45	795.9
40	800	0.01	0.000e+00	1.658	7.941e-04	0.001925	0.45	662.2
40	800	0.1	0.000e+00	0.7286	3.997e-04	9.891e-04	0.45	624.9
40	800	1.0	0.000e+00	0.7324	6.832e-05	1.566e-04	0.45	624.9

H Code Availability and Build Statement

This report is built from the LaTeX sources rooted at `final-report.tex`. The compiled PDF should be treated as the rendered form of those sources. The report package used for this build is the repository root containing `final-report.tex`. Because no public remote or archival release is declared here, this appendix supports internal reproducibility of the local report package only; it does not claim a public repository or archival URL.

The Julia simulation code is packaged as `StenosisHemodynamics` in the root Julia project. The tracked source inventory is:

Tracked source	Role in the report package
`Project.toml, Manifest.toml`	Julia package identity, dependency declarations, compatibility bounds, and resolved dependency versions.
`src/StenosisHemodynamics.jl`	Package entry point and public export surface for the solver, studies, resolved-3D comparison, and report figure export helpers.
`src/StenosisHemodynamics/**/*.jl`	Solver components organized under adapter, core, CLI, IO, numerics, and workflow subdirectories: model descriptors, geometry, state layout, finite-volume and DG operators, native and SciML backends, Stokes initial condition, OpenBF-style adapter, studies, refinement, resolved-3D diagnostics, output writers, package benchmarks, and geometry-export helpers.
`scripts/stenosis-hemodynamics, scripts/stenosis-hemodynamics.jl`	Primary Julia command surface. The dispatcher exposes simulation, OpenBF adapter, studies, stationary-Stokes refinement, resolved-3D comparison, benchmarks, and report asset export through `StenosisHemodynamics.run_cli`.
`scripts/render_*.py, scripts/audit_*.py`	Python auxiliary tooling for report figure/table rendering and TeX/reference audits. Python is not a simulation surface in this package.
`test/runtests.jl, test/test_*.jl`	Julia test orchestrator and feature-specific regression tests for model closures, backends, comparison diagnostics, CLI parsing, studies, benchmarks, and exports.
`scripts/julia-release`	Repository launcher that selects Julia 1.12 or newer and binds the root project environment.

The standard package import surface used by report regeneration commands is:

```julia
using StenosisHemodynamics

r = run_available_resolved3d_comparison(
    output_dir="simulations/output/3d_comparison/full_t1_native_nx400",
    overwrite=true,
)
publish_resolved3d_report_assets(r; overwrite=true)
```

The package-benchmark suite is regenerated separately from the tracked PDF. A smoke run writes deterministic schema checks; an overnight run expands the same schemas to the full benchmark matrix:

```
./scripts/stenosis-hemodynamics benchmark \
  --profile overnight \
```

```
  --output-dir simulations/output/package_benchmark/<run_id> \
  --overwrite \
  --include-resolved3d \
  --publish-report-assets

pipenv run python scripts/render_package_benchmark_figures.py \
  --benchmark-dir simulations/output/package_benchmark/<run_id>
```

The tracked package-benchmark assets used in Section 4.2 are copied under `figures/static/static/data/package-benchmark/` and rendered under `figures/static/static/rendered/`. The tracked assets were generated from the `overnight-20260618-quadrature/` benchmark run and then pruned to the Julia-authoritative benchmark stages. The recorded Julia runtime context was Julia `1.12.6`, with generation commit `24d311c0ddd169f943169b05736d439d9b877b57`. That commit identifies the benchmark/data-generation base state, not by itself the complete integrated manuscript state after the final report edits. The integrated report state is identified by the tracked source tree containing this appendix together with the file hashes below. The published benchmark CSVs contain 243 status rows across the seven retained report stages, all marked `ok`; this is an internal status summary, not an external validation claim.

The tracked package-benchmark data and table fragment have the following SHA-256 hashes:

Tracked package-benchmark file	SHA-256
`figures/static/static/data/package-benchmark/case_results.csv`	`0843a3e3fa986e0ec1227bedab1ff8f` `616e3d3652b9e5e609c64a171e6d124fe`
`figures/static/static/data/package-benchmark/refinement.csv`	`de20c481fbbfe4a5930d996887a55a2c` `3fe3734de40422ff9fc83f4f4fc59e7e`
`figures/static/static/data/package-benchmark/backend_parity.` `csv`	`1a52b83cebb750cca68ac68f1a508dda` `11b1c649ed62f71fe239582ee4ddf671`
`figures/static/static/data/package-benchmark/stokes_ic.csv`	`2379d100d1b01943b1afcd79718f298` `10bad0fec9762796c72d78158963d6adb`
`figures/static/static/data/package-benchmark/rheology_` `profile.csv`	`8c83c6119d54dd3465eb8c3ec1e96897` `2a981bd41afdbe34861ea4151c4fb62a`
`figures/static/static/data/package-benchmark/boundary_openbf.` `csv`	`90fcac4d39e1e0b705837597ff15b929` `cb165a84fc56853a70839d53914c993d`
`figures/static/static/data/package-benchmark/resolved3d.csv`	`96443ae2dd54f9a0132ac7979209b256` `0353195ee7ebb87d07a4778b66d1a53a`
`figures/static/static/tables/package-benchmark/` `package-benchmark-summary.tex`	`f256ac7e5b3072587f5a3f3863a4181d` `fdf923e6374e117f2fcb1c0161d12386`

The tracked package-benchmark rendered figures used by the PDF have the following SHA-256 hashes:

Tracked rendered file	SHA-256
figures/static/static/rendered/package-benchmark-convergence.pdf	91cc997692d3975b30f20a0ed4a9b904 5e0a9440d0d53211d09371bfb548a9fd
figures/static/static/rendered/package-benchmark-backend-parity.pdf	4e6a065a89942ea4c79f1f4795e3e70c 3504e949f8d7da72703cf8ec0b08ae57
figures/static/static/rendered/package-benchmark-rheology-profile.pdf	ab5f8979dfdce9d84d7514330a3085aa 75e7d4a89fc4e6408975498dc5ba86cd
figures/static/static/rendered/package-benchmark-resolved3d.pdf	913363fda4ae76489017eb3a83dadc3a 3b9febccfb99713b2ddfe278965c6482

The Section 5 comparison tables and plots in this build are the tracked report assets under `figures/static/static/data/stenosis-comparison/`. Scratch sensitivity runs for the revision use ignored directories under `simulations/output/revision-*` and `tmp/revision-evidence/`; those scratch runs are evidence inputs, not the manuscript assets used by this build. The command shape for regenerating the tracked comparison assets, when the raw 3D inputs are restored, is:

```
./scripts/stenosis-hemodynamics compare-3d \
  --data-root simulations/data/3d/canic_case3 \
  --output-dir simulations/output/3d_comparison/full_t1_native_nx400 \
  --nx 400 \
  --target-time 1.0 \
  --time-atol 1e-3 \
  --overwrite \
  --publish-report-assets \
  --report-assets-dir figures/static/static/data/stenosis-comparison \
  --radial-bin-counts 10,20,40 \
  --radial-radius-modes current,reference
```

The raw upstream XDMF/HDF5 inputs used for Figure 12 and Section 5 were restored locally from `https://github.com/qcutexu/Extended-1D-AQ-system`, commit `056a9da2b36b480691f18025d242d2c00f6e7180`, under `case3_all_3d_results/`. The local copies under `simulations/data/3d/canic_case3/` are ignored by git and are not archived in this report package. Rerunning the raw XDMF/HDF5 extraction therefore requires those local inputs or the upstream repository. The observed local raw velocity-input hashes are:

Ignored local input	SHA-256
simulations/data/3d/canic_case3/77/velocity.xdmf	8cbb61e8d6a8b9fbfe9662bea7544eb8 9ef58390cdcc93a5d0f68fbb19c110b7
simulations/data/3d/canic_case3/77/velocity.h5	315157b3c9a9a92d1e99b26c3e9c393f 22303a676b3b24f52287d0b8cd22d168
simulations/data/3d/canic_case3/60/velocity.xdmf	765adb3f603b2306144dfb9be6ffec5573 23686867172ea9cc7b6e3f246284e0
simulations/data/3d/canic_case3/60/velocity.h5	1dee423991e35ced10a051e20b0598ee 0cbea3d59521b0fc3cb490b49bb4fe41

The tracked static data copied into the PDF has the following SHA-256 hashes:

Tracked file	SHA-256
figures/static/static/data/stenosis-comparison/ section-quadrature.dat	5f3b8f2be5fa4bbb1035abf7acca995e a38d2667642ed1cfe9c6ab39ee16b431
figures/static/static/data/stenosis-comparison/ radial-quadrature-C23.dat	a2ce3e183c87946d80984186ebd20815 87e702a77854404c5dc715303727f70a
figures/static/static/data/stenosis-comparison/ radial-quadrature-C40.dat	bec0f0d6658809dae1cf4af5dff260be 69a19c347c1d5ce0a0a494330ed8b3e2
figures/static/static/data/stenosis-comparison/ node-slab-sensitivity.csv	7af64a08ef4e84bcfe4770ab2b37be82 9b394a9cb9fcf3ad2ebd94951fbc9306
figures/static/static/data/stenosis-comparison/area-audit.dat	98e1fbb5b06ebe8cd928784e8c48fdc3 9cc07b0dde228c58edaf82107a738696
figures/static/static/data/stenosis-geometry/resolved_ velocity_nodes_manifest.csv	372796413873c09581ced890f8de657e f39e9ff69eabca8ce46b51b0cf758d3f
figures/static/static/data/stenosis-geometry/resolved_ velocity_nodes_case77_sev23.csv	0c0451dcee29216d967cecb11d36f518 a95c469fbcf18a1b472d7b9f01d211d0
figures/static/static/data/stenosis-geometry/resolved_ velocity_nodes_case60_sev40.csv	eb4b2d5149128740bfbb89ff219e92a47 d0535f3a7fd45078732b837619f68d1
figures/static/static/data/stenosis-geometry/resolved_ envelope_manifest.csv	d8236b5549851ff97eb9d9628b2bfbc6 60f0e594738b634509d0538476ef25a7
figures/static/static/data/stenosis-geometry/resolved_ envelope_case77_sev23.csv	7ffbbdbf52e4cab5c27e7aebc54bc47d 8aa0e7e2ab2b7058c61da8f06bb40192
figures/static/static/data/stenosis-geometry/resolved_ envelope_case60_sev40.csv	e661114841a0b3f8abccb0dc244e2d4 b4409291c8f02dcd5a4da2398892a7c1d
figures/static/static/rendered/resolved-3d-flow-field.pdf	dbee4f2bc37c980c8ce99f172e26edbf 90eaf8cb1b23e1fb787a77298a4e8aca

The tracked targeted-verification tables introduced for the major revision have the following SHA-256 hashes. They were generated with the package CLI:

```
./scripts/stenosis-hemodynamics verify mms \
  --nxs 50,100,200,400 \
  --dt-values 2e-5,1e-5,5e-6,2.5e-6 \
  --output-dir figures/static/static/tables/verification \
  --overwrite

./scripts/stenosis-hemodynamics verify rest \
  --severities 23,40 \
  --nxs 100,200,400,800 \
  --elapsed-times 0,0.001,0.01,0.1,1.0 \
  --output-dir figures/static/static/tables/verification \
  --overwrite

./scripts/stenosis-hemodynamics stokes refine \
  --severities 23,40 \
  --meshes 8x2x8,16x4x16 \
  --pressure-drop-pa 40 \
  --summary-csv figures/static/static/tables/verification/stationary_stokes_refinement.csv \
```

```
  --overwrite \
  --parallel-workers 0
```

Tracked verification file	SHA-256
figures/static/static/tables/verification/mms_verification.csv	071f8fd53bfdd49dbd58f42a6613bdfb576e509b3ee0ffc23d43191bc218e408
figures/static/static/tables/verification/mms_verification.tex	c0c768fee47bb33d52cc6043b67dec079bbd17e9e1abc45f0ed591699f10ec02
figures/static/static/tables/verification/rest_state_drift.csv	4ef2126c5d1edb34e7d27e217bba40bc718cef008f85cef1bba2cdea10fedf3f
figures/static/static/tables/verification/rest_state_drift.tex	3e7dd97f6c1b4437ba2ca49107a56d6fc61c88857f9164a30ae28354fb1efe41
figures/static/static/tables/verification/rest_state_drift_full.tex	f3ff7f2931ff191958f3c811ce3f1faf16007797456a18c6250e98996f2b4405
figures/static/static/tables/verification/stationary_stokes_refinement.csv	cdc3ee48b4f693a019258d7de05188774c91b1e4591b21aef58226a75cc404f4
figures/static/static/tables/verification/stationary_stokes_refinement.tex	3396042c43c389a6589091061565872ebfbcd58230c009fd15934da27940c51a

The non-archived comparison artifact observed for this build contains the following output hashes; these files are provenance evidence, not tracked archive contents:

Generated artifact file	SHA-256
comparison_summary.csv	a3009b5672ca52ac555c3e7a26c7fe32e2ea3567d8d5e457293365615c1ba69b
section_mean_case77_sev23.csv	5f6874c2e5086e91c693f6de9584f68fbf95a21aa182640236445dede4c5fcf4
section_mean_case60_sev40.csv	b4f6500000c854c7caf33699b1c156c7f822c0a17c62800ab52081f6149593da
radial_profile_case77_sev23.csv	be6dec4f08d13ebb68bce007bcedbb003df7ee8eeaab4c6b220738d9f3b92b4f
radial_profile_case60_sev40.csv	6af2f16af0ab6ed195d9298a68413b9fe148034d8eaba8ef1dfbfe453c5f7ea0
node_slab_sensitivity.csv	7af64a08ef4e84bcfe4770ab2b37be829b394a9cb9fcf3ad2ebd94951fbc9306
section_quadrature_overlay.svg	119d73f560cf4e44009002d1f033e318efa63002476bfe58aa9b4e932e358937

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